Sufficiency-aware representation learning for microstructure-to-failure prediction: Certified Lipschitz barriers, cross-fitted McShane regularization, and transfer to phase-field fracture

A.E. Gómez-Ovalle[a,*], C. Ramírez Ovalle[b]

[a] *University of Burgos, Escuela Politécnica Superior, 09006 Burgos, Spain*

[b] *Department of Natural Sciences & Mathematics, Pontificia Universidad Javeriana – Cali, Cali, Colombia*

Abstract

Surrogate models for microstructure–property relations are typically evaluated by average prediction error (R^2 or RMSE), conflating three distinct properties of an encoder $z = E_\theta(x)$: functional sufficiency for a mechanical response $R(x)$, metric regularity with respect to a normalized latent distance d_Ψ, and the accuracy of a specific readout or interpolant. We introduce a diagnostic benchmark and a representation-learning objective grounded in the exact finite-sample Lipschitz error barrier $e^\star_\theta(\Lambda) = \frac{1}{2}\max_{i,j}[|R_i - R_j| - \Lambda d_z(i,j)]_+$, attained on sample anchors by the McShane–Whitney central midpoint extension M_A. To regularize worst-case latent neighborhoods, we propose a cross-fitted McShane barrier regularizer (`SBR-CF`) combined with cross-batch anchor memory, hard-query mining, periodic-translation and dihedral D_2 invariance, generalized-mean (GeM) pooling, and a spectrally normalized Λ-Lipschitz head. Across a 3,300-cell two-phase elastic corpus and an externally frozen AT1 phase-field fracture benchmark, we show that: (i) high-fidelity residual autoencoders (`ConvAE-HiFi`, IoU = 0.728) still exhibit high worst-case insufficiency on nonlocal initiation hazard $R_H(\beta = 20)$ ($\Omega/\Omega^{\text{null}} = 0.882$ under d_{white}); (ii) relative to standard MSE supervision ($\Omega/\Omega^{\text{null}} = 0.273$, $R^2_{\text{McShane}} = 0.947$), `SBR-CF` lowers whitened 1%-ball insufficiency to 0.188–0.192, raises out-of-sample McShane R^2_{McShane} to 0.962, halves the certified training head excess gap ($0.513 \to 0.248$), and mitigates high-dimensional whitening dilution at $d = 16$ ($0.433 \to 0.238$), while exact global barriers $e^\star_\theta(\Lambda)$ depend on both metric and budget (`RNC` and `Tail-MSE` achieve lower ordinates under d_{white}, whereas `SBR-CF` achieves the lowest ordinates in $d_{\mathcal{N}}$ at $\Lambda/\sigma_R \in \{2, 3\}$); (iii) under coordinate standardization and regularized whitening, Rank-N-Contrast (`RNC`) achieves the strongest single-target alignment ($\Omega/\Omega^{\text{null}} = 0.085$–$0.148$) and linear contrast/biaxial transfer via a 1D embedding ($d_{\text{eff}} = 1.004$), whereas `SBR-CF` preserves a multi-dimensional geometry ($d_{\text{eff}} = 5.21$–5.77) that is empirically stable across whitening regimes and transfers better across hazard exponents under d_{white}; and (iv) augmenting the critical-distance fracture descriptor $E_1 + J_{1.5\ell}$ with frozen `SBR-CF` embeddings lifts prospective high-multiplicity peak-strength R^2 from 0.679–0.689 to 0.939 (mean relative peak-stress error 0.94%, vs. 0.922 for +`MSE`), whereas frozen pre-critical elastic embeddings paired with linear ridge readouts do not resolve the post-critical percolating crack-path cost R_{path} ($R^2 \leq 0.12$).

Keywords: representation sufficiency, Lipschitz error barrier, McShane–Whitney extension, microstructure homogenization, phase-field fracture, equivariant neural networks

*Corresponding author

Email address: `aegovalle@ubu.es` (A.E. Gómez-Ovalle)

1. Introduction

1.1. Motivation: Separating representation adequacy from readout specification

A microstructure-to-property surrogate typically follows the composite architecture

$$x \xrightarrow{E_\theta} z \xrightarrow{g_\phi} \widehat{R}, \tag{1}$$

where $x \in \mathcal{X}$ is a high-dimensional microstructure image, $z = E_\theta(x) \in \mathbb{R}^d$ is a reduced latent representation, and $R : \mathcal{X} \to \mathbb{R}$ is a target mechanical response. Standard machine-learning protocols optimize and evaluate the end-to-end composition $g_\phi \circ E_\theta$ using average error metrics such as mean squared error (MSE), mean absolute error (MAE), or the coefficient of determination R^2. While effective for bulk constitutive properties, evaluating only average end-to-end error obscures why a surrogate fails on extreme or localized mechanical responses. Specifically, an end-to-end error statistic conflates two distinct questions:

1. Does the representation $z = E_\theta(x)$ organize microstructures so that mechanically distinct cells are separated in latent space?
2. Given $z = E_\theta(x)$, does the chosen readout g_ϕ accurately approximate the response surface?

This distinction is acute in damage initiation and fracture mechanics. A deep encoder trained with MSE may achieve $R^2 > 0.96$ over a population while mapping two microstructures with markedly different local ligament spacing or stress concentrations to nearly identical latent coordinates. Such near-collisions have negligible impact on mean squared error across thousands of samples, yet they impose a strict lower bound on the worst-case error of any regular prediction head.

In a companion study on explicit physical descriptors for periodic two-phase microstructures [1], we introduced a counterexample-driven insufficiency diagnostic based on an exact finite-sample Lipschitz error barrier. On the same microstructure ensemble, different mechanical responses selected fundamentally different descriptors: phase-wise mean strains were sufficient for homogenized elastic energy R_E, whereas nonlocal initiation hazard $R_H(\beta)$ and AT1 phase-field peak fracture strength R_σ required tail- and process-zone-sensitive descriptors ($J_{1.5\ell}$) capturing inter-inclusion ligaments. The present work investigates whether *learned* neural representations exhibit the same target-dependent geometry and whether cross-fitted Lipschitz extension operators can be used both to audit and to regularize deep microstructure embeddings.

1.2. Three distinct levels of representation evaluation

A central conceptual premise of this paper is that informational sufficiency, metric regularity, and observed predictor performance are distinct properties that must not be conflated. Let $E_\theta : \mathcal{X} \to \mathbb{R}^d$ be an encoder and $R : \mathcal{X} \to \mathbb{R}$ a deterministic mechanical response.

Definition 1 (Functional / informational sufficiency)**.** *An encoder E_θ is* functionally sufficient *for R on $\mathcal{X}$ if there exists a measurable map $g : \mathbb{R}^d \to \mathbb{R}$ such that*

$$R(x) = g(E_\theta(x)) \qquad \forall x \in \mathcal{X}. \tag{2}$$

Functional sufficiency asks only whether $E_\theta(x_1) = E_\theta(x_2) \implies R(x_1) = R(x_2)$ (or $R \perp\!\!\!\perp X \mid E_\theta(X)$ in the stochastic setting of sufficient dimension reduction [17]); it is invariant to any injective reparameterization of z and is unaffected by appending uninformative coordinates. For example, if $E_A(x) = u \in \mathbb{R}^2$ is functionally sufficient for $R(x) = f(u)$, then the augmented embedding $E_B(x) = (u, \xi_1, \ldots, \xi_{30}) \in \mathbb{R}^{32}$ formed by concatenating 30 independent noise coordinates is *also* functionally sufficient, because the coordinate projection $\pi(z) = (z_1, z_2)$ recovers u exactly. However, if E_B is evaluated under an isotropic Euclidean or Mahalanobis distance that weights all 32 coordinates equally, pairwise distances become dominated by $(\xi_1, \ldots, \xi_{30})$: nearest neighbors in E_B

no longer share similar values of u, the Lipschitz constant required to interpolate R with respect to the isotropic distance grows sharply, and nonparametric distance-based interpolants degrade.

To avoid attributing metric or readout degradation to "information loss," we separate three evaluation levels throughout the manuscript, summarized in Table 1.

Table 1: Three distinct levels of representation evaluation separated in this work.

Level	Mathematical Question	Operational Diagnostics in This Paper
1. Functional sufficiency	Does there exist *any* function g such that $R(x) = g(E_\theta(x))$?	Exact latent collisions ($z_i = z_j$ with $R_i \neq R_j$), symmetry-orbit barriers $b_G(x)$ under built-in invariances.
2. Metric regularity	Can R be approximated on the sample within error ε by a function that is Λ-Lipschitz with respect to a normalized metric d_Ψ?	Exact finite-sample barrier curve $e^\star_\theta(\Lambda)$, finite-sample interpolation slope Λ_{interp}, and equal-coverage worst-case ratio $\Omega_\Psi(q)/\Omega^{\text{null}}_\Psi(q)$.
3. Observed readout accuracy	How accurately does a specific readout or extension rule predict R in-sample and out-of-sample?	Linear ridge R^2_{ridge}, spectrally normalized neural head error e_∞ and excess gap Δ_{head}, and out-of-sample McShane midpoint metrics (R^2_{McShane}, $\text{CF}_{\max}/\sigma_R$).

1.3. Positioning relative to materials informatics, Lipschitz interpolation, and regression representation learning

Machine learning for microstructure–property linkage has progressed from two-point spatial correlations and Materials Knowledge Systems (MKS) [3] to data-driven computational mechanics frameworks [4], sparse descriptor selection [2, 7], variational autoencoders coupled with property regressors [5], graph contrastive autoencoders [6], and pretrained vision transformers [8]; see Peng et al. [9] for a recent review. While these studies demonstrate strong average predictive accuracy, they evaluate learned embeddings primarily through downstream regression R^2 or image reconstruction fidelity rather than worst-case Lipschitz regularity.

From the perspective of approximation theory, our finite-sample barrier and midpoint interpolant build directly on classical McShane–Whitney Lipschitz extension [10], on optimal central interpolants in metric spaces developed by Beliakov [11], and on Calliess's Lipschitz optimisation for Lipschitz interpolation [12], which optimizes the Lipschitz constants and metric parameters of a McShane–Whitney interpolant by minimizing leave-out validation error. In machine learning, deep nonlinear sufficient dimension reduction has been analyzed statistically by Chen et al. [17], while continuous-label representation learning has been addressed by contrastive and ranking regularizers that align latent distances with target distances, notably RankSim [14], Rank-N-Contrast (`RNC`) [13], ConR [15], ACCon [16], and spectrally normalized neural networks [24, 25], alongside hard-example / top-k tail losses [18]. Our proposed cross-fitted McShane barrier regularizer (`SBR-CF`) extends the validation-driven Lipschitz interpolation paradigm of Calliess [12] from tuning metric parameters on fixed inputs to end-to-end training of deep equivariant encoders. What distinguishes the present framework is:

1. the explicit separation of functional sufficiency, metric regularity ($e^\star_\theta(\Lambda)$ and $\Omega/\Omega^{\text{null}}$ across whitening regimes), and readout excess error (Δ_{head});
2. the use of the cross-fitted McShane–Whitney midpoint $M_A(z)$ both as a nonparametric evaluation rule (without a trained parametric readout) and as a differentiable training loss with cross-batch anchor memory and leave-one-out hard-query mining; and
3. a controlled mechanics benchmark in which identical periodic cells carry bulk elastic energy R_E, tunable nonlocal stress-tail hazards $R_H(\beta)$, OOD contrast/biaxial loads, and externally frozen AT1 phase-field fracture responses ($R_\sigma, R_{\text{path}}$).

1.4. Working hypotheses, post-hoc controls, and manuscript organization

To structure the empirical campaign across the 3,300-cell elastic corpus and the frozen 216-cell fracture benchmark, six working hypotheses were formulated and subsequently tested against additional post-hoc controls (`Tail-MSE`, `ConvAE-HiFi`, regularized whitening d_{reg}, 254×254 FFT grid refinement, and AT1 shift simulations):

Hypothesis H1 (Unsupervised reconstruction vs. failure regularity)**.** *High morphological reconstruction accuracy of an unsupervised autoencoder (measured by pixel accuracy, inclusion IoU, Dice, and boundary F1) does not guarantee low worst-case metric insufficiency for tail-dominated initiation hazards.*

Hypothesis H2 (Decoupling of average accuracy and worst-case collisions)**.** *Supervised encoders trained with MSE can achieve high average R^2 while retaining severe latent near-collisions ($\Omega/\Omega^{\mathrm{null}}$ and cross-fitted McShane maximum error $\mathrm{CF}_{\max}/\sigma_R$).*

Hypothesis H3 (Target dependence and generalized-mean pooling)**.** *The latent geometry required for bulk energy R_E differs systematically from that required for tail-dominated hazard $R_H(\beta)$, with the divergence and the learned generalized-mean (*GeM*) pooling exponent p increasing monotonically with β across the hazard family.*

Hypothesis H4 (Cross-fitted McShane barrier regularization)**.** *At fixed latent dimension and training budget, adding the cross-fitted McShane barrier (`SBR-CF`) lowers whitened $\Omega/\Omega^{\mathrm{null}}$, lowers positive barrier ordinates $e^{\star}_{\theta}(\Lambda)$, and reduces out-of-sample McShane errors relative to standard `MSE`, un-cross-fitted `SBR-in`, and direct top-k residual `Tail-MSE`, while maintaining multi-dimensional covariance conditioning across whitening normalizations.* (As shown in Section 4.4, this hypothesis is partially supported: `SBR-CF` improves $\Omega/\Omega^{\mathrm{null}}$, cross-fitted McShane errors, and high-budget $d_{\mathcal{N}}$ barriers relative to `MSE` and `Tail-MSE`, whereas under d_{white} the global barrier curve $e^{\star}_{\theta}(\Lambda)$ ranks `RNC` first and `Tail-MSE` ahead of `SBR-CF`.)

Hypothesis H5 (Physical interpretability of latent counterexamples)**.** *Worst-case latent near-collisions of MSE encoders correspond to identifiable mechanical configurations (such as local ligament clustering at matched volume fraction) whose computed stress-hazard differences remain stable upon doubling the FFT solver resolution on the fixed voxelized masks and are separated in latent space by `SBR-CF`.*

Hypothesis H6 (Conditional sufficiency and transfer on phase-field fracture)**.** *Conditional on the solved-field critical-distance descriptor $E_1 + J_{1.5\ell}$, frozen elastic embeddings exhibit weak local-swap conditional signal on R_σ within the 150-cell design set while improving linear ridge prediction when combined with $E_1 + J_{1.5\ell}$ on prospective high-inclusion-count microstructures (Set C2); the post-critical normalized percolating crack-path cost R_{path} remains unresolved by frozen pre-critical elastic embeddings paired with linear ridge readouts.*

The remainder of the paper is organized into six sections: Section 2 develops the mathematical framework, exact sample barrier, McShane midpoint properties, out-of-sample counterexample, training objectives, metric normalizations, and synthetic dilution pilot; Section 3 details the elastic and AT1 phase-field mechanical formulations, dataset splits, architectures, and evaluation protocol; Section 4 presents the main elastic benchmark results (Experiments A–E, barrier curves, and whitening/metric sensitivity); Section 5 analyzes physical counterexamples, 254×254 FFT grid convergence, and architecture/objective ablations; Section 6 reports cross-horizon transfer, 90° load rotations, OOD-physics shifts, seed-combined conditional permutation tests, and AT1 phase-field fracture transfer; and Section 7 synthesizes the conclusions and scope limitations.

2. Finite-Sample Lipschitz Barriers and Cross-Fitted McShane Regularization

2.1. Exact finite-sample Lipschitz barrier and pairwise feasibility

Let $\{x_i\}_{i=1}^n \subset \mathcal{X}$ be a finite sample with scalar responses $R_i = R(x_i) \in \mathbb{R}$ and latent embeddings $z_i = E_\theta(x_i) \in \mathbb{R}^d$. Let $d_z(i,j) = d_z(z_i, z_j)$ denote a metric or pseudo-metric on $\mathbb{R}^d$. A prediction head $g : \mathbb{R}^d \to \mathbb{R}$ is Λ-Lipschitz with respect to d_z ($\Lambda \geq 0$) if

$$|g(z) - g(z')| \leq \Lambda\, d_z(z, z') \qquad \forall z, z' \in \mathbb{R}^d. \tag{3}$$

On the finite sample, the uniform (sup-norm) approximation error of g is

$$e_\infty(g) = \max_{1 \leq i \leq n} |R_i - g(z_i)|. \tag{4}$$

Theorem 1 (Exact finite-sample Lipschitz barrier)**.** *For a fixed sample $\{(z_i, R_i)\}_{i=1}^n$ and Lipschitz budget $\Lambda \geq 0$, define*

$$\boxed{e^\star(\Lambda) = \frac{1}{2} \max_{1 \leq i,j \leq n} \left[|R_i - R_j| - \Lambda d_z(i,j) \right]_+ ,} \tag{5}$$

where $[u]_+ = \max(u, 0)$. Every function $g : \mathbb{R}^d \to \mathbb{R}$ that is Λ-Lipschitz with respect to d_z satisfies

$$e_\infty(g) \geq e^\star(\Lambda). \tag{6}$$

Moreover, there exists a globally Λ-Lipschitz function $g : \mathbb{R}^d \to \mathbb{R}$ with respect to d_z that attains $e_\infty(g) = e^\star(\Lambda)$ on the sample.

Theorem 1 is proved in Appendix Appendix A via the triangle inequality and McShane–Whitney envelopes [10, 11, 12]. It admits an immediate pairwise feasibility interpretation:

Proposition 1 (Pairwise feasibility condition)**.** *A sample $\{(z_i, R_i)\}_{i=1}^n$ admits a Λ-Lipschitz head with uniform error $e_\infty(g) \leq \varepsilon$ if and only if $e^\star(\Lambda) \leq \varepsilon$, which is equivalent to*

$$|R_i - R_j| \leq 2\varepsilon + \Lambda d_z(i,j) \qquad \forall i, j \in \{1, \ldots, n\}. \tag{7}$$

Remark 1 (Sample geometry, finite interpolation slope Λ_{interp}, metrics vs. pseudo-metrics)**.** *The curve $\Lambda \mapsto e^\star(\Lambda)$ is non-increasing, convex, and piecewise linear on $[0, \infty)$, starting at $e^\star(0) = \frac{1}{2} \max_{i,j} |R_i - R_j|$. Its asymptotic behavior as $\Lambda \to \infty$ separates two regimes:*

1. ***Zero-distance collisions** ($d_z(i,j) = 0$ **for some** $R_i \neq R_j$)**:** In this case,*

$$e^\star(\infty) = \lim_{\Lambda \to \infty} e^\star(\Lambda) = \frac{1}{2} \max \{ |R_i - R_j| : d_z(i,j) = 0 \} > 0. \tag{8}$$

No function g that is Lipschitz with respect to d_z can reduce the sample uniform error below $e^\star(\infty)$ (because $\Lambda < \infty$ and $d_z(z_i, z_j) = 0$ force $g(z_i) = g(z_j)$). If d_z is a strict metric on $\mathbb{R}^d$ ($d_z(z_i, z_j) = 0 \iff z_i = z_j$), this is an unconditional Level-1 violation of functional sufficiency for $E_\theta(x)$. If d_z is a pseudo-metric induced by a rank-k_{eff} orthogonal projector $P_{\text{keep}} = V_{\text{keep}} V_{\text{keep}}^\top \in \mathbb{R}^{d \times d}$ (such as truncated-spectrum whitening d_{trunc} when $k_{\text{eff}} < d$), then $d_z(z_i, z_j) = 0$ means $P_{\text{keep}}(z_i - \mu_z) = P_{\text{keep}}(z_j - \mu_z)$: it establishes Level-1 insufficiency of the projected *representation $P_{\text{keep}} \circ E_\theta$, while the discarded orthogonal component $(I_d - P_{\text{keep}})(z_i - z_j) \in \mathbb{R}^d$ may still separate z_i and z_j.*

2. ***Finite sample without zero-distance collisions** ($d_z(i,j) > 0$ **whenever** $R_i \neq R_j$)**:** The finite-sample exact interpolation constant*

$$\Lambda_{\text{interp}} = \max_{i \neq j,\ d_z(i,j) > 0} \frac{|R_i - R_j|}{d_z(i,j)} < \infty \tag{9}$$

is finite, and $e^\star(\Lambda) = 0$ for all $\Lambda \geq \Lambda_{\text{interp}}$. Numerical or physical noise in R_i among close pairs does not *keep $e^\star(\Lambda) > 0$ for all Λ; rather, it inflates Λ_{interp} to a large finite value. Conversely, in a high-dimensional embedding where whitened distances concentrate around $\sqrt{2}$ as d grows, $\min_{i \neq j} d_z(i,j)$ is bounded away from zero, artificially lowering Λ_{interp} and $e^\star(\Lambda)$ even when the extra coordinates are pure noise. Consequently, raw values of $e^\star(\Lambda)$ cannot be compared across dimensions without null normalization and cross-fitting.*

2.2. Why the in-sample barrier fails as a training loss and what the McShane midpoint measures

Minimizing the in-sample pairwise residuals $r_{ij} = |R_i - R_j| - \Lambda d_z(i,j)$ directly on training batches (`SBR-in`) suffers from a fundamental geometric degeneracy: an encoder can reduce r_{ij} either by grouping cells with similar responses and separating cells with different responses, or by spreading points into high-dimensional directions so that no pair is close ($d_z(i,j)$ large for all $i \neq j$). Under covariance whitening, the average squared distance is fixed, but the lower tail of the pairwise distance distribution is not; penalizing only in-sample close pairs encourages uniform repulsion rather than predictive neighborhood alignment.

To obtain a two-sided criterion that rewards informative neighborhoods, we use the McShane–Whitney central interpolant fit on an anchor set A and evaluated on a disjoint query set Q [11, 12].

Definition 2 (McShane–Whitney envelopes and central midpoint interpolant)**.** *Given an anchor set $A = \{(z_i, R_i)\}_{i \in A}$ and slope budget $\Lambda \geq 0$, define the upper and lower Λ-Lipschitz envelopes and their midpoint [10, 11, 12]:*

$$U_A(z) = \min_{i \in A} \left[R_i + \Lambda d_z(z, z_i)\right], \qquad L_A(z) = \max_{i \in A} \left[R_i - \Lambda d_z(z, z_i)\right], \qquad M_A(z) = \tfrac{1}{2}\big(U_A(z) + L_A(z)\big). \tag{10}$$

Proposition 2 (Anchor optimality and uncertainty band of M_A)**.** *For any anchor set A and $\Lambda \geq 0$:*

1. *U_A, L_A, and M_A are globally Λ-Lipschitz on $(\mathbb{R}^d, d_z)$.*
2. *On the anchor set itself, M_A attains the exact finite-sample barrier of Theorem 1:*
$$\max_{i \in A} \big|M_A(z_i) - R_i\big| = e^\star_A(\Lambda). \tag{11}$$
3. *If A is Λ-compatible ($\Lambda \geq \Lambda_{\text{interp}}(A)$, so $e^\star_A(\Lambda) = 0$), then every Λ-Lipschitz function g interpolating A ($g(z_i) = R_i \; \forall i \in A$) satisfies $L_A(z) \leq g(z) \leq U_A(z)$ for all $z \in \mathbb{R}^d$, and $M_A(z)$ is the optimal central interpolant minimizing the worst-case deviation $\sup_g \big|M_A(z) - g(z)\big| = \frac{1}{2}(U_A(z) - L_A(z))$ at each query z [11].*

For a query set Q disjoint from A, we define the cross-fitted McShane residuals and maximum error:

$$\rho_j = |M_A(z_j) - R_j| \quad (j \in Q), \qquad e^{\text{cf}}_{A \to Q}(\Lambda) = \max_{j \in Q} \rho_j. \tag{12}$$

Remark 2 (Scope of $e^{\text{cf}}_{A \to Q}(\Lambda)$: Out-of-sample error of the central extension rule vs. impossibility bound)**.** *Whereas $e^\star_A(\Lambda)$ in Theorem 1 is a predictor-independent lower bound over* all Λ*-Lipschitz functions on A, the cross-fitted quantity $e^{\text{cf}}_{A \to Q}(\Lambda)$ in Eq. (12) is the out-of-sample prediction error of a* specific *nonparametric extension rule—namely, the central midpoint M_A. A large value of $e^{\text{cf}}_{A \to Q}(\Lambda)$ outside the convex hull of A does not imply that no Λ-Lipschitz function can fit $A \cup Q$.*

Consider the exact one-dimensional counterexample ($d = 1$) with anchors $A = \{(0,0), (1,1)\}$, slope budget $\Lambda = 1$, and a single out-of-sample query $(z_q, R_q) = (2,2)$. Evaluating Eq. (10) at $z_q = 2$ gives

$$U_A(2) = \min(0{+}1{\cdot}2,\ 1{+}1{\cdot}1) = 2, \qquad L_A(2) = \max(0{-}1{\cdot}2,\ 1{-}1{\cdot}1) = 0, \qquad M_A(2) = \tfrac{1}{2}(2{+}0) = 1.$$

Hence the cross-fitted McShane error at $z_q = 2$ is $\rho_q = |M_A(2) - R_q| = 1$, even though the linear function $g(z) = z$ is globally 1-Lipschitz and interpolates both A and (z_q, R_q) with zero error ($e^\star_{A \cup Q}(1) = 0$). In $d = 1$, outside the anchor interval $[0,1]$, the upper envelope U_A grows with slope $+\Lambda$ while the lower envelope L_A decays with slope $-\Lambda$, so their midpoint M_A flattens to constant extrapolation (in $d > 1$, $M_A(z)$ outside the anchor hull depends on $d_z(z, z_{i_+}) - d_z(z, z_{i_-})$ between the active upper and lower anchors and is not generally constant, but still cannot extrapolate linear trends unboundedly beyond $\max_{i \in A} R_i + \frac{1}{2}\Lambda \operatorname{diam}(A)$).

Accordingly, throughout this paper we refer to $e^{\text{cf}}(\Lambda)$, $\text{CF}_{\text{rmse}}/\sigma_R$, $\text{CF}_{\max}/\sigma_R$, and R^2_{McShane} strictly as cross-fitted McShane error metrics *(Level 3 in Table 1, evaluated without a trained parametric readout), while reserving the term* certified Lipschitz barrier *for the exact sample quantity $e^\star(\Lambda)$ (Level 2).*

2.3. Proposed cross-fitted objective (`SBR-CF`) and extreme-residual baselines

During encoder training, scalar responses are standardized to zero mean and unit variance on the training split, $\tilde{R}_i = (R_i - \mu_R)/\sigma_R$ (with $\sigma_R = 4.331$ for $R_H(\beta = 20)$ and $\sigma_R = 0.331$ for R_E), and raw embeddings $z_i = E_\theta(x_i) \in \mathbb{R}^d$ pass through a parameter-free coordinate standardization layer $\tilde{z}_i = \mathcal{N}(z_i)$ (`LatentNorm`, implemented as a non-affine `BatchNorm1d` layer whose running training statistics are frozen at evaluation) with scaled Euclidean distance $d_{\mathcal{N}}(i,j) = \|\tilde{z}_i - \tilde{z}_j\|_2 / \sqrt{d}$. Using the dimensionless training slope budget $\widetilde{\Lambda}_{\mathrm{cf}} = \Lambda_{\mathrm{cf}}/\sigma_R = 4.0$ (in σ_R units per $d_{\mathcal{N}}$-unit; Table 2), the standardized McShane–Whitney envelopes on an anchor set A are $\widetilde{U}_A(\tilde{z}) = \min_{i\in A}[\tilde{R}_i + \widetilde{\Lambda}_{\mathrm{cf}} d_{\mathcal{N}}(\tilde{z}, \tilde{z}_i)]$, $\widetilde{L}_A(\tilde{z}) = \max_{i\in A}[\tilde{R}_i - \widetilde{\Lambda}_{\mathrm{cf}} d_{\mathcal{N}}(\tilde{z}, \tilde{z}_i)]$, and $\widetilde{M}_A(\tilde{z}) = \frac{1}{2}(\widetilde{U}_A(\tilde{z}) + \widetilde{L}_A(\tilde{z}))$. Each mini-batch $\mathcal{B}$ of size B is partitioned into two disjoint halves, or paired against a cross-batch memory queue A of recent normalized anchor embeddings (with stale copies of the current batch cells masked out). The cross-fitted McShane loss penalizes the k largest query residuals above a standardized target tolerance $\varepsilon_{\mathrm{target}}$:

$$\boxed{\mathcal{L}_{\mathrm{cf}} = \frac{1}{2k} \sum_{(A,Q)\in\{(A,Q),(Q,A)\}} \sum_{j\in\mathcal{H}_k(Q)} \Big[\big|\widetilde{M}_A(\tilde{z}_j) - \tilde{R}_j\big| - \varepsilon_{\mathrm{target}}\Big]_+^2,} \tag{13}$$

where $\mathcal{H}_k(Q) \subset Q$ indexes the k queries with the largest standardized residuals $\rho_j = |\widetilde{M}_A(\tilde{z}_j) - \tilde{R}_j|$. Because

$$\widetilde{M}_A(\tilde{z}_j) = \frac{1}{2}(\tilde{R}_{i_+} + \tilde{R}_{i_-}) + \frac{\widetilde{\Lambda}_{\mathrm{cf}}}{2\sqrt{d}}\Big(\|\tilde{z}_j - \tilde{z}_{i_+}\|_2 - \|\tilde{z}_j - \tilde{z}_{i_-}\|_2\Big)$$

depends on the active upper and lower anchors $i_+, i_- \in A$ in Eq. (10), when A spans a diverse range of responses ($|A| \gg 1$), $\mathcal{L}_{\mathrm{cf}}$ provides a two-sided geometric mechanism: if $\tilde{z}_j$ is placed far from all anchors of similar response, $\widetilde{U}_A(\tilde{z}_j)$ and $\widetilde{L}_A(\tilde{z}_j)$ are determined by distant anchors and $\widetilde{M}_A(\tilde{z}_j)$ typically drifts away from $\tilde{R}_j$; if $\tilde{z}_j$ is placed too close to an anchor $\tilde{z}_i$ with a very different response $\tilde{R}_i$, either $\widetilde{U}_A(\tilde{z}_j)$ or $\widetilde{L}_A(\tilde{z}_j)$ is pulled toward $\tilde{R}_i$, inflating ρ_j. Thus, across multi-anchor batches, $\mathcal{L}_{\mathrm{cf}}$ encourages queries to stay near compatible anchors while separating incompatible pairs relative to the dimensionless slope budget $\widetilde{\Lambda}_{\mathrm{cf}}$ (note that in the degenerate single-anchor case $|A| = 1$, $i_+ = i_-$ and $\widetilde{M}_A(z) \equiv \tilde{R}_1$ has zero gradient).

The auxiliary spectrally normalized regression head g_ϕ acts directly on the normalized embedding $\tilde{z}_i = \mathcal{N}(E_\theta(x_i))$:

$$\mathcal{L}_{\mathrm{pred}} = \frac{1}{B}\sum_{i\in\mathcal{B}} (\tilde{R}_i - \widehat{R}_i)^2, \qquad \widehat{R}_i = g_\phi(\tilde{z}_i), \tag{14}$$

where predictions in physical response units are recovered by $\mu_R + \sigma_R \widehat{R}_i$. The total training objective for `SBR-CF` is

$$\boxed{\mathcal{L}_{\text{SBR-CF}} = \mathcal{L}_{\mathrm{pred}} + \lambda_{\mathrm{cf}}\mathcal{L}_{\mathrm{cf}} + \lambda_{\mathrm{cov}}\mathcal{L}_{\mathrm{cov}},} \tag{15}$$

where $\mathcal{L}_{\mathrm{cov}} = \frac{1}{d(d-1)}\sum_{a\neq b} \mathrm{Cov}(\tilde{z})_{ab}^2$ penalizes off-diagonal correlations as in Barlow Twins [19] and VICReg [20] so that the standardized training distance $d_{\mathcal{N}}$ remains close to the Mahalanobis whitened distance d_{white} (because $\mathcal{N}$ enforces unit per-coordinate variance by construction, no separate variance hinge $\mathcal{L}_{\mathrm{var}}$ is needed, i.e., $\lambda_{\mathrm{var}} = 0$ and $\lambda_{\mathrm{cov}} = 0.1$).

To isolate whether the empirical gains of `SBR-CF` come from organizing the latent geometry via M_A or merely from emphasizing large errors during training, we compare `SBR-CF` against two direct ablations sharing the exact same architecture, memory/mining schedule, covariance penalty, and 60-epoch training budget:

1. **In-sample pairwise barrier (`SBR-in`):** Replaces $\lambda_{\mathrm{cf}}\mathcal{L}_{\mathrm{cf}}$ with the un-cross-fitted top-k pairwise hinge loss on in-sample standardized residuals $r_{ij} = |\tilde{R}_i - \tilde{R}_j| - \widetilde{\Lambda}_{\mathrm{cf}} d_{\mathcal{N}}(i,j)$:

$$\mathcal{L}_{\text{topk-bar}} = \frac{1}{k}\sum_{(i,j)\in\mathcal{H}_k} [r_{ij} - 2\varepsilon_{\mathrm{target}}]_+^2 . \tag{16}$$

As noted in Proposition 3 below, we use the exact top-k hinge rather than a log-sum-exp smooth maximum $\mathcal{L}_{\text{bar}} = \frac{\tau}{2}\ln\left(1+\sum_{i<j} e^{r_{ij}/\tau}\right)$ to avoid smooth-maximum bias.

2. **Direct top-k extreme-residual head loss (`Tail-MSE`) [18]:** Replaces $\lambda_{\text{cf}}\mathcal{L}_{\text{cf}}$ with a direct penalty on the $k = 16$ largest squared head prediction residuals in each mini-batch:

$$\mathcal{L}_{\text{Tail-MSE}} = \mathcal{L}_{\text{pred}} + \lambda_{\text{tail}}\left(\frac{1}{k}\sum_{i\in\mathcal{H}_k(\mathcal{B})}\left(\tilde{R}_i - \widehat{R}_i\right)^2\right) + \lambda_{\text{cov}}\mathcal{L}_{\text{cov}}. \tag{17}$$

`Tail-MSE` penalizes extreme errors through the parametric head g_ϕ without using the McShane envelope M_A or pairwise latent distances.

Proposition 3 (Bias bound of the log-sum-exp smooth maximum)**.** *For any P pairwise residuals $r_1, \ldots, r_P$ and temperature $\tau > 0$,*

$$\frac{1}{2}\left[\max_{1\le p\le P} r_p\right]_+ \;\le\; \frac{\tau}{2}\ln\left(1+\sum_{p=1}^{P} e^{r_p/\tau}\right) \;\le\; \frac{1}{2}\left[\max_{1\le p\le P} r_p\right]_+ + \frac{\tau}{2}\ln(1+P). \tag{18}$$

For $P = \binom{256}{2} = 32{,}640$, the additive bias reaches 5.2τ (0.26 at $\tau = 0.05$, exceeding the numerical noise floor of R_E by two orders of magnitude), motivating our use of exact top-k selection in Eqs. (13)–(17).

2.4. Global pair coverage, metric normalizations, and spectrally normalized Lipschitz head

Global hard-query mining and cross-batch memory.. Under uniform random mini-batching of size $B = 64$ from $N_{\text{train}} = 1{,}400$ cells, any specific pair of cells co-occurs in a batch with probability $(B-1)/(N_{\text{train}}-1) = 4.5\%$ per epoch. To ensure global coverage of worst-case pairs, our training loop combines two mechanisms after a 5-epoch warmup: (i) at the start of each mining cycle (every epoch, `mine_every` $= 1$, in the main 60-epoch campaign), all N_{train} embeddings are computed and leave-one-out McShane residuals $\rho_i = |M_{A\setminus\{i\}}(\tilde{z}_i) - \tilde{R}_i|$ are evaluated across the training set to identify the $K_{\text{mine}} = 128$ hardest queries, from which $K_b = 16$ hard cells are injected into each mini-batch of $B = 64$; and (ii) a first-in-first-out cross-batch memory queue [21] stores the latest $M = 1{,}024$ detached normalized embeddings $\tilde{z}/\sqrt{d}$ and standardized responses $\tilde{R}$ as anchors A for Eq. (13), masking out any stored anchor whose cell ID matches a current query $j \in Q$.

Latent metric normalizations and regularized whitening.. Without a scale constraint, an encoder can trivially shrink $e^\star(\Lambda)$ by multiplying z by a large constant. Given raw training embeddings $\{z_i\}_{i=1}^{N_{\text{train}}}$ with sample mean $\mu_z \in \mathbb{R}^d$ and sample covariance $\Sigma_z = V\,\text{diag}(\lambda_1, \ldots, \lambda_d)V^\top$ (with eigenvalues $\lambda_1 \ge \cdots \ge \lambda_d \ge 0$, orthogonal eigenvectors $V = [v_1, \ldots, v_d]$, and mean eigenvalue $\bar{\lambda} = \frac{1}{d}\sum_{k=1}^{d}\lambda_k$), we evaluate five scale-normalized distance families to test metric sensitivity rigorously:

1. **Full Mahalanobis / ZCA whitening (d_{white}):** Implemented in `WhiteningTransform` with absolute diagonal loading $\delta = 10^{-9}$:

$$\widetilde{z} = V\,\text{diag}\big((\lambda_k+\delta)^{-1/2}\big)V^\top(z-\mu_z), \qquad d_{\text{white}}(i,j) = \frac{\|\widetilde{z}_i - \widetilde{z}_j\|_2}{\sqrt{d}}, \qquad \delta = 10^{-9}. \tag{19}$$

2. **Relative-ridge regularized whitening ($d_{\text{reg}}(\alpha_{\text{reg}})$):** To prevent noise amplification along near-zero eigendirections when a representation has an anisotropic or low-rank spectrum (such as `RNC`), `RegWhiteningTransform` regularizes each eigenvalue proportionally to the mean eigenvalue $\bar{\lambda} = \text{tr}(\Sigma_z)/d$ and scales by $1/\sqrt{d}$:

$$z^{(\alpha_{\text{reg}})} = V\,\text{diag}\big((\lambda_k + \alpha_{\text{reg}}\bar{\lambda})^{-1/2}\big)V^\top(z-\mu_z), \qquad d_{\text{reg}}(\alpha_{\text{reg}})(i,j) = \frac{\left\|z_i^{(\alpha_{\text{reg}})} - z_j^{(\alpha_{\text{reg}})}\right\|_2}{\sqrt{d}}, \tag{20}$$

over $\alpha_{\text{reg}} \in \{10^{-5}, 10^{-3}, 10^{-2}, 10^{-1}\}$, with $\alpha^{\star}_{\text{reg}}$ selected strictly on held-out validation data ($d^{\star}_{\text{reg}}$). Comparing Eqs. (19) and (20), exact equivalence holds at

$$d_{\text{white}} = d_{\text{reg}}\left(\frac{\delta}{\bar{\lambda}}\right), \tag{21}$$

so α_{reg} and δ coincide numerically only when $\bar{\lambda} = 1$. As $\alpha_{\text{reg}} \to 0$, $d_{\text{reg}}(\alpha_{\text{reg}})$ approaches unregularized ZCA whitening; as $\alpha_{\text{reg}} \to \infty$, $d_{\text{reg}}(\alpha_{\text{reg}})$ becomes proportional to the raw Euclidean distance $\|z_i - z_j\|_2 / \sqrt{d}$ with scale factor $(\alpha_{\text{reg}}\bar{\lambda})^{-1/2} \to 0$.

3. **Truncated-spectrum whitening (d_{trunc}, a pseudo-metric when $k_{\text{eff}} < d$):** Let $K = \{k \in \{1, \ldots, d\} : \lambda_k \geq 10^{-2}\lambda_1\}$, $k_{\text{eff}} = |K|$, $V_{\text{keep}} = [v_k]_{k \in K} \in \mathbb{R}^{d \times k_{\text{eff}}}$, and $P_{\text{keep}} = V_{\text{keep}} V_{\text{keep}}^{\top} \in \mathbb{R}^{d \times d}$. We project $z - \mu_z$ onto the active subspace $\text{Im}(P_{\text{keep}})$, whiten only those k_{eff} directions, and divide by $\sqrt{k_{\text{eff}}}$:

$$z^{(\text{trunc})} = \sum_{k \in K} \lambda_k^{-1/2} v_k v_k^{\top}(z - \mu_z), \qquad d_{\text{trunc}}(i,j) = \frac{\left\| z_i^{(\text{trunc})} - z_j^{(\text{trunc})} \right\|_2}{\sqrt{k_{\text{eff}}}}. \tag{22}$$

4. **Per-coordinate standardization (d_{std}):** Implemented in `Standardizer`, which divides each coordinate $z_a - \mu_{\text{tr},a}$ by its exact full-training-set sample standard deviation $\sigma_{\text{tr},a} = \sqrt{(\Sigma_z)_{aa}}$ and divides the Euclidean norm by $\sqrt{d}$.

5. **Frozen training normalization layer ($d_{\mathcal{N}}$):** Evaluates the scaled Euclidean distance on the frozen `LatentNorm` outputs $\tilde{z}_i = \mathcal{N}(z_i)$ saved at the end of training:

$$d_{\mathcal{N}}(i,j) = \frac{\|\tilde{z}_i - \tilde{z}_j\|_2}{\sqrt{d}}. \tag{23}$$

In our implementation (`src/suffml/train/train_objectives.py`), every training mini-batch executes the forward pass $\widehat{R}, z, \tilde{z} = \text{model}(x)$ for *all* objective families (including `RNC`, where the contrastive loss acts on raw z while `model.norm` updates its non-affine `BatchNorm1d` running mean μ_{bn} and running variance σ^2_{bn}). At evaluation, $d_{\mathcal{N}}$ uses these frozen running `BatchNorm1d` moments $(\mu_{\text{bn}}, \sigma^2_{\text{bn}})$, whereas d_{std} uses the exact full-training-set sample moments $(\mu_{\text{tr}}, \sigma^2_{\text{tr}})$. Thus $d_{\mathcal{N}}$ and d_{std} are defined by distinct moment estimators (running mini-batch exponential moving averages vs. exact full-sample moments), although their resulting test metrics agree closely (to within 0.000–0.026 in Table 14).

Spectrally normalized Lipschitz head and summary of Lipschitz constants.. To make Theorem 1 operational on the trained neural pipeline, the regression head g_ϕ acts on the normalized embedding $\tilde{z} = \mathcal{N}(E_\theta(x))$ and is parameterized as a $d \to 64 \to 32 \to 1$ network with spectral normalization [24] on every linear weight matrix W_l ($l = 1, 2, 3$) and 1-Lipschitz GroupSort activations [25]. Letting $u = \tilde{z}/\sqrt{d} \in \mathbb{R}^d$ so that $\|u - u'\|_2 = d_{\mathcal{N}}(z, z')$, the head is defined in standardized target units (σ_R-units) as

$$g_\phi(\tilde{z}) = \widetilde{\Lambda}_{\text{head}}\, f_\phi\left(\frac{\tilde{z}}{\sqrt{d}}\right), \qquad \|f_\phi\|_{\text{Lip},\,2} \leq \prod_{l=1}^{3} \|W_l\|_2\,, \qquad \|g_\phi\|_{\text{Lip},\,d_{\mathcal{N}}} \leq \widetilde{\Lambda}_{\text{head}} \prod_{l=1}^{3} \|W_l\|_2\,, \tag{24}$$

so that $\mathcal{N}$ is applied *once* in the composite predictor $\widehat{R} = g_\phi(\mathcal{N}(E_\theta(x)))$. Here $\|f_\phi\|_{\text{Lip},2}$ is the Lipschitz constant of f_ϕ with respect to the Euclidean norm $\|\cdot\|_2$ on its input $u = \tilde{z}/\sqrt{d}$, which directly yields the Lipschitz bound $\|g_\phi\|_{\text{Lip},d_{\mathcal{N}}}$ with respect to $d_{\mathcal{N}}$. During training, power-iteration spectral normalization keeps each effective weight matrix near unit spectral norm ($\|W_l\|_2 \approx 1$). At evaluation, 100 power-iteration updates (`finalize_spectral`) converge the spectral normalization vectors, and `certified_lambda` computes the exact SVD operator 2-norm product $\widetilde{\Lambda}_{\text{cert}} = \widetilde{\Lambda} \prod_{l=1}^{3} \|W_l\|_2$ (`torch.linalg.matrix_norm(..., ord=2)`) without assuming $\prod_{l=1}^{3} \|W_l\|_2 \leq 1$. Multiplying by

Table 2: Summary of Lipschitz slope budgets and empirical slope quantities used throughout this work. Dimensionless slopes $\widetilde{\Lambda} = \Lambda/\sigma_R$ act on standardized targets $\tilde{R} = (R - \mu_R)/\sigma_R$ ($\sigma_R = 4.331$ on $R_H(20)$) and convert to physical response units via $\Lambda = \sigma_R\widetilde{\Lambda}$.

Symbol	Value / Range	Mathematical Definition and Role	Units
$\widetilde{\Lambda}_{\text{cf}}$	4.0 ($\Lambda_{\text{cf}} = 17.32$)	Dimensionless slope budget inside the training cross-fitted McShane midpoint M_A in Eq. (13) (selected on `val`).	σ_R per $d_{\mathcal{N}}$-unit
$\widetilde{\Lambda}_{\text{head}}$	16.0 ($\Lambda_{\text{head}} = 69.30$)	Permissive upper bound on the auxiliary head g_ϕ during encoder training (Eq. (24)).	σ_R per $d_{\mathcal{N}}$-unit
$\widetilde{\Lambda}_{\text{cert}}$	$\widetilde{\Lambda}\prod_{l=1}^{3}\|W_l\|_2$	Exact SVD-certified upper bound on $\|g_\phi\|_{\text{Lip},d_{\mathcal{N}}}$ for post-hoc heads trained over $\widetilde{\Lambda} \in \{1, 2, 4, 8, 16\}$ on frozen $\tilde{z}$.	σ_R per $d_{\mathcal{N}}$-unit
$\Lambda_{\text{interp}}/\sigma_R$	Eq. (9)	Exact finite-sample interpolation threshold $\max_{i\neq j}\|\tilde{R}_i - \tilde{R}_j\|/d_\Psi(i,j)$ where $e^\star(\Lambda)$ reaches 0.	σ_R per d_Ψ-unit
$\Lambda^\star(\varepsilon)$	$\inf\{\Lambda : e^\star(\Lambda) \leq \varepsilon\}$	Minimum Lipschitz slope required to fit the sample within uniform tolerance ε (Section 2.5).	Raw or σ_R units

σ_R converts any dimensionless slope $\widetilde{\Lambda}$ in σ_R-units to physical response units $\Lambda = \sigma_R\widetilde{\Lambda}$. Table 2 summarizes all slope quantities used across the paper.

For any post-hoc head with certified dimensionless upper bound $\widetilde{\Lambda}_{\text{cert}} = \Lambda_{\text{cert}}/\sigma_R$ on frozen normalized embeddings $d_{\mathcal{N}}$, Theorem 1 yields the exact training decomposition (in σ_R-units):

$$e_{\infty,\text{train}} = \max_{1\leq i\leq N_{\text{train}}} \left|\tilde{R}_i - g_\phi(\tilde{z}_i)\right| = \frac{e^\star_{\text{train}}(\Lambda_{\text{cert}})}{\sigma_R} + \Delta_{\text{head}}(\widetilde{\Lambda}_{\text{cert}}), \qquad \Delta_{\text{head}}(\widetilde{\Lambda}_{\text{cert}}) \geq 0. \tag{25}$$

Here $e^\star_{\text{train}}(\Lambda_{\text{cert}})/\sigma_R$ is the predictor-independent representation barrier at budget $\widetilde{\Lambda}_{\text{cert}}$, whereas $\Delta_{\text{head}}(\Lambda_{\text{cert}})$ combines: (i) the approximation gap of the $d \to 64 \to 32 \to 1$ GroupSort architecture relative to all $\widetilde{\Lambda}_{\text{cert}}$-Lipschitz functions, (ii) training with mean squared error rather than L_∞ minimax optimization, (iii) stochastic optimization suboptimality, and (iv) conservatism of the layerwise spectral-norm product $\prod_{l=1}^{3}\|W_l\|_2$ as an upper bound on the true global Lipschitz constant of g_ϕ.

2.5. Synthetic pilot: Disentangling functional information from isotropic metric regularity

Before training deep networks, a self-contained synthetic experiment ($N = 600$) illustrates how appending independent noise coordinates preserves Level-1 functional sufficiency while degrading Level-2 isotropic metric regularity and Level-3 McShane interpolation. Let $u = (u_1, u_2) \sim U(-1, 1)^2$ and

$$R = \sin(3u_1) + u_2^2 + 0.02\,\eta, \qquad \eta \sim \mathcal{N}(0, 1) \tag{26}$$

(sample range $\Lambda_R = 2.94$, standard deviation $\sigma_R = 0.77$). We compare three representations evaluated under d_{white}: (A) the true latent variable $E_A = u \in \mathbb{R}^2$; (B) the noise-diluted embedding $E_B = (u, \xi_1, \ldots, \xi_{30}) \in \mathbb{R}^{32}$ with $\xi_k \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$; and (C) pure noise $E_C = (\xi_1, \ldots, \xi_{32}) \in \mathbb{R}^{32}$. As noted in Section 1.2, E_A and E_B carry the *exact same* functional information about u; what changes from E_A to E_B is the isotropic metric d_{white}, which weights 30 uninformative directions equally with (u_1, u_2).

Tables 3 and 4 demonstrate two essential facts: (i) raw $e^\star(\Lambda_R)$ and raw $\Lambda^\star(\varepsilon)$ are confounded by high-dimensional distance concentration, assigning $e^\star(\Lambda_R) = 0.000$ to the 32-dimensional noise-diluted representation E_B while assigning 0.015 to the true 2D variable E_A; and (ii) in this synthetic setting and in the neural dimension sweep of Section 4.5, null-normalized $\Omega/\Omega^{\text{null}}$ and cross-fitted McShane error $\text{CF}_{\text{rmse}}/\sigma_R$ detect the isotropic dilution of u monotonically across $k \in [0, 30]$.

Table 3: Synthetic dilution pilot ($N = 600$): comparison of raw and null-normalized metric regularity and cross-fitted McShane interpolation under d_{white}. Both E_A and E_B are functionally sufficient for the noiseless part of R, but isotropic whitening across 30 noise dimensions destroys local neighborhood alignment in E_B.

Representation	$\Omega/\Omega^{\text{null}}$ ($q = 1\%$)	$e^\star(\Lambda_R)$	$\Lambda^\star(0.05)$	$\Lambda^\star/\Lambda^\star_{\text{null}}$	$\text{CF}_{\text{rmse}}/\sigma_R$	$\text{CF}^{\text{null}}_{\text{rmse}}/\sigma_R$
A: true latent u ($d = 2$)	**0.14**	0.015	2.58	**0.007**	**0.09**	1.00
B: $u + 30$ noise coords. ($d = 32$)	0.98	**0.000**	2.62	0.916	0.89	0.99
C: pure Gaussian noise ($d = 32$)	0.92	0.015	2.86	0.995	1.01	1.00

Table 4: Progressive dilution of the 2D latent variable u by k independent Gaussian noise coordinates ($d = 2 + k$, $N = 600$). Raw $e^\star(\Lambda_R)$ is non-monotone and vanishes at $k = 30$ due to distance concentration, whereas $\Omega/\Omega^{\text{null}}$, $\Lambda^\star/\Lambda^\star_{\text{null}}$, and $\text{CF}_{\text{rmse}}/\sigma_R$ track the degradation of isotropic metric regularity monotonically.

k noise coordinates ($d = 2 + k$)	$\Omega/\Omega^{\text{null}}$	$\Lambda^\star(0.05)$	$\Lambda^\star/\Lambda^\star_{\text{null}}$	$e^\star(\Lambda_R)$	$\text{CF}_{\text{rmse}}/\sigma_R$
0	0.14	2.58	0.008	0.015	0.09
1	0.29	3.08	0.032	0.081	0.24
2	0.48	3.57	0.170	0.194	0.38
4	0.67	3.62	0.378	0.244	0.53
8	0.91	3.62	0.689	0.279	0.69
16	0.95	2.97	0.799	0.060	0.83
30	0.99	2.66	0.926	0.000	0.92

3. Mechanical Benchmark, Datasets, and Architectures

3.1. Linear-elastic homogenization targets and generalized-mean identity

We consider periodic two-phase linear-elastic unit cells $Y = [0, 1)^2$ discretized on a 127×127 uniform grid, with matrix phase Y_m ($E_m = 1$, $\nu_m = 0.3$) and stiffer circular or elliptical inclusion phase Y_i ($E_i = \kappa E_m$, default stiffness contrast $\kappa = 10$, $\nu_i = 0.3$) subjected to macroscopic uniaxial tensile strain $\bar{\varepsilon} = \bar{\varepsilon}_{11}\, e_1 \otimes e_1$ (solved to machine precision via the Moulinec–Suquet FFT method [29]). Two elastic response families are defined on every cell [1]:

1. **Normalized homogenized elastic energy (R_E):**

$$R_E = \frac{W_{\text{hom}}}{W_m(\bar{\varepsilon})}, \qquad W_{\text{hom}} = \frac{1}{2}\int_Y \varepsilon(x) : \mathbb{C}(x) : \varepsilon(x)\, \mathrm{d}x, \tag{27}$$

   where $W_m(\bar{\varepsilon})$ is the strain energy of the homogeneous matrix under $\bar{\varepsilon}$. Under linear elasticity, R_E is exactly determined by the volume fraction $f = |Y_i|$ and the phase-averaged strain tensor (`E0`, $d = 4$).

2. **Nonlocal matrix initiation hazard family ($R_H(\beta)$):** Let $\sigma_1(x)$ be the maximum principal tensile stress in the matrix phase Y_m, nonlocally averaged over a periodic disk of radius $\ell = 0.025$ (3.175 px) and normalized by the homogeneous matrix tensile stress $\sigma_m(\bar{\varepsilon})$ to form the dimensionless driving field $s(x) \geq 0$. For tail exponent $\beta \in \{2, 5, 10, 20, 40\}$,

$$B_\ell(\beta) = \int_{Y_m} s(x)^\beta\, \mathrm{d}x, \qquad R_H(\beta) = \ln B_\ell(\beta), \tag{28}$$

   with $\beta = 20$ serving as the primary benchmark target. Writing the power mean $\text{GeM}_p(s)$ as $\left(|Y_m|^{-1}\int_{Y_m} s(x)^p\, \mathrm{d}x\right)^{1/p}$, $R_H(\beta)$ satisfies the exact identity

$$R_H(\beta) = \beta \ln \text{GeM}_\beta(s) + \ln(1 - f). \tag{29}$$

   Equation (29) motivates equipping neural encoders with a learnable GeM_p spatial pooling layer [26], whose learned exponent p can be tracked across targets of increasing tail localization β.

3.2. Self-contained summary of the AT1 phase-field fracture benchmark

To test whether representations learned on fast linear-elastic solves transfer to nonlinear fracture initiation and propagation, we use the externally frozen AT1 phase-field fracture benchmark generated in the companion study [1]. Each periodic unit cell $Y = [0,1)^2$ is modeled under 2D plane-strain kinematics with periodic boundary conditions on both the displacement fluctuation $\tilde{u}$ and the scalar damage field $d \in [0,1]$, governed by the AT1 variational fracture functional:

$$\mathcal{E}_{\mathrm{AT1}}(u,d) = \int_Y (1-d)^2 \psi_+(\varepsilon(u))\,\mathrm{d}x + \int_Y \psi_-(\varepsilon(u))\,\mathrm{d}x + \int_Y \frac{3G_c(x)}{8}\left(\frac{d}{\ell} + \ell|\nabla d|^2\right)\mathrm{d}x, \tag{30}$$

where $\psi_\pm(\varepsilon)$ denotes the Amor volumetric–deviatoric tensile/compressive strain-energy split, $\ell = 0.025$ is the phase-field regularization length, $G_c^m = 1.0$ in the matrix, and $G_c^i = 10\,G_c^m$ inside inclusions (suppressing trans-inclusion cracking). In the AT1 model, the linear-elastic state $d \equiv 0$ remains a local minimizer until the local tensile energy density $\psi_+(x)$ reaches the local nucleation threshold

$$\psi_c(x) = \frac{3G_c(x)}{16\ell}. \tag{31}$$

In the AT1 solver (`PhaseFieldRVE.run` in `src/suffml/physics/pf_fft.py`), the exact same numerical implementation and parameter settings generated all phase-field datasets and tables in this work (`design_cells.csv`, blind `Set B`, `Set O`, prospective `Set C1`, `Set C2`, and the strip-shift verification `phase5_pf_strip_shifts.csv`). First, the undamaged unit-strain elastic solution ($\bar{\varepsilon}_{11} = 1$, solved by Galerkin FFT conjugate gradients to relative residual tolerance 10^{-9}) determines the exact AT1 elastic onset strain $\bar{\varepsilon}_0 = \min_{x\in Y}\sqrt{\psi_c(x)/\psi_1^+(x)}$, up to which $d \equiv 0$ everywhere. From $\bar{\varepsilon}_0$ onward, monotone uniaxial tension advances in uniform relative strain increments

$$\Delta\bar{\varepsilon}_{11} = \delta_\varepsilon \bar{\varepsilon}_0, \qquad \delta_\varepsilon = \texttt{dE_rel} = 0.01 \tag{32}$$

in both the benchmark campaign and the strip-translation experiments (`scripts/run_phase5b_pf_strip_shifts.py` invokes `rve.run()` with the identical default `dE_rel` $= 0.01$, `max_steps` $= 800$, and strain cap $\bar{\varepsilon}_{\max} = 6\bar{\varepsilon}_0$). At each load step s, staggered alternate minimization (`max_alt`=60) alternates Galerkin FFT conjugate-gradient elasticity (relative tolerance 10^{-6}) and exact primal-dual active-set bound-constrained quadratic programming for the irreversibility-constrained damage update $d^{(k)} \in [d_{\mathrm{prev}}, 1]$ until $\left\|d^{(k)} - d^{(k-1)}\right\|_\infty < 10^{-3}$. Stepping continues past peak load until either the macroscopic secant modulus drops below 10% of its initial undamaged value ($\bar{\sigma}_{11}^{(s)}/\bar{\varepsilon}_{11}^{(s)} < 0.10\,M_0$, complete macroscopic unloading, `separated` $=$ `True`) or the strain cap $\bar{\varepsilon}_{11}^{(s)} = 6\bar{\varepsilon}_0$ is reached.

Two dimensionless logarithmic fracture responses are evaluated on each solved trajectory:

$$R_\sigma = \ln\frac{\sigma_{\mathrm{peak}}}{\sigma_{\mathrm{peak}}^{\mathrm{hom}}}, \qquad R_{\mathrm{path}} = \ln\frac{C_{\mathrm{path}}(d_{\mathrm{final}})}{C_{\mathrm{path}}^{\mathrm{hom}}}, \tag{33}$$

where $\sigma_{\mathrm{peak}} = \max_s \bar{\sigma}_{11}^{(s)}$ is extracted directly as the discrete maximum of the macroscopic axial stress over the computed load steps (`ipk` $= \mathrm{argmax}_s\,\bar{\sigma}_{11}^{(s)}$). Whereas `PhaseFieldRVE.fracture_energy` records the bulk regularized AT1 surface-energy integral $\Gamma_\ell(d_{\mathrm{final}}) = \int_Y \frac{3G_c(x)}{8}(d_{\mathrm{final}}/\ell + \ell|\nabla d_{\mathrm{final}}|^2)\,\mathrm{d}x$ over the entire cell (which includes diffuse damage and arrested secondary branches), the response R_{path} used across Ref. [1] and all tables of the present study is defined from the *minimum-cost percolating crack-path functional* $C_{\mathrm{path}}(d_{\mathrm{final}})$ computed by `min_path(Gc,` $d_{\mathrm{final}} \geq 0.9$) (`scripts/run_phase5b_pf_strip_shifts.py`). Specifically, on the $N \times (N+1)$ cylinder graph $\mathcal{V}_h = \{0,\ldots,N-1\} \times \{0,\ldots,N\}$ ($N = 127$, grid spacing $h = 1/N$, periodic in $i \in \mathbb{Z}_N$ along x_1 and unwrapped across one full period $j \in \{0,\ldots,N\}$ along the transverse crack direction x_2), vertices are restricted to the severely damaged set $A(d_{\mathrm{final}}) = \{(i,j) \in \mathcal{V}_h : d_{\mathrm{final}}(i,\, j \bmod N) \geq 0.9\}$. Undirected edges $e = ((i,j),(i',j'))$ connect active vertices $A_{i,j} = A_{i',j'} = 1$ ($i' = (i+\Delta i) \bmod N$, $0 \leq j' = j + \Delta j \leq N$) across the 16-direction stencil generated by the 8 half-space offsets

$\mathcal{O}_+ = \{(1,0),(0,1),(1,1),(-1,1),(1,2),(-1,2),(2,1),(-2,1)\}$, with trapezoidal toughness-length weight

$$w_e = h\sqrt{\Delta i^2 + \Delta j^2}\,\frac{G_c(i,\, j \bmod N) + G_c(i',\, j' \bmod N)}{2}, \qquad C_{\mathrm{path}}(d_{\mathrm{final}}) = \min_{\gamma \in \mathcal{P}_h(d_{\mathrm{final}};0.9)} \sum_{e\in\gamma} w_e, \tag{34}$$

where $\mathcal{P}_h(d_{\mathrm{final}};0.9)$ is the set of all paths $\gamma \subset A(d_{\mathrm{final}})$ connecting any bottom vertex $(i_0,0) \in A(d_{\mathrm{final}})$ to its periodic counterpart $(i_0,N) \in A(d_{\mathrm{final}})$ across one vertical period in x_2. The function `min_path` returns the raw Dijkstra shortest-path cost $C_{\mathrm{path}}(d_{\mathrm{final}})$ ($\approx \min_\gamma \int_\gamma G_c(s)\,\mathrm{d}s$), which is then normalized in Eq. (33) by the homogeneous reference cost $C^{\mathrm{hom}}_{\mathrm{path}} = \texttt{min_path}(G^{\mathrm{hom}}_c, d_{\mathrm{hom}} \geq 0.9) = 0.98\,G^m_c = 0.98$ along the 2%-weakened vertical strip. Every cell across all phase-field sets (180/180 non-laminate design cells, 45/45 in `Set B`, 12/12 in `Set O`, 30/30 in `Set C1`, 12/12 in `Set C2`, and all 38 strip-shift configurations) achieved macroscopic secant unloading `separated` = `True` ($\bar\sigma_{11}/\bar\varepsilon_{11} < 0.10\,M_0$) well before the $6\bar\varepsilon_0$ strain cap. However, `separated` = `True` and single-path percolation $\mathcal{P}_h(d_{\mathrm{final}};0.9) \neq \emptyset$ are distinct conditions: in 1/45 cells of `Set B` (`bid` = 72, `ellipses_0.2_9`, unloading at $1.90\bar\varepsilon_0$) and 1/12 cells of `Set O` (`bid` = 106, `ellipses6_0.2_4`, unloading at $2.32\bar\varepsilon_0$), the cell unloads below $0.10\,M_0$ via two vertically overlapping *en-echelon* crack branches whose inter-tip bridge remains at $d_{\mathrm{final}} < 0.9$ at termination. On those two cells, $\mathcal{P}_h(d_{\mathrm{final}};0.9) = \emptyset$ and `min_path` returns `NaN`; both cells retain valid peak strength R_σ and are excluded from R_{path} evaluations across all rows of Table 20, giving effective sample sizes $n_{\mathrm{eff}}(R_{\mathrm{path}}) = 180$ (Design CV), 44 (`Set B`), and 11 (`Set O`).

Critical-distance onset descriptor $E_1 + J_{1.5\ell}$.. In Ref. [1], we showed that peak fracture strength R_σ is governed by a nonlocal critical-distance onset intensity J_r: let $\psi_+(x)/\psi_c(x)$ be the intact elastic driving-to-threshold ratio at unit macroscopic strain, averaged over a periodic disk $D_r(x)$ of radius r and maximized over the unit cell:

$$J_r = \ln \max_{x\in Y}\left(\frac{1}{|D_r|}\int_{D_r(x)} \frac{\psi_+(y)}{\psi_c(y)}\,\mathrm{d}y\right). \tag{35}$$

At radius $r = 1.5\ell$ (4.76 px), combining the 6-feature energy descriptor E_1 (comprising f, $\bar\varepsilon^m_{11}$, $\bar\varepsilon^m_{22}$, $\bar\varepsilon^m_{12}$, J^m_{gap}, and J^i_{gap}) with $J_{1.5\ell}$ forms a 7-feature solved-field baseline ($E_1 + J_{1.5\ell}$) that requires an FFT elastic solve at test time and achieves $R^2 = 0.879$–0.881 on the 180 non-laminate design cells, 0.930–0.932 on blind Set B ($n = 45$), 0.941–0.945 on high-aspect-ratio Set O ($\alpha = 6$, $n = 12$), 0.905–0.906 on prospective Set C1 ($n = 30$), and 0.679–0.689 on high-inclusion-count prospective Set C2 ($n_{\mathrm{inc}} \in \{16, 25\}$, $n = 12$).

Weakened localization strip and translation symmetry.. To break the trivial translation degeneracy of homogeneous and laminate cells during phase-field crack localization, the AT1 solver reduces $G_c(x)$ by 2% inside a vertical strip of width 2/127 centered at $x_1 = 0.95$ [1]. Consequently, the phase-field boundary value problem is exactly invariant to periodic shifts along e_2 and reflections $x_2 \mapsto -x_2$, but only approximately invariant to periodic horizontal shifts $T_{\Delta x_1} \in G_{x_1}$ along e_1. For any translation-invariant representation ($E(Tx) = E(x)$ for all $T \in G_{x_1}$), the exact full-orbit Level-1 translation barrier on cell x is $b_{G_{x_1}}(x) = \frac{1}{2}[\max_{T\in G_{x_1}} R(Tx) - \min_{T\in G_{x_1}} R(Tx)]$. For any finite set of tested shifts $S \subset G_{x_1}$, the empirical shift barrier

$$b_S(x) = \frac{1}{2}\left[\max_{T\in S} R(Tx) - \min_{T\in S} R(Tx)\right] \leq b_{G_{x_1}}(x) \tag{36}$$

provides a rigorous lower bound on the full-orbit translation barrier $b_{G_{x_1}}(x)$. In Section 6.4, we evaluate $b_S(x)$ by running full AT1 phase-field solves across grid translations and the targeted adversarial shift T_{adv} that aligns each cell's maximum elastic hotspot $\arg\max_x(\psi_+/\psi_c)$ directly with the weakened strip $x_1 = 0.95$.

3.3. Stage 0 verification, canonical hash isolation, and the 3,300-cell elastic corpus

Prior to neural training, Stage 0 reproduced all 216 design cells of Ref. [1] from raw phase masks, matching all published elastic responses and descriptors (W_{hom}, $\ln B$, f, ρ_I, $\bar{\varepsilon}^m$, $\bar{\varepsilon}^i$, J^m_{gap}, J^i_{gap}, A_{lig}, S_2, J_0, J_1, and J_2) to machine precision ($\max|\Delta| = 0.0$ on tabular descriptors and $\max|\Delta R_H(\beta)| \leq 4.58 \times 10^{-6}$ against single-precision stress archives). All 162 canonical geometries in the external frozen phase-field sets (Sets A/B, O, C1, C2) and all 216 design cells were registered by their dihedral-translation canonical SHA-256 hash

$$h_{\mathrm{canon}}(x) = \min_{g \in \mathbb{Z}^2_{127} \rtimes D_2} \mathrm{SHA256}(g \cdot x), \tag{37}$$

and excluded from the Stage 1 elastic corpus with zero collisions. Table 5 summarizes all dataset splits used in this study.

Table 5: Summary of the Stage 1 elastic benchmark splits ($N = 3{,}300$ cells) and the externally frozen AT1 phase-field fracture splits. All splits are strictly disjoint up to periodic translations and D_2 reflections.

Dataset / Split	Role	n	Morphology and Physics Specification
Elastic `train`	Encoder/head training	1,400	734 disk, 666 ellipse cells; $f \in [0.050, 0.350]$, $n_{\mathrm{inc}} \in \{1, \ldots, 25\}$, $\alpha \in [1.00, 3.99]$, $\kappa = 10$, uniaxial $\bar{\varepsilon} = e_1 \otimes e_1$.
Elastic `val`	Model/hyperparam. selection	300	158 disk, 142 ellipse cells from training morphology ranges ($\kappa = 10$).
Elastic `test1`	Secondary IID & physics-shift test	300	137 disk, 163 ellipse IID cells; also solved under 90°-rotated load $e_2 \otimes e_2$, contrasts $\kappa \in \{5, 20\}$, and biaxial $\bar{\varepsilon} = (1, 0.5, 0)$.
Elastic `test2`	Primary IID benchmark	1,000	495 disk, 505 ellipse IID cells; never accessed during training or tuning.
Elastic `ood_morphology`	OOD morphology transfer	300	162 disk, 138 ellipse cells with unseen multiplicity $n_{\mathrm{inc}} \in \{30, 36\}$ and extreme aspect ratios $\alpha \in [1.00, 6.98]$.
PF Design (non-lam.)	5-fold CV readout fitting	180	120 disk ($n_{\mathrm{inc}} \in \{1, 4, 9, 16\}$) and 60 ellipse ($\alpha = 3, n_{\mathrm{inc}} \in \{4, 9\}$) cells ($n = 150$ multi-inclusion cells for conditional tests `T1`–`T6`).
PF `Set B`	Blind IID fracture test	45	27 disk, 18 ellipse cells ($f \in [0.10, 0.30], n_{\mathrm{inc}} \in \{4, 9, 16\}$).
PF `Set O`	Blind OOD aspect-ratio test	12	Slender ellipses with $\alpha = 6$, $f \in [0.10, 0.20]$, $n_{\mathrm{inc}} \in \{4, 9, 16\}$.
PF `Set C1`	Prospective design-family test	30	18 disk, 12 ellipse cells generated after model freeze in Ref. [1].
PF `Set C2`	Prospective high-n_{inc} test	12	6 disk ($n_{\mathrm{inc}} = 25$) and 6 ellipse ($n_{\mathrm{inc}} = 16, \alpha = 3$) cells generated after freeze.

3.4. Compared representations, neural architectures, and evaluation protocol

Physics, statistical, and unsupervised autoencoder baselines.. We compare learned supervised representations against: (1) explicit physics descriptors `E0` ($d = 4$), `E1` ($d = 6$), geometric ligament descriptor `GL` ($d = 3$: f, interface density ρ_I, ligament amplification A_{lig}), and critical-distance `E1+J_1.5` ($d = 7$, for R_σ); (2) statistical baselines `MKS10` ($d = 10$, PCA of two-point autocovariance S_2) and `PCA10` ($d = 10$, PCA of raw binary masks); and (3) two families of unsupervised convolutional autoencoders:

- `ConvAE-BCE` ($d \in \{8, 32\}$): standard strided convolutional autoencoder ($127 \times 127 \to 8 \times 8$ via 4 strided conv layers, linear bottleneck d, symmetric upsampling decoder) trained with unweighted binary cross-entropy (BCE). Because the mean inclusion volume fraction on `test2` is $\bar{f} = 0.2026$ ($\bar{f} = 0.1920$ on `val`), predicting the constant all-matrix image $x \equiv 0$ already achieves a trivial pixel accuracy of $1 - \bar{f} = 79.7\%$ on `test2` (80.8% on `val`) with zero inclusion overlap (IoU = 0).
- `ConvAE-HiFi` ($d \in \{32, 128\}$): a high-capacity residual convolutional autoencoder ($127 \times 127 \to 8 \times 8$ spatial feature map of shape $128 \times 8 \times 8$ via 4 stride-2 GroupNorm residual downsampling blocks, linear bottleneck $d \in \{32, 128\}$, and symmetric residual upsampling decoder, trained for 80 epochs with positive-weighted BCE $w_{\mathrm{pos}} = (1 - \bar{f})/\bar{f} \approx 4.0$ plus soft Dice loss). We evaluate all autoencoders by pixel accuracy relative to the $1 - \bar{f}$ baseline, inclusion Intersection-over-Union (IoU), inclusion Dice coefficient, and 1-pixel morphological boundary F1 score.

Equivariant neural backbones.. To ensure that representation quality rather than symmetry mismatch is tested, our default encoder `E-circ` uses 6 full-resolution circularly padded dilated convolutional layers ($1 \to 16 \to 32 \to 32 \to 64 \to 64 \to 64$, kernel 5 then 3, dilations $1, 2, 4, 8, 16, 32$, receptive field $129\,\text{px} > 127\,\text{px}$) with whole-cell GroupNorm, GELU, softplus activation, target-matched spatial pooling (`GAP` $p = 1$ or learnable GeM_p), and linear projection to $\mathbb{R}^d$. Exact invariance under the dihedral reflection group $D_2 = \{e, \sigma_x, \sigma_y, \sigma_x\sigma_y\}$ is enforced by averaging the 4 reflection views at evaluation and hard-query mining (`d2_embed`). Two additional backbones are evaluated in Section 5.2: a spectral Fourier Neural Operator encoder (`E-FNO`, 4 FNO blocks retaining 12×12 Fourier modes [28]) and a non-equivariant zero-padded dilated CNN (`E-std`) with and without a soft invariance penalty $\lambda_{\text{inv}}\mathcal{L}_{\text{inv}}$. All primary supervised models (`MSE`, `Tail-MSE`, `SBR-in`, `RankSim`, `RNC`, and `SBR-CF`) are trained for 60 epochs with per-epoch hard-query mining (`mine_every` $= 1$) across 3 locked random seeds ($s \in \{0, 1, 2\}$); the auxiliary component-ablation suite in Section 5.2 uses a 45-epoch schedule with `mine_every` $= 5$ across matched pairs.

Evaluation metrics and 1D calibration references.. Every representation Ψ is evaluated by: (i) the equal-coverage worst-case response difference among the $q = 1\%$ closest pairs,

$$\Omega_\Psi(q) = \max\{|R_i - R_j| : (i, j) \text{ among the } q\text{-fraction closest pairs in } d_\Psi\}, \tag{38}$$

normalized by its median $\Omega_\Psi^{\text{null}}(q)$ over $n_{\text{perm}} = 200$ random response permutations; (ii) the exact finite-sample barrier curve $e^\star(\Lambda)/\sigma_R$ and finite-sample interpolation slope $\Lambda_{\text{interp}}/\sigma_R$; (iii) out-of-sample McShane midpoint metrics (R^2_{McShane}, $\text{CF}_{\text{rmse}}/\sigma_R$, and $\text{CF}_{\max}/\sigma_R$) with anchor set $A =$ `train` and slope Λ selected on `val`; and (iv) linear ridge readout R^2_{ridge} and post-hoc certified Lipschitz head error e_∞. As 1D calibration references (not universal lower bounds), we report `ORACLE_noise`$(x) = R(x) + \varepsilon_{\text{num}}$ ($\sigma_{\text{num}} = 0.0025$ for R_E, 0.08 for $R_H(20)$) and `ORACLE_10pct`$(x) = R(x) + 0.10\sigma_R\eta$ ($\eta \sim \mathcal{N}(0, 1)$, $d = 1$). Because a 1D noisy copy of R has no orthogonal coordinates, multi-dimensional embeddings ($d = 8$) are not mathematically constrained to stay above `ORACLE_10pct`, which serves purely as a physical scale reference on the 0–1 insufficiency axis.

4. Empirical Results on the Elastic Benchmark (Experiments A–E)

4.1. Experiments A–C: Physics descriptors, high-IoU autoencoders, and MSE-supervised encoders

Table 6 and Figures 1–2 compare explicit physics descriptors, statistical baselines, unsupervised autoencoders, and MSE-supervised equivariant encoders on homogenized energy R_E and nonlocal initiation hazard $R_H(\beta = 20)$ under Mahalanobis whitening d_{white}. To address Hypothesis H1 rigorously, Table 7 reports the full morphological reconstruction metrics (pixel accuracy vs. the $1 - \bar{f} = 0.797$ all-matrix baseline, inclusion IoU, Dice, and boundary F1) alongside mechanical sufficiency for both standard `ConvAE-BCE` ($d = 8, 32$) and high-fidelity `ConvAE-HiFi` ($d = 32, 128$).

Three conclusions follow from Tables 6 and 7:

1. **Autoencoder reconstruction vs. hazard regularity (H1):** As Table 7 shows, standard `ConvAE-BCE` at $d = 8$ (81.5% pixel accuracy on `test2`) barely exceeds the trivial all-matrix baseline $1-\bar{f} = 79.7\%$ (IoU = 0.147), so its poor performance alone would not test Hypothesis H1 fairly. However, when we train the residual `ConvAE-HiFi` ($d = 128$) with positive-weighted BCE and soft Dice loss, morphological reconstruction reaches **93.7**% pixel accuracy, inclusion IoU = **0.728**, Dice = **0.825**, and boundary F1 = **0.857**. Despite this high reconstruction fidelity, its latent metric exhibits $\Omega/\Omega^{\text{null}} = 0.882$ and $R^2_{\text{McShane}} = -0.714$ on $R_H(20)$ (plausibly because pixel-wise and overlap reconstruction losses allocate latent capacity primarily to global inclusion positions and translation phase rather than narrow inter-inclusion ligaments). Thus Hypothesis H1 is confirmed even in the high-IoU regime.
2. **Closure of bulk energy R_E vs. hazard gap (H2):** On R_E, the 6-feature physics descriptor `E1` achieves $\Omega/\Omega^{\text{null}} = 0.038$ and $R^2_{\text{McShane}} = 0.983$, while supervised `E-circ` reaches $\Omega/\Omega^{\text{null}} =$

Table 6: Experiments A–C (3-seed medians for learned models under d_{white}): comparison of explicit descriptors, statistical baselines, unsupervised autoencoders, and MSE-supervised equivariant encoders on R_E and $R_H(\beta = 20)$, alongside 1D calibration references (`ORACLE_noise`, `ORACLE_10pct`).

Target	Representation	d	IID Test (`test2`, $n = 1{,}000$)				OOD (`ood`, $n = 300$)	
			$\Omega/\Omega^{\text{null}}$	R^2_{McShane}	R^2_{ridge}	$\text{CF}_{\max}/\sigma_R$	$\Omega/\Omega^{\text{null}}$	R^2_{McShane}
R_E	`ORACLE_noise` (1D num. ref.)	1	0.007	0.997	1.000	1.639	0.016	1.000
	`ORACLE_10pct` (1D 10% noise ref.)	1	0.046	0.988	0.991	1.618	0.103	0.990
	`E0` (volume frac. + matrix strains)	4	0.044	0.966	0.970	1.618	0.045	0.942
	`E1` (`E0` + Jensen gaps)	6	**0.038**	**0.983**	**0.984**	2.032	0.054	0.918
	`GL` ($f, \rho_I, A_{\text{lig}}$)	3	0.178	0.921	0.847	3.155	0.337	0.810
	`MKS10` (2-point stats PCA)	10	0.230	0.750	0.782	2.346	0.291	−0.192
	`PCA10` (image PCA)	10	0.184	0.783	0.826	5.270	0.174	0.878
	`ConvAE-BCE` (IoU = 0.147)	8	0.282	0.463	0.000	5.270	0.251	−0.182
	`ConvAE-BCE` (IoU = 0.437)	32	1.047	0.315	0.725	5.270	0.802	0.201
	`ConvAE-HiFi` (IoU = 0.728)	128	1.414	−1.307	0.366	5.270	0.891	−0.912
	`Sup E-circ (GAP, p=1)`	8	**0.085**	0.970	**0.981**	**1.618**	0.170	0.948
	`Sup E-circ (GeM, learnable p)`	8	0.090	**0.970**	0.971	**1.618**	**0.154**	**0.973**
$R_H(20)$	`ORACLE_noise` (1D num. ref.)	1	0.019	1.000	1.000	0.067	0.027	1.000
	`ORACLE_10pct` (1D 10% noise ref.)	1	0.106	0.989	0.990	0.330	0.137	0.984
	`E0` (volume frac. + matrix strains)	4	0.335	0.792	0.740	2.582	0.231	0.685
	`E1` (`E0` + Jensen gaps)	6	0.335	0.853	0.833	1.734	0.220	−0.431
	`GL` ($f, \rho_I, A_{\text{lig}}$)	3	0.549	0.848	0.828	2.108	0.327	0.671
	`MKS10` (2-point stats PCA)	10	0.355	0.687	0.766	2.269	0.377	0.575
	`PCA10` (image PCA)	10	0.459	0.613	0.677	3.018	**0.178**	0.904
	`ConvAE-BCE` (IoU = 0.147)	8	0.580	0.385	−0.009	3.002	0.299	0.412
	`ConvAE-BCE` (IoU = 0.437)	32	0.822	0.258	0.582	3.379	0.678	−0.553
	`ConvAE-HiFi` (IoU = 0.728)	128	0.882	−0.714	0.378	3.410	0.402	−0.418
	`Sup E-circ (GAP, p=1)`	8	0.396	0.933	0.965	**1.575**	0.258	0.922
	`Sup E-circ (GeM, learnable p)`	8	**0.273**	**0.947**	**0.969**	1.677	0.229	**0.957**

Table 7: Morphological reconstruction metrics vs. mechanical regularity for unsupervised convolutional autoencoders on `test2` ($n = 1{,}000$, 3-seed medians; trivial all-matrix constant predictor $x \equiv 0$ achieves pixel accuracy $1 - \bar{f} = 0.797$ on `test2` and 0.808 on `val` with IoU = 0). Improving inclusion segmentation to IoU = 0.728 and Dice = 0.825 (`ConvAE-HiFi`, $d = 128$) does not organize the latent metric around stress-tail hazards.

Autoencoder Family	d	Morphological Reconstruction (`test2`)				$R_H(\beta = 20)$ Regularity			R_E
		Px. Acc. (vs. 0.797)	Incl. IoU	Incl. Dice	Bound. F1	$\Omega/\Omega^{\text{null}}$	R^2_{Mc}	R^2_{ridge}	$\Omega/\Omega^{\text{null}}$
All-matrix baseline ($x \equiv 0$)	—	0.797 (`val` 0.808)	0.000	0.000	0.000	—	—	—	—
`ConvAE-BCE` ($d = 8$)	8	0.815 (`val` 0.827)	0.147	0.206	0.140	0.580	0.385	−0.009	0.282
`ConvAE-BCE` ($d = 32$)	32	0.859 (`val` 0.862)	0.437	0.556	0.449	0.822	0.258	0.582	1.047
`ConvAE-HiFi` ($d = 32$)	32	0.841 (`val` 0.844)	0.444	0.571	0.514	0.625	−0.140	−0.000	0.533
`ConvAE-HiFi` ($d = 128$)	128	**0.937** (`val` **0.937**)	**0.728**	**0.825**	**0.857**	0.882	−0.714	0.378	1.414

0.085–0.090. On $R_H(\beta = 20)$, `E1` degrades to $\Omega/\Omega^{\text{null}} = 0.335$ ($R^2_{\text{McShane}} = -0.431$ on OOD). Standard MSE supervision (`Sup E-circ (GeM, d=8)`) attains high average accuracy ($R^2_{\text{ridge}} = 0.969$, $R^2_{\text{McShane}} = 0.947$), yet retains worst-case insufficiency $\Omega/\Omega^{\text{null}} = 0.273$ (seed range $[0.265, 0.375]$) and $\text{CF}_{\max}/\sigma_R = 1.677$.

4.2. Hypothesis H3: Target dependence and learnable GeM *pooling exponent*

Table 8 reports the learned GeM_p pooling exponent across targets and objectives (initialized at $p_0 = 3.0$ with dedicated log-exponent learning rate $\eta_q = 0.05$). Across the initiation hazard family $R_H(\beta)$, the 3-seed median exponent grows monotonically with tail sharpness β from $p = 3.84$ ($\beta = 2$) to $p = 8.64$ ($\beta = 20$; 8.84–9.61 under `SBR-CF`) and $p = 10.31$ ($\beta = 40$), and replacing GeM_p with fixed `GAP` ($p = 1$) on $R_H(20)$ degrades `MSE` insufficiency from 0.273 to 0.396. On smooth bulk energy R_E, downstream affine layers compensate moderate power-mean distortions, leaving a flat valley around $p = 3.90$, whereas removing the MSE regression head under unanchored `RNC` drives

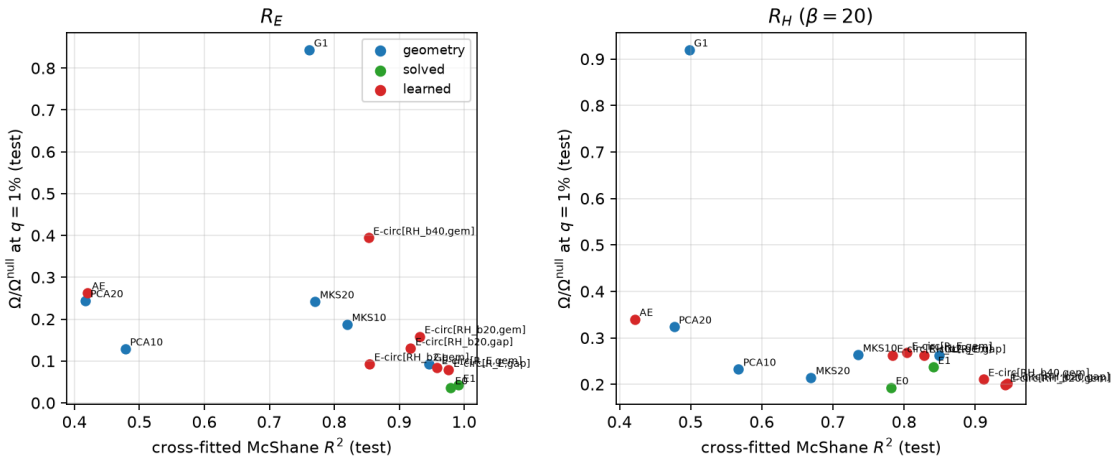


Figure 1: Initial gate benchmark (`test1`, $n = 300$, d_{white}): cross-fitted McShane R^2 vs. null-normalized worst-case insufficiency $\Omega/\Omega^{\text{null}}$ ($q = 1\%$) for R_E (left) and $R_H(\beta = 20)$ (right).

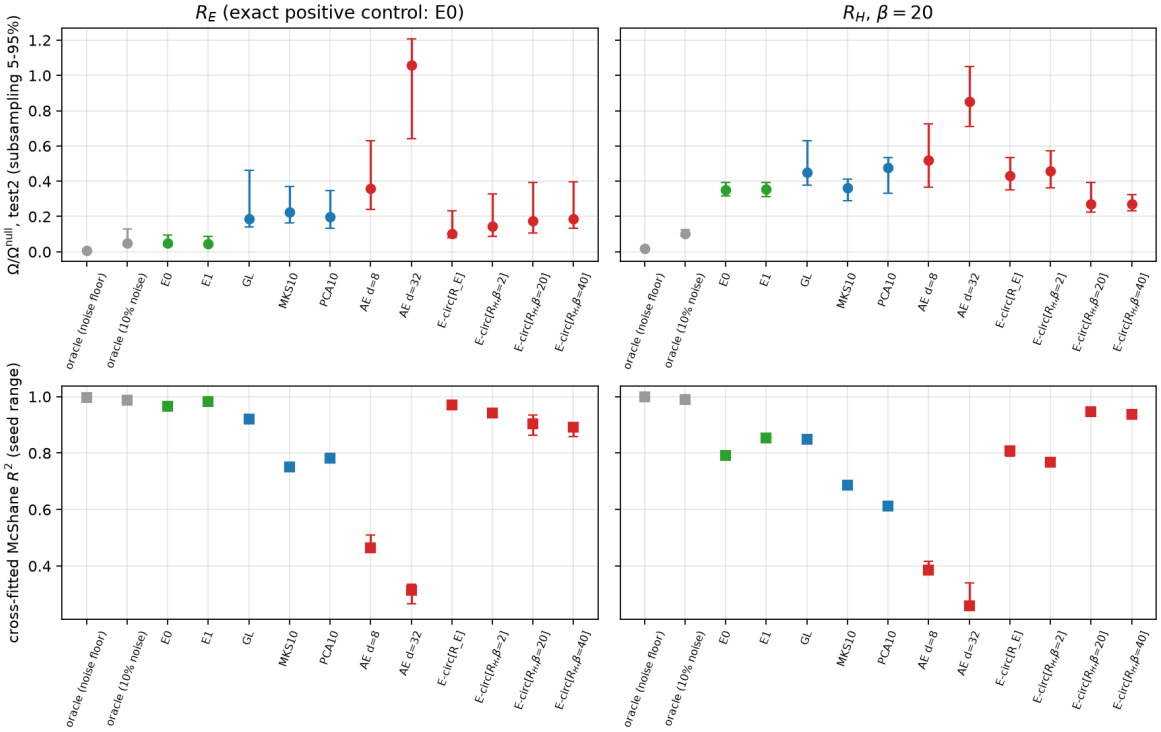


Figure 2: Consolidated 3-seed comparison (`test2`, $n = 1{,}000$) showing 5–95% paired subsampling intervals (SI_{90}) for $\Omega/\Omega^{\text{null}}$ (top) and 3-seed min/max ranges for R^2_{McShane} (bottom) on R_E (left) and $R_H(\beta = 20)$ (right).

$p \to 841$–2032.

4.3. Experiments D–E: Cross-fitted barrier sweep, `Tail-MSE`, `SBR-in`, and matched hyperparameter selection

To ensure transparent, symmetric hyperparameter selection across all objective families before inspecting `test2`, Table 9 reports the 3-seed median validation metrics on held-out `val` ($n = 300$, d_{white}) across all 15 hyperparameter candidates. On `val`, `RankSim` selects $\gamma = 1.0$ over $\gamma \in \{10, 100\}$ under all three criteria ($\Omega/\Omega^{\text{null}}$, $\text{CF}_{\text{rmse}}/\sigma_R$, and $\text{CF}_{\text{max}}/\sigma_R$); `RNC` selects $\tau = 1.0$ over $\tau \in \{0.5, 2.0\}$ under all three criteria; `Tail-MSE` selects $\lambda_{\text{tail}} = 3.0$ over 1.0 under all three criteria; `SBR-in` selects $\lambda_{\text{bar}} = 1.0$ by $\Omega/\Omega^{\text{null}}$ and $\text{CF}_{\text{rmse}}/\sigma_R$ (while $\lambda_{\text{bar}} = 3.0$ has lower `val` $\text{CF}_{\text{max}}/\sigma_R$); and within

Table 8: Learned generalized-mean (GeM_p) pooling exponent p across mechanical targets of increasing tail sharpness β and training objectives ($d = 8$, 3 seeds, initialized at $p_0 = 3.0$).

Target	Objective	Seed 0	Seed 1	Seed 2	3-Seed Median p
R_E (bulk homogenized energy)	`MSE`	3.90	3.35	5.15	3.90
$R_H(\beta = 2)$ (mild heterogeneity)	`MSE`	3.84	3.24	5.18	**3.84**
$R_H(\beta = 20)$ (sharp tail hazard)	`MSE`	8.64	4.58	10.21	**8.64**
$R_H(\beta = 40)$ (extreme tail hazard)	`MSE`	10.31	5.41	10.84	**10.31**
$R_H(\beta = 20)$	`SBR-CF` ($\lambda_{\mathrm{cf}} = 1.0$)	8.84	7.47	9.35	**8.84**
$R_H(\beta = 20)$	`SBR-CF` ($\lambda_{\mathrm{cf}} = 3.0$)	9.61	8.67	9.78	**9.61**
$R_H(\beta = 20)$	`RankSim` ($\gamma = 1.0$)	8.38	3.13	7.80	7.80
$R_H(\beta = 20)$	`RNC` ($\tau = 1.0$, no MSE head)	841.1	1065.5	1108.8	1065.5†
$R_H(\beta = 20)$	`RNC` ($\tau = 2.0$, no MSE head)	1633.3	2032.4	1909.4	1909.4†

†Without an MSE regression head, pure `RNC` gradients drive $p \to \infty$ (hard max-pooling regime).

`SBR-CF`, $\lambda_{\mathrm{cf}} = 1.0$ achieves the lowest validation $\mathrm{CF}_{\mathrm{rmse}}/\sigma_R$ (0.186), whereas $\lambda_{\mathrm{cf}} = 3.0$ achieves the lowest validation $\Omega/\Omega^{\mathrm{null}}$ (0.144) and $\mathrm{CF}_{\max}/\sigma_R$ (1.061). We therefore report both $\lambda_{\mathrm{cf}} = 1.0$ and $\lambda_{\mathrm{cf}} = 3.0$ alongside all baseline hyperparameter settings in Table 10.

Table 9: Validation hyperparameter sweep on held-out `val` ($n = 300$, $d = 8$, 3-seed medians under d_{white}). Bold entries mark the best hyperparameter within each multi-parameter family under each validation criterion.

Method Family	Hyperparameter Grid	`val` $\Omega/\Omega^{\mathrm{null}}$ (↓)	`val` $\mathrm{CF}_{\mathrm{rmse}}/\sigma_R$ (↓)	`val` $\mathrm{CF}_{\max}/\sigma_R$ (↓)	Selected Setting(s)
`MSE`	$\lambda_{\mathrm{cf}} = 0$	0.244	0.223	0.799	$\lambda_{\mathrm{cf}} = 0$ (ref.)
`SBR-CF`	$\lambda_{\mathrm{cf}} \in \{0.1, 0.3, 1.0, 3.0\}$	$0.213, 0.211, 0.158, \mathbf{0.144}$	$0.232, 0.221, \mathbf{0.186}, 0.193$	$1.164, 1.064, 1.089, \mathbf{1.061}$	$\lambda_{\mathrm{cf}} = 1.0, 3.0$
`Tail-MSE`	$\lambda_{\mathrm{tail}} \in \{1.0, 3.0\}$	$0.228, \mathbf{0.188}$	$0.264, \mathbf{0.260}$	$1.230, \mathbf{1.220}$	$\lambda_{\mathrm{tail}} = 3.0$
`SBR-in`	$\lambda_{\mathrm{bar}} \in \{1.0, 3.0\}$	$\mathbf{0.238}, 0.253$	$\mathbf{0.286}, 0.294$	$1.243, \mathbf{1.058}$	$\lambda_{\mathrm{bar}} = 1.0, 3.0$
`RankSim`	$\gamma \in \{1.0, 10.0, 100.0\}$	$\mathbf{0.220}, 0.276, 0.237$	$\mathbf{0.265}, 0.293, 0.288$	$\mathbf{0.953}, 1.219, 1.157$	$\gamma = 1.0$
`RNC`	$\tau \in \{0.5, 1.0, 2.0\}$	$0.226, \mathbf{0.206}, 0.273$	$0.311, \mathbf{0.266}, 0.288$	$1.206, \mathbf{0.900}, 1.243$	$\tau = 1.0$

Because $\Omega_\Psi(q)$ is an extreme-value order statistic over $\binom{m}{2}$ pairs, we quantify paired finite-sample stability on `test2` ($n = 1{,}000$) by drawing $B = 300$ paired random half-samples of size $m = 500$ without replacement and recording the empirical 5th–95th percentile range $[q_5, q_{95}]$ of the seed-median difference $\Delta_{\mathrm{paired}} = \Omega_{\mathrm{method}}/\Omega^{\mathrm{null}}_{\mathrm{method}} - \Omega_{\mathrm{MSE}}/\Omega^{\mathrm{null}}_{\mathrm{MSE}}$. Because overlapping half-samples of a fixed $n = 1{,}000$ test set are positively correlated, we denote $[q_5, q_{95}]$ explicitly as a 90% paired Subsample Interval (90% SI), distinguishing it from an asymptotic population confidence interval.

Table 10: Experiments D–E on $R_H(\beta = 20)$ ($d = 8$, 3-seed medians under d_{white}, all trained for 60 epochs): comparison of `SBR-CF` against `MSE`, top-k extreme-residual `Tail-MSE` ($\lambda_{\mathrm{tail}} \in \{1, 3\}$), un-cross-fitted `SBR-in` ($\lambda_{\mathrm{bar}} \in \{1, 3\}$), `RankSim`, and `RNC`. Δ_{paired} reports the median and 90% paired subsample interval ($[q_5, q_{95}]$, 90% SI) over 300 paired half-samples ($m = 500$) relative to `MSE`.

Objective ($d = 8$, 60 epochs)	IID Test (`test2`, $n = 1{,}000$, d_{white})						OOD (`ood`, $n = 300$)	
	$\Omega/\Omega^{\mathrm{null}}$	Δ_{paired} [90% SI]	% < MSE	R^2_{Mc}	R^2_{ridge}	$\mathrm{CF}_{\max}/\sigma_R$	$\Omega/\Omega^{\mathrm{null}}$	R^2_{Mc}
`MSE` ($\lambda_{\mathrm{cf}} = 0$)	0.273	0.000 (sub. 0.275)	—	0.947	0.969	1.677	0.229	0.957
`SBR-CF` ($\lambda_{\mathrm{cf}} = 0.1$)	0.269	-0.033 $[-0.117, +0.035]$	82.3%	0.948	0.964	1.504	0.241	0.938
`SBR-CF` ($\lambda_{\mathrm{cf}} = 0.3$)	0.268	-0.046 $[-0.178, +0.139]$	81.7%	0.951	0.970	1.683	0.216	0.962
`SBR-CF` ($\lambda_{\mathrm{cf}} = 1.0$)*	**0.192**	$\mathbf{-0.092}$ $[-0.199, -0.015]$	**98.3%**	**0.962**	0.975	1.402	0.199	**0.963**
`SBR-CF` ($\lambda_{\mathrm{cf}} = 3.0$)*	**0.188**	$\mathbf{-0.090}$ $[-0.213, -0.025]$	**99.7%**	0.961	**0.977**	**1.165**	**0.169**	**0.963**
`Tail-MSE` ($\lambda_{\mathrm{tail}} = 1.0$, top-$k$)	0.306	$+0.021$ $[-0.088, +0.114]$	36.0%	0.935	0.974	1.572	0.229	0.953
`Tail-MSE` ($\lambda_{\mathrm{tail}} = 3.0$, top-$k$)*	0.240	-0.034 $[-0.142, +0.048]$	74.3%	0.941	0.979	1.797	0.183	0.950
`SBR-in` ($\lambda_{\mathrm{bar}} = 1.0$)	0.303	$+0.037$ $[-0.095, +0.105]$	26.7%	0.926	0.961	1.775	0.228	0.943
`SBR-in` ($\lambda_{\mathrm{bar}} = 3.0$)	0.314	$+0.044$ $[-0.081, +0.132]$	21.3%	0.909	0.949	1.779	0.263	0.902
`RankSim` ($\gamma = 10$, default)	0.294	$+0.036$ $[-0.087, +0.125]$	26.7%	0.911	0.962	1.621	0.232	0.886
`RankSim` ($\gamma = 1.0$, val-selected)*	0.261	-0.012 $[-0.147, +0.045]$	59.0%	0.924	0.974	1.598	0.205	0.949
`RNC` ($\tau = 2.0$, default)	0.298	$+0.002$ $[-0.102, +0.072]$	47.7%	0.903	0.989	1.797	0.328	0.893
`RNC` ($\tau = 1.0$, val-selected)*	0.225	-0.036 $[-0.165, +0.018]$	89.0%	0.915	**0.990**	2.065	0.284	0.915

*Selected on held-out `val` ($n = 300$, Table 9): $\lambda_{\mathrm{cf}} = 1.0$ minimizes validation $\mathrm{CF}_{\mathrm{rmse}}/\sigma_R$, while $\lambda_{\mathrm{cf}} = 3.0$ minimizes validation $\Omega/\Omega^{\mathrm{null}}$ and $\mathrm{CF}_{\max}/\sigma_R$.

Table 10 isolates the mechanism of SBR-CF: (i) Penalizing the top-$k = 16$ squared residuals of the parametric head directly (Tail-MSE) improves linear ridge fit ($R^2_{\text{ridge}} = 0.979$ at $\lambda_{\text{tail}} = 3.0$) and lowers $\Omega/\Omega^{\text{null}}$ modestly to 0.240, but degrades R^2_{McShane} ($0.947 \to 0.941$) and inflates out-of-sample McShane maximum error to $\text{CF}_{\max}/\sigma_R = 1.797$. Thus emphasizing extreme head residuals is not equivalent to regularizing cross-fitted McShane neighborhoods. (ii) Increasing the un-cross-fitted in-sample barrier weight from $\lambda_{\text{bar}} = 1.0$ to 3.0 (SBR-in) further worsens $\Omega/\Omega^{\text{null}}$ ($0.303 \to 0.314$) and R^2_{McShane} ($0.926 \to 0.909$), confirming Section 2.2. (iii) SBR-CF ($\lambda_{\text{cf}} = 1.0, 3.0$) cuts whitened $\Omega/\Omega^{\text{null}}$ to **0.192** and **0.188** (98.3%–99.7% paired subsample wins over MSE), raises R^2_{McShane} to **0.962**, and lowers $\text{CF}_{\max}/\sigma_R$ to **1.165** on test2 and $\Omega/\Omega^{\text{null}}$ to **0.169** on ood_morphology.

4.4. Full exact barrier curves $e^\star_\theta(\Lambda)/\sigma_R$, $\Lambda_{\text{interp}}/\sigma_R$, and certified head decomposition

In Table 12, post-hoc spectrally normalized Lipschitz heads g_ϕ are evaluated at their validation-selected dimensionless budget $\widetilde{\Lambda}_{\text{cert}} = \Lambda_{\text{cert}}/\sigma_R$. Because $\widetilde{\Lambda}_{\text{cert}} = 4.0$ exceeds the finite-sample training interpolation constant $\Lambda_{\text{interp}}/\sigma_R$ in $d_{\mathcal{N}}$ for well-conditioned $d = 8$ models (e.g., $\Lambda_{\text{interp}}/\sigma_R = 3.700 < 4.0$ for MSE and $2.797 < 4.0$ for SBR-CF $\lambda_{\text{cf}} = 1.0$ on train in Table 11), $e^\star_{\text{train}}(4\sigma_R) = 0$ in Table 12, so $\Delta_{\text{head}} = e_{\infty,\text{train}}$ at that budget measures the improvement in maximum training fit of the Lipschitz head ($0.513 \to 0.248$). To compare the predictor-independent representation barriers $e^\star_\theta(\Lambda)$ directly across budgets where the barriers are strictly positive, Table 11 reports the complete exact barrier curves $e^\star_\theta(\Lambda)/\sigma_R$ across $\Lambda/\sigma_R \in \{0.25, 0.5, 1.0, 1.5, 2.0, 3.0, 4.0\}$ together with $\Lambda_{\text{interp}}/\sigma_R$ on both train ($n = 1{,}400$) and test2 ($n = 1{,}000$) in $d_{\mathcal{N}}$ and d_{white}.

Table 11: Exact finite-sample Lipschitz barrier curves $e^\star_\theta(\Lambda)/\sigma_R$ and finite-sample interpolation slopes $\Lambda_{\text{interp}}/\sigma_R$ (Eq. (9)) across common Lipschitz budgets $\Lambda/\sigma_R \in \{0.25, 0.5, 1.0, 1.5, 2.0, 3.0, 4.0\}$ on $R_H(\beta = 20)$ ($d = 8$, 3-seed medians) in frozen LatentNorm space $d_{\mathcal{N}}$ and full Mahalanobis whitening d_{white} ($\delta = 10^{-9}$). Bold entries mark the exact column minimum within each split and metric space. Under d_{white} on test2, RNC ($\tau = 1.0$) and Tail-MSE ($\lambda_{\text{tail}} = 3.0$) achieve lower positive barriers than SBR-CF, whereas in $d_{\mathcal{N}}$ on test2, SBR-CF ($\lambda_{\text{cf}} = 3.0$) achieves the lowest barrier at higher budgets $\Lambda/\sigma_R \in \{2.0, 3.0\}$.

Split / Space	Objective ($d = 8$)	$\Lambda_{\text{interp}}/\sigma_R$	Exact Sample Barrier $e^\star_\theta(\Lambda)/\sigma_R$ at $\Lambda/\sigma_R =$						
			0.25	0.50	1.00	1.50	2.00	3.00	4.00
train ($d_{\mathcal{N}}$)	MSE ($\lambda_{\text{cf}} = 0$)	3.700	3.040	2.724	2.122	1.520	0.937	0.237	**0.000**
	Tail-MSE ($\lambda_{\text{tail}} = 3.0$)	3.044	2.989	2.597	1.869	1.148	0.427	0.003	**0.000**
	SBR-in ($\lambda_{\text{bar}} = 1.0$)	3.960	3.068	2.762	2.182	1.664	1.115	0.255	**0.000**
	SBR-in ($\lambda_{\text{bar}} = 3.0$)	4.364	3.077	2.762	2.212	1.663	1.143	0.331	0.150
	RankSim ($\gamma = 1.0$)	3.356	3.055	2.690	2.028	1.400	0.785	0.024	**0.000**
	RNC ($\tau = 1.0$)	4.769	**2.666**	**1.913**	**0.571**	**0.244**	**0.172**	0.086	0.035
	SBR-CF ($\lambda_{\text{cf}} = 1.0$)	**2.797**	3.094	2.768	2.117	1.465	0.842	**0.000**	**0.000**
	SBR-CF ($\lambda_{\text{cf}} = 3.0$)	3.650	3.048	2.677	1.987	1.324	0.662	0.013	**0.000**
test2 ($d_{\mathcal{N}}$)	MSE ($\lambda_{\text{cf}} = 0$)	4.249	2.554	2.252	1.706	1.189	0.777	0.267	0.037
	Tail-MSE ($\lambda_{\text{tail}} = 3.0$)	**3.719**	2.492	2.132	1.455	0.863	0.477	0.180	**0.000**
	SBR-in ($\lambda_{\text{bar}} = 1.0$)	5.006	2.560	2.266	1.730	1.213	0.810	0.288	0.086
	SBR-in ($\lambda_{\text{bar}} = 3.0$)	4.563	2.563	2.277	1.768	1.283	0.855	0.305	0.080
	RankSim ($\gamma = 1.0$)	4.531	2.552	2.229	1.672	1.217	0.885	0.340	0.067
	RNC ($\tau = 1.0$)	32.242	**2.261**	**1.632**	**0.992**	**0.859**	0.802	0.688	0.573
	SBR-CF ($\lambda_{\text{cf}} = 1.0$)	4.398	2.571	2.251	1.632	1.068	0.584	0.148	0.034
	SBR-CF ($\lambda_{\text{cf}} = 3.0$)	5.255	2.562	2.233	1.610	1.016	**0.444**	**0.059**	0.024
test2 (d_{white})	MSE ($\lambda_{\text{cf}} = 0$)	3.865	2.554	2.255	1.692	1.242	0.812	0.307	**0.000**
	Tail-MSE ($\lambda_{\text{tail}} = 3.0$)	3.822	2.497	2.138	1.575	1.106	0.652	0.123	**0.000**
	SBR-in ($\lambda_{\text{bar}} = 1.0$)	4.768	2.560	2.265	1.743	1.235	0.803	0.363	0.069
	RankSim ($\gamma = 1.0$)	3.619	2.530	2.205	1.571	1.187	0.808	0.269	**0.000**
	RNC ($\tau = 1.0$)	**3.039**	**2.438**	**2.037**	**1.481**	**0.964**	**0.546**	**0.010**	**0.000**
	SBR-CF ($\lambda_{\text{cf}} = 1.0$)	3.731	2.562	2.234	1.618	1.121	0.782	0.189	**0.000**
	SBR-CF ($\lambda_{\text{cf}} = 3.0$)	3.571	2.581	2.271	1.652	1.188	0.772	0.163	**0.000**

Table 12: Post-hoc certified Lipschitz head decomposition (Eq. (25), 3-seed medians on $R_H(\beta = 20)$, $d = 8$, in standardized target units σ_R). At $\widetilde{\Lambda}_{\text{cert}} = \Lambda_{\text{cert}}/\sigma_R = 4.0$, $e^\star_{\text{train}}(\Lambda_{\text{cert}})/\sigma_R = 0$ because $\Lambda_{\text{interp}}/\sigma_R < 4.0$ on `train` in $d_{\mathcal{N}}$ (Table 11), so the reduction of $\Delta_{\text{head}} = e_{\infty,\text{train}}$ from 0.513 (`MSE`) to 0.248 (`SBR-CF`) reflects improved uniform fit of the spectrally normalized head on the regularized embedding.

Objective	Sel. $\widetilde{\Lambda}_{\text{cert}}$	Train $e^\star_\theta(\Lambda_{\text{cert}})/\sigma_R$	Train e_∞	Head Gap Δ_{head}	Test RMSE	Test $e^{\text{cf}}(\Lambda_{\text{cert}})/\sigma_R$
`MSE` ($\lambda_{\text{cf}} = 0$)	4.000	0.000	0.513	0.513	0.158	1.751
`SBR-CF` ($\lambda_{\text{cf}} = 0.1$)	4.000	0.000	0.291	0.291	0.165	1.541
`SBR-CF` ($\lambda_{\text{cf}} = 1.0$)	4.000	0.000	**0.248**	**0.248**	0.124	1.615
`SBR-in` ($\lambda_{\text{bar}} = 1.0$)	4.000	0.000	0.676	0.676	0.185	1.806
`RankSim` ($\gamma = 10$)	8.000	0.185	0.563	0.327	0.170	1.358
`RNC` ($\tau = 2.0$)	2.000	0.183	0.354	0.136	**0.106**	**1.236**

Table 11 reveals an important distinction between the exact global sample barrier $e^\star_\theta(\Lambda)$ and the local neighborhood metrics ($\Omega/\Omega^{\text{null}}$, e^{cf}) in Table 10, refining the evaluation of Hypothesis H4:

1. **Ordering under d_{white} vs. $d_{\mathcal{N}}$:** Under d_{white} on `test2`, `RNC` ($\tau = 1.0$) achieves the lowest exact barrier across all six evaluated budgets where the barriers are positive ($\Lambda/\sigma_R \in [0.25, 3.00]$ in Table 11, e.g., $1.481, 0.546, 0.010$ at $\Lambda/\sigma_R = 1, 2, 3$), and `Tail-MSE` ($\lambda_{\text{tail}} = 3.0$) achieves lower positive barriers ($1.575, 0.652, 0.123$) than `SBR-CF` ($\lambda_{\text{cf}} = 1.0$: $1.618, 0.782, 0.189$; $\lambda_{\text{cf}} = 3.0$: $1.652, 0.772, 0.163$), while `SBR-CF` improves consistently over standard `MSE` ($1.692, 0.812, 0.307$ for $\Lambda/\sigma_R \geq 1.0$) and un-cross-fitted `SBR-in` ($1.743, 0.803, 0.363$). In `LatentNorm` space $d_{\mathcal{N}}$ on `test2`, `RNC` dominates at small budgets $\Lambda/\sigma_R \in [0.25, 1.50]$ (0.859 at 1.50, vs. 0.863 for `Tail-MSE` and 1.016 for `SBR-CF` $\lambda_{\text{cf}} = 3.0$), whereas `SBR-CF` ($\lambda_{\text{cf}} = 3.0$) achieves the lowest barrier at higher budgets $\Lambda/\sigma_R \in \{2.0, 3.0\}$ (0.444 and 0.059, vs. 0.477 and 0.180 for `Tail-MSE` and 0.802 and 0.688 for `RNC`, whose interpolation slope in $d_{\mathcal{N}}$ reaches $\Lambda_{\text{interp}}/\sigma_R = 32.24$). On `train` in $d_{\mathcal{N}}$, `SBR-CF` ($\lambda_{\text{cf}} = 1.0$) achieves the lowest zero-barrier interpolation threshold $\Lambda_{\text{interp}}/\sigma_R = \mathbf{2.797}$.
2. **Why $\Omega/\Omega^{\text{null}}$, e^{cf}, and $e^\star_\theta(\Lambda)$ rank models differently:** By Theorem 1, the exact barrier $e^\star_\theta(\Lambda)$ is governed by a single global worst-case pair (i, j) at a fixed slope budget Λ, whereas $\Omega_\Psi(q)/\Omega^{\text{null}}_\Psi(q)$ measures worst-case collisions restricted to the $q = 1\%$ closest pairs relative to the distance distribution, and e^{cf} measures anchor-to-query midpoint extension from `train` to `test2`. `Tail-MSE` directly penalizes the largest individual head residuals, which separates the single most extreme global outlier pair at moderate Λ without preventing 1%-ball nearest-neighbor collisions ($\Omega/\Omega^{\text{null}} = 0.240$) or out-of-sample McShane spikes ($\text{CF}_{\max}/\sigma_R = 1.797$). Consequently, Hypothesis H4 is *partially supported*: `SBR-CF` improves $\Omega/\Omega^{\text{null}}$, R^2_{McShane}, $\text{CF}_{\max}/\sigma_R$, certified head excess Δ_{head}, and high-budget barriers in $d_{\mathcal{N}}$, whereas exact barriers $e^\star_\theta(\Lambda)$ under d_{white} favor `RNC` and `Tail-MSE`.

4.5. Latent dimension sweep ($d \in \{4, 8, 16\}$) and whitening/metric sensitivity (`RNC` vs. `SBR-CF`)

Table 13 and Figure 3 evaluate `MSE` and `SBR-CF` ($\lambda_{\text{cf}} = 1.0$) across latent dimensions $d \in \{4, 8, 16\}$. Under standard `MSE`, linear ridge accuracy remains invariant ($R^2_{\text{ridge}} = 0.961 \to 0.969 \to 0.967$), showing that functional predictive information is preserved as d grows; however, isotropic Mahalanobis whitening d_{white} amplifies weakly constrained extra coordinates at $d = 16$, degrading $\Omega/\Omega^{\text{null}}$ from 0.273 to 0.433 and R^2_{McShane} from 0.947 to 0.834 (the exact neural analogue of E_B in the synthetic pilot of Section 2.5). `SBR-CF` mitigates this high-dimensional whitening dilution, cutting $\Omega/\Omega^{\text{null}}$ at $d = 16$ from 0.433 to **0.238** (100.0% of paired $m = 500$ subsamples below `MSE`; $R^2_{\text{McShane}} = 0.910$ vs. 0.834) and OOD $\Omega/\Omega^{\text{null}}$ from 0.336 to **0.140**.

To provide a balanced, comprehensive comparison between `SBR-CF` and Rank-N-Contrast (`RNC`), Table 14 and Figure 4 evaluate all methods across the five metric normalizations of Section 2.4: full Mahalanobis whitening d_{white} ($\delta = 10^{-9}$, Eq. (19)), trace-scaled regularized whitening $d_{\text{reg}}(\alpha_{\text{reg}})$ across $\alpha_{\text{reg}} \in \{10^{-5}, 10^{-3}, 10^{-2}, 10^{-1}\}$ and validation-selected $d^\star_{\text{reg}}$ (Eq. (20)), eigenvalue-truncated whitening d_{trunc} ($\lambda_k \geq 10^{-2}\lambda_1$), exact training-set coordinate standardization d_{std}, and frozen `LatentNorm` running-moment standardization $d_{\mathcal{N}}$.

Table 13: Latent dimension sweep $d \in \{4, 8, 16\}$ on $R_H(\beta = 20)$ (3-seed medians under d_{white}). Standard `MSE` preserves $R^2_{\text{ridge}} \approx 0.96$–$0.97$ while suffering isotropic whitening dilution at $d = 16$ ($\Omega/\Omega^{\text{null}} = 0.433$); `SBR-CF` improves metric regularity at every dimension.

d	Objective	IID Test (`test2`, $n = 1{,}000$)					OOD (`ood`, $n = 300$)	
		$\Omega/\Omega^{\text{null}}$ (sub.)	Δ_{paired} [90% SI]	% < MSE	R^2_{Mc}	R^2_{ridge}	$\Omega/\Omega^{\text{null}}$	R^2_{Mc}
4	`MSE`	0.301 (0.238)	0.000	—	0.968	0.961	0.206	0.968
4	`SBR-CF` ($\lambda_{\text{cf}} = 1$)	**0.218** (**0.173**)	−0.064 [−0.173, +0.040]	84.3%	**0.980**	**0.973**	**0.173**	0.963
8	`MSE`	0.273 (0.275)	0.000	—	0.947	0.969	0.229	0.957
8	`SBR-CF` ($\lambda_{\text{cf}} = 1$)	**0.192** (**0.190**)	**−0.092** [−0.199, −0.015]	**98.3%**	**0.962**	**0.975**	**0.199**	**0.963**
16	`MSE`	0.433 (0.422)	0.000	—	0.834	0.967	0.336	0.834
16	`SBR-CF` ($\lambda_{\text{cf}} = 1$)	**0.238** (**0.239**)	**−0.181** [−0.313, −0.106]	**100.0%**	**0.910**	**0.971**	**0.140**	**0.899**

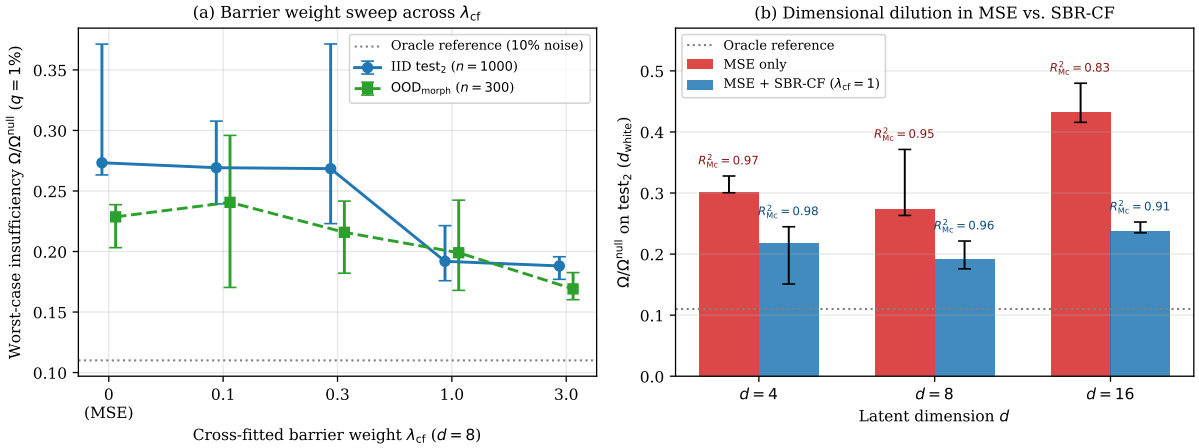


Figure 3: Left: Full-sample 3-seed median and min/max range of $\Omega/\Omega^{\text{null}}$ on `test2` ($n = 1{,}000$) and `ood_morphology` ($n = 300$) across $\lambda_{\text{cf}} \in \{0, 0.1, 0.3, 1.0, 3.0\}$. Right: Dimensional sweep $d \in \{4, 8, 16\}$ under d_{white}.

Table 14: Metric and whitening regularization sensitivity on `test2` ($n = 1{,}000$, $d = 8$, 3-seed medians) across full whitening d_{white} ($\delta = 10^{-9}$), trace-scaled regularized whitening $d_{\text{reg}}(\alpha_{\text{reg}})$ ($\alpha_{\text{reg}} \in \{10^{-5}, 10^{-3}\}$ and validation-selected $d^{\star}_{\text{reg}}$), truncated-spectrum whitening d_{trunc}, exact training-set coordinate standardization d_{std}, and frozen `LatentNorm` $d_{\mathcal{N}}$. Participation ratio $d_{\text{eff}} = (\sum_k \lambda_k)^2 / \sum_k \lambda_k^2$ and top eigenvalue share $\lambda_1 / \sum_k \lambda_k$ quantify embedding dimensionality.

Objective ($d = 8$)	d_{eff}	$\frac{\lambda_1}{\sum \lambda_k}$	$\Omega/\Omega^{\text{null}}$ (↓) on `test2`						R^2_{McShane} (↑) on `test2`			$\text{CF}_{\max}/\sigma_R$ (↓)	
			d_{white}	$d_{\text{reg}}(10^{-5})$	$d_{\text{reg}}(10^{-3})$	$d^{\star}_{\text{reg}}$	d_{trunc}	d_{std} / $d_{\mathcal{N}}$	d_{white}	$d^{\star}_{\text{reg}}$	d_{std}	d_{white}	d_{std}
`MSE` ($\lambda_{\text{cf}} = 0$)	5.03	0.341	0.273	0.271	0.271	0.271	0.271	0.264 / 0.258	0.947	0.948	0.952	1.677	1.650
`Tail-MSE` ($\lambda_{\text{tail}} = 1.0$)	4.71	0.358	0.306	0.306	0.306	0.258	0.288	0.257 / 0.257	0.935	0.943	0.959	1.572	1.503
`Tail-MSE` ($\lambda_{\text{tail}} = 3.0$)	3.68	0.456	0.240	0.240	0.240	0.237	0.240	0.222 / 0.222	0.941	0.955	0.967	1.797	1.797
`SBR-in` ($\lambda_{\text{bar}} = 1.0$)	5.42	0.279	0.303	0.303	0.303	0.304	0.303	0.300 / 0.274	0.926	0.915	0.937	1.775	1.926
`SBR-in` ($\lambda_{\text{bar}} = 3.0$)	5.86	0.242	0.314	0.314	0.314	0.314	0.314	0.314 / 0.295	0.909	0.915	0.931	1.779	1.765
`RankSim` ($\gamma = 1.0$)	1.32	0.868	0.261	0.258	0.258	0.219	0.266	0.236 / 0.237	0.924	0.957	0.950	1.598	1.598
`RNC` ($\tau = 1.0$)	1.004	0.998	0.225	0.224	0.166	**0.085**	0.281	**0.148** / **0.148**†	0.915	**0.989**	**0.991**	2.065	**1.028**
`SBR-CF` ($\lambda_{\text{cf}} = 1.0$)	5.77	0.283	0.192	0.193	0.193	0.193	0.193	0.192 / 0.192	**0.962**	0.965	0.976	1.402	1.520
`SBR-CF` ($\lambda_{\text{cf}} = 3.0$)	5.21	0.312	**0.188**	**0.188**	**0.188**	0.187	**0.186**	0.150 / 0.150	0.961	0.966	0.981	**1.165**	1.239

†As detailed in Section 2.4, d_{std} uses exact full-training-set coordinate moments $(\mu_{\text{tr}}, \sigma_{\text{tr}})$, whereas $d_{\mathcal{N}}$ uses the frozen running `BatchNorm1d` moments $(\mu_{\text{bn}}, \sigma_{\text{bn}})$ accumulated during training forward passes; for `RNC`, both yield 3-seed medians $\Omega/\Omega^{\text{null}} = 0.148$, $R^2_{\text{McShane}} = 0.991$, and $\text{CF}_{\max}/\sigma_R = 1.028$. Under $d_{\text{reg}}(10^{-5})$, `RNC` attains $R^2_{\text{McShane}} = 0.922$ and 3-seed median $\text{CF}_{\max}/\sigma_R = 1.527$ (seed range $[1.203, 2.065]$), vs. 0.915 and 2.065 under d_{white} ($\delta = 10^{-9}$). Across $\alpha_{\text{reg}} \in \{10^{-5}, 10^{-3}, 10^{-2}, 10^{-1}\}$, `RNC`'s $\Omega/\Omega^{\text{null}}$ moves from $0.224 \to 0.166 \to 0.112 \to 0.085$, whereas `SBR-CF` ($\lambda_{\text{cf}} = 1, 3$) remains within $[0.192, 0.193]$ and $[0.187, 0.188]$.

Table 14 and Figure 4 reveal a fundamental structural trade-off between 1D rank-contrastive embeddings (`RNC`) and full-rank barrier-regularized embeddings (`SBR-CF`):

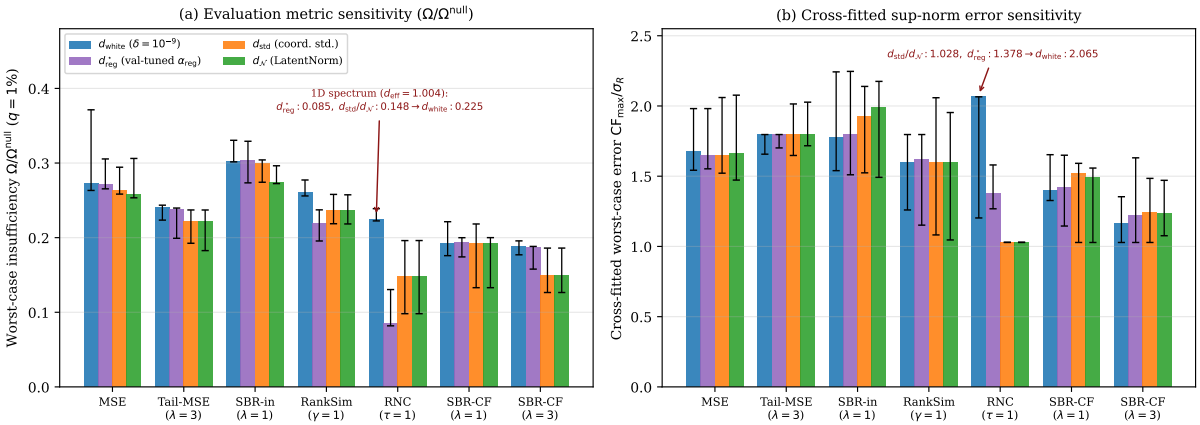


Figure 4: Evaluation metric sensitivity on `test2` ($d = 8$; bars show 3-seed medians, whiskers give 3-seed min/max ranges) across full Mahalanobis whitening d_{white} ($\delta = 10^{-9}$), validation-selected regularized whitening $d^{\star}_{\text{reg}}$, coordinate standardization d_{std}, and frozen `LatentNorm` $d_{\mathcal{N}}$, including both `RNC`'s $d_{\mathcal{N}}$ bar and the `Tail-MSE` ($\lambda = 3$) baseline.

1. **Strengths of `RNC` under coordinate standardization and regularized whitening:** Because $R_H(\beta = 20)$ is a scalar target, collapsing the 8-dimensional embedding onto a nearly one-dimensional curve ($d_{\text{eff}} = 1.004$, with 99.8% of variance in the leading eigenvalue; recall that an ideal scalar map $E(x) = R(x)$ is 1D) is geometrically natural for single-target prediction. Under coordinate standardization d_{std}, frozen `LatentNorm` $d_{\mathcal{N}}$, or validation-regularized whitening $d^{\star}_{\text{reg}}$ ($\alpha_{\text{reg}} = 10^{-1}$, which downweights the 7 minor eigendirections), `RNC` ($\tau = 1.0$) achieves the strongest single-target metrics in Table 14: $\Omega/\Omega^{\text{null}} = 0.148$ (d_{std} and $d_{\mathcal{N}}$) and 0.085 ($d^{\star}_{\text{reg}}$), $R^2_{\text{McShane}} = 0.991$ and 0.989, $\text{CF}_{\max}/\sigma_R = 1.028$, and $R^2_{\text{ridge}} = 0.990$.
2. **Distinct effects of full whitening d_{white} vs. 1D spectral truncation d_{trunc} on `RNC`, and metric stability of `SBR-CF`:** Under full Mahalanobis whitening d_{white} ($\delta = 10^{-9}$) or nearly unregularized $d_{\text{reg}}(10^{-5})$, rescaling the 7 minor eigendirections ($\lambda_2/\lambda_1 \leq 3.13 \times 10^{-3}$) to unit variance amplifies transverse noise, raising `RNC`'s $\Omega/\Omega^{\text{null}}$ to 0.225 (0.224 under $d_{\text{reg}}(10^{-5})$) and increasing its cross-fitted maximum error to $\text{CF}_{\max}/\sigma_R = 2.065$ (1.527 median, 2.065 max under $d_{\text{reg}}(10^{-5})$). Under hard eigenvalue truncation d_{trunc} ($\lambda_k \geq 10^{-2}\lambda_1$), the opposite mechanism occurs: because $\lambda_2/\lambda_1 < 10^{-2}$, d_{trunc} sets $k_{\text{eff}} = 1$ and projects `RNC` orthogonally onto its single leading principal axis v_1 (discarding the 7 minor directions rather than amplifying them). This 1D projection preserves global McShane interpolation ($R^2_{\text{McShane}} = 0.991$, $\text{CF}_{\max}/\sigma_R = 1.278$), but increases the local 1%-ball collision ratio $\Omega/\Omega^{\text{null}}$ to 0.281 because projecting out small transverse separations folds nearly co-projected cells into the $q = 1\%$ nearest-neighbor set. By contrast, `SBR-CF` maintains a well-conditioned multi-dimensional covariance ($d_{\text{eff}} = 5.77$ at $\lambda_{\text{cf}} = 1.0$ and 5.21 at $\lambda_{\text{cf}} = 3.0$, $k_{\text{eff}} = 8$), yielding empirically stable insufficiency across all spectral transformations (0.192–0.193 across $d_{\text{white}}, d_{\text{reg}}(10^{-5}), d^{\star}_{\text{reg}}, d_{\text{trunc}}, d_{\text{std}}, d_{\mathcal{N}}$ for $\lambda_{\text{cf}} = 1.0$; 0.186–0.188 across whitened/truncated metrics and 0.150 under $d_{\text{std}}/d_{\mathcal{N}}$ for $\lambda_{\text{cf}} = 3.0$). As shown in Section 6, this multi-dimensional structure also enables `SBR-CF` to transfer across distinct mechanical targets ($R_E, R_H(2)$) under whitened McShane evaluation (d_{white}) and standalone prospective fracture (`Set C2`) where a 1D embedding specialized to $R_H(20)$ degrades.

5. Physical Counterexamples, Grid Convergence, and Component Ablations

5.1. *Experiment F: Physical counterexample pairs, collision resolution across seeds, and* 254×254 *FFT grid refinement*

To test Hypothesis H5, we inspect the exact worst-case `MSE` collision pairs at $q = 1\%$ across each seed on `test2` ($\sigma_R = 4.33$) and across the full ensemble of 227 severe `MSE` collisions on `test2` (and 9 severe collisions on `ood_morphology`) satisfying $q_{\text{MSE}} \leq 1\%$ and $|\Delta R_H|$ in the top decile of 1%-ball

pair differences ($|\Delta R_H| \geq 2.47$ on `test2`). Table 15 and Figure 5 report seed-matched and 3-seed median tracking.

To verify that these severe hazard differences $|\Delta R_H(20)|$ on the fixed voxelized masks are not artifacts of FFT or nonlocal-kernel under-resolution on the 127×127 grid (where minimum gaps of 2.2–3.0 px span 2–3 voxels and nonlocal averaging radius $\ell = 0.025$ spans 3.18 px), we refine all 20 microstructures comprising the top-10 consensus `MSE` counterexample pairs onto a $2\times$ finer 254×254 periodic grid by exact 2×2 sub-voxel splitting (`np.repeat(..., 2)` along both axes) of the 127×127 binary phase masks, re-solve the Moulinec–Suquet FFT elasticity problem at 254×254 with doubled kernel resolution ($\ell = 6.35$ px), and recompute R_E and $R_H(20)$. Keeping the discrete voxelized phase boundary fixed while doubling the FFT and nonlocal-averaging resolution tests the numerical stability of the FFT stress solve and nonlocal integral on the fixed pixelated geometry (note that this check does not re-rasterize smooth analytical circle/ellipse boundaries and therefore does not bound continuum boundary-discretization error).

Table 15: Experiment F: resolution of physical latent counterexamples on $R_H(\beta = 20)$ and 254×254 FFT solver stability on the fixed voxelized masks. Top rows track each seed's exact worst-case `MSE` $\Omega(0.01)$ pair and the top-3 consensus pairs across `MSE` and `SBR-CF` ($\lambda_{\text{cf}} = 1, 3$). Bottom rows report aggregate resolution across all 227 severe `MSE` collisions on `test2` and 9 severe collisions on `ood_morphology`.

Seed / Split	Pair / Ensemble	$\lvert\Delta R_H\rvert_{127}$ (127^2)	$\lvert\Delta R_H\rvert_{254}$ (254^2 FFT)	Δf	Geom. Contrast (gaps / counts)	`MSE` ($\lambda = 0$) d_{white} (q%) / Slope	`SBR-CF` ($\lambda = 1$) d_{white} (q%) / Slope	`SBR-CF` ($\lambda = 3$) d_{white} (q%) / Slope
Seed 0 (`test2`)	(2401, 3205)	5.88	—	0.097	12 vs 9 disks (3.0 px)	0.403 (0.75%) / 14.6	**0.884** (**21.1%**) / **6.6**	**0.870** (**16.7%**) / **6.8**
Seed 1 (`test2`)	(3214, 3270)	6.06	—	0.018	12 vs 12 ell. (2.24 vs 3.0 px)	0.404 (0.95%) / 15.0	**0.724** (**11.6%**) / **8.4**	**1.101** (**31.5%**) / **5.5**
Seed 2 (`test2`)	(2981, 3270)	8.26	—	0.111	16 disks vs 12 ell. (3.0 px)	0.435 (0.94%) / 19.0	**1.143** (**37.2%**) / **7.2**	**1.266** (**45.6%**) / **6.5**
Cons. #1	(2751, 3270)	4.512	4.577 (1.4%)	0.011	16 disks vs 12 ell. (3.0 px)	0.333 (0.43%) / 13.6	**0.503** (**3.8%**) / **9.0**	**0.651** (**8.0%**) / **6.9**
Cons. #2	(2723, 3270)	4.487	4.514 (0.6%)	0.036	16 disks vs 12 ell. (3.0 px)	0.341 (0.49%) / 13.2	**0.542** (**5.1%**) / **8.3**	**0.678** (**9.4%**) / **6.6**
Cons. #3	(2340, 3207)	4.092	3.840 (6.6%)	0.008	9 disks vs 12 ell. (2.5 px)	0.352 (0.58%) / 11.6	**0.489** (**3.3%**) / **8.4**	**0.612** (**6.4%**) / **6.7**
Aggregate across all 227 severe `MSE` collisions on `test2` ($q_{\text{MSE}} \leq 1\%$, top 10% $\lvert\Delta R_H\rvert$):								
Median whitened distance d_{white} (median quantile q%)						0.380 (0.60%)	**0.412** (**2.37%**)	**0.496** (**3.53%**)
Median local Lipschitz slope $\lvert\Delta R_H\rvert/d_{\text{white}}$						6.43	**5.91**	**4.85**
Fraction pushed outside $q = 1\%$ ball (vs. `SBR-in` 50.7%, `RankSim` 71.4%)						8.8%†	**85.5%**	**94.7%**
Aggregate across all 9 severe `MSE` collisions on `ood_morphology` ($q_{\text{MSE}} \leq 1\%$, top 10% $\lvert\Delta R_H\rvert$):								
Fraction of OOD severe collisions pushed outside $q = 1\%$ ball						11.1%†	**77.8%**	**88.9%**

†A pair inside $q = 1\%$ in one MSE seed may lie just outside 1% under `MSE`'s cross-seed median embedding. Across all top-10 consensus pairs (Appendix Appendix A.3), the median relative change between $|\Delta R_H(20)|_{127}$ and $|\Delta R_H(20)|_{254}$ upon 2×2 sub-voxel splitting is 2.55% (maximum 8.13%).

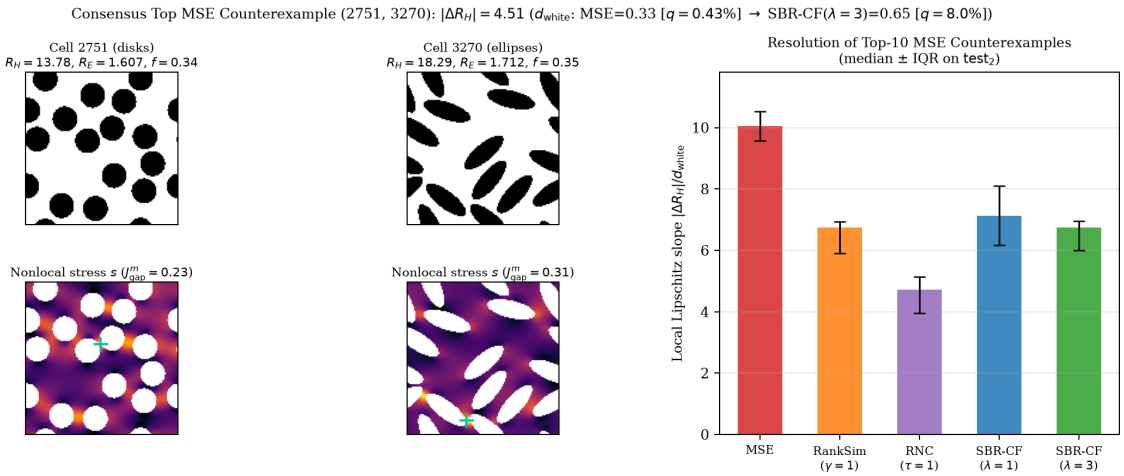


Figure 5: Physical counterexample resolution on `test2` (Experiment F, Hypothesis H5). Left/Center: Consensus rank-#1 `MSE` collision pair (2751, 3270), pairing a 16-disk cell ($f = 0.337, R_H = 13.78$ at 127^2; 13.92 at 254^2) and a 12-ellipse cell ($f = 0.348, R_H = 18.29$ at 127^2; 18.50 at 254^2, $\Delta J^m_{\text{gap}} = 0.081$, $|\Delta R_H| = 4.51$ at 127^2 vs. 4.58 at 254^2). Under `MSE` they collide at $d_{\text{white}} = 0.33$ ($q = 0.43\%$, slope 13.6), whereas `SBR-CF` ($\lambda_{\text{cf}} = 3$) separates them to $d_{\text{white}} = 0.65$ ($q = 8.0\%$, slope 6.9). Right: Top-10 `MSE` collision slopes (median ± IQR).

Table 15 and Appendix Appendix A.3 confirm two points: (i) Across the top-10 consensus counterexample pairs, doubling the FFT resolution from 127×127 to 254×254 via 2×2 sub-voxel splitting

changes $|\Delta R_H(20)|$ by a median of **2.55**% (maximum 8.13%; for the rank-#1 pair $(2751, 3270)$, $|\Delta R_H(20)| = 4.512$ at 127^2 vs. 4.577 at 254^2, a 1.42% change), confirming that the computed hazard differences are numerically stable with respect to the FFT grid and nonlocal kernel resolution on the fixed voxelized geometry; and (ii) across all 227 severe `MSE` collisions on `test2`, `SBR-CF` pushes **85.5**% ($\lambda_{\text{cf}} = 1$) and **94.7**% ($\lambda_{\text{cf}} = 3$) outside the 1% nearest-neighbor ball (and 77.8%–88.9% on OOD).

5.2. Architecture, pooling, and exploratory component ablations

To examine the roles of cross-batch anchor memory, leave-one-out hard-query mining, matched GeM_p pooling, spectral operator backbones (`E-FNO`), and built-in periodic/dihedral equivariance (`E-circ` vs. zero-padded `E-std`), we evaluate the 12-family, 3-seed ablation suite on $R_H(\beta = 20)$ at $d = 8$. In this suite, the primary objective rows (`E-circ + GeM + MSE`, `SBR-in`, and full `SBR-CF` $\lambda_{\text{cf}} = 1, 3$) use the 60-epoch primary schedule (`mine_every` $= 1$), whereas the auxiliary ablation rows (`no mem`, `no mine`, `GAP + SBR-CF`, `E-FNO`, and `E-std`) were trained under a shorter 45-epoch schedule with `mine_every` $= 5$. Consequently, the internal 45-epoch pairs—`E-FNO` with `MSE` vs. `SBR-CF`, and `E-std` with vs. without $\mathcal{L}_{\text{inv}}$—are strictly schedule-matched, whereas comparisons of the 45-epoch `no mem` and `no mine` rows against 60-epoch full `SBR-CF` are presented as exploratory comparisons because they vary both the ablated component and the epoch/mining schedule. In addition to sufficiency metrics, we record the empirical periodic-translation and reflection invariance defect

$$\delta_{\text{inv}}(E_\theta) = \left(\frac{1}{nd} \sum_{i=1}^{n} \left\| \frac{E_\theta(g \cdot x_i) - E_\theta(x_i)}{\sigma_z} \right\|_2^2 \right)^{1/2} \tag{39}$$

under shift $(17, -23)$ composed with horizontal reflection across $n = 200$ `test2` cells. Table 16 and Figure 6 summarize the results.

Table 16: Architecture, pooling, and exploratory objective-component ablations on `test2` (n = 1,000) and `ood_morphology` (n = 300, d = 8, 3-seed medians under d_{white}): objective decomposition (no cross-batch memory $M = 0$, no hard-query mining $K_{\text{mine}} = 0$), pooling (`GAP` vs. GeM_p), and backbone / equivariance (`E-FNO` and zero-padded `E-std` with and without $\lambda_{\text{inv}} = 1.0$). Primary campaign rows use 60 epochs (`mine_every` $= 1$); auxiliary ablation rows marked ‡ use 45 epochs (`mine_every` $= 5$).

Group	Configuration ($d = 8$)	δ_{inv}	p_{med}	IID Test (`test2`, n = 1,000)			OOD (`ood`, n = 300)	
				$\Omega/\Omega^{\text{null}}$	R^2_{Mc}	$\text{CF}_{\max}/\sigma_R$	$\Omega/\Omega^{\text{null}}$	R^2_{Mc}
Obj.	`E-circ + GeM + MSE`	$< 10^{-4}$	8.64	0.273	0.947	1.677	0.229	0.957
	`E-circ + GeM + SBR-in` ($\lambda = 1$)	$< 10^{-4}$	6.49	0.303	0.926	1.775	0.228	0.943
	`E-circ + GeM + SBR-CF (no mem)`‡	$< 10^{-4}$	8.38	0.227	0.953	1.361	0.202	0.956
	`E-circ + GeM + SBR-CF (no mine)`‡	$< 10^{-4}$	8.04	0.189	0.962	1.377	0.232	0.948
	`E-circ + GeM + SBR-CF (full,` $\lambda = 1$)	$< 10^{-4}$	8.84	0.192	**0.962**	1.402	0.199	**0.963**
	`E-circ + GeM + SBR-CF (full,` $\lambda = 3$)	$< 10^{-4}$	9.61	**0.188**	0.961	**1.165**	**0.169**	**0.963**
Pool	`E-circ + GAP + MSE` ($p = 1$)	$< 10^{-4}$	1.00	0.396	0.933	1.575	0.258	0.922
	`E-circ + GAP + SBR-CF` ($\lambda = 1$)‡	$< 10^{-4}$	1.00	0.255	0.960	**1.085**	0.210	0.941
Arch.	`E-FNO + GeM + MSE`‡	$< 10^{-4}$	4.01	0.338	0.922	1.797	0.306	0.882
	`E-FNO + GeM + SBR-CF` ($\lambda = 1$)‡	$< 10^{-4}$	5.16	0.250	0.959	1.576	0.233	0.959
	`E-std (no sym) + GeM + MSE`‡	**1.073**	6.51	0.459	0.805	2.187	0.405	0.878
	`E-std (no sym) + GeM + MSE +` $\mathcal{L}_{\text{inv}}$‡	0.174	3.95	0.343	0.901	2.619	0.241	0.930

Table 16 supports three observations: (i) In the schedule-matched 45-epoch comparisons, `SBR-CF` transfers cleanly to the spectral `E-FNO` backbone, cutting $\Omega/\Omega^{\text{null}}$ from 0.338 to 0.250 on `test2` and from 0.306 to 0.233 on OOD while raising R^2_{McShane} from 0.922 to 0.959; (ii) the 45-epoch zero-padded configuration `E-std (no sym) + GeM + MSE` exhibits a large invariance defect $\delta_{\text{inv}} = 1.073$ and $\Omega/\Omega^{\text{null}} = 0.459$ ($R^2_{\text{McShane}} = 0.805$); within the strictly schedule-matched 45-epoch pair, adding the soft invariance penalty $\mathcal{L}_{\text{inv}}$ reduces δ_{inv} from 1.073 to 0.174, lowers $\Omega/\Omega^{\text{null}}$ from 0.459 to 0.343, and raises R^2_{McShane} from 0.805 to 0.901 (compared with $\Omega/\Omega^{\text{null}} = 0.338$, $R^2_{\text{McShane}} = 0.922$ for 45-epoch

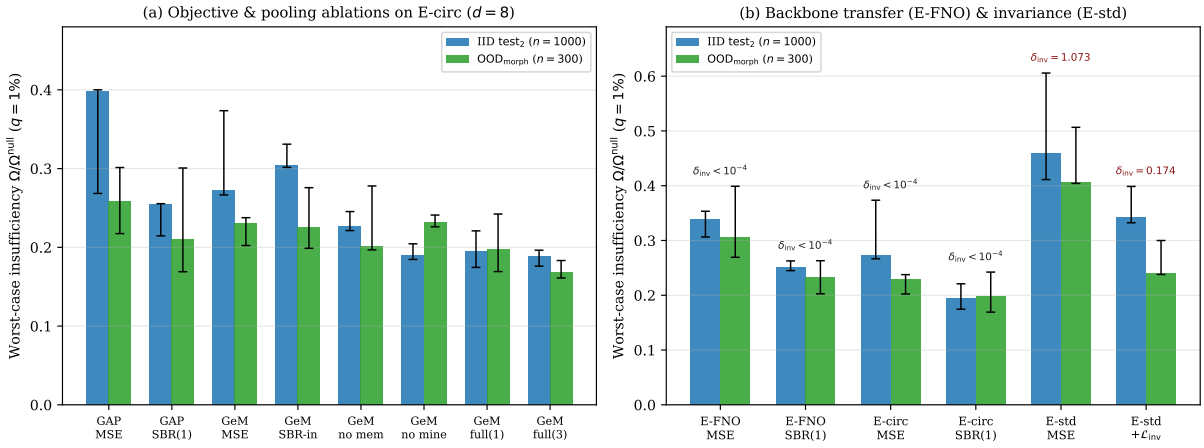


Figure 6: Component ablations ($d = 8$, 3-seed medians and min/max ranges on `test2` and `ood_morphology`). Left: Exploratory objective and pooling decomposition. Right: Schedule-matched spectral `E-FNO` backbone transfer vs. symmetry breakdown in zero-padded `E-std`.

equivariant `E-FNO + GeM + MSE` and $\Omega/\Omega^{\text{null}} = 0.273$, $R^2_{\text{McShane}} = 0.947$ for 60-epoch `E-circ + GeM + MSE`); and (iii) in the exploratory 45-epoch objective-component runs, both `SBR-CF (no mem)` (0.227) and `SBR-CF (no mine)` (0.189) outperform 60-epoch `MSE` (0.273) and `SBR-in` (0.303) on IID `test2`, whereas on `ood_morphology` `no mem` lowers $\Omega/\Omega^{\text{null}}$ to 0.202 while `no mine` gives 0.232 (vs. 0.229 for `MSE`, 0.228 for `SBR-in`, and 0.199 for full `SBR-CF` $\lambda_{\text{cf}} = 1$), suggesting that cross-batch memory primarily aids IID coverage while hard-query mining aids OOD coverage (subject to the 45- vs. 60-epoch schedule difference).

6. Cross-Horizon, OOD-Physics, and Phase-Field Fracture Transfer (Experiments G–H)

6.1. Experiment G (Part 1): Cross-horizon transfer matrix across R_E and $R_H(\beta)$

Table 17 and Figure 7(a) evaluate frozen-encoder transfer across R_E, $R_H(2)$, $R_H(20)$, and $R_H(40)$ on `test2` ($n = 1{,}000$) under d_{white}, refitting only the whitening transform and the McShane readout on each target. Encoders trained on bulk energy (R_E) or mild heterogeneity ($R_H(2)$) fail to resolve sharp stress-tail hazards ($\Omega/\Omega^{\text{null}} = 0.480$–$0.534$ on $R_H(20), R_H(40)$). Conversely, under d_{white} and McShane interpolation, `SBR-CF` trained solely on $R_H(20)$ transfers both forward to $R_H(40)$ ($\Omega/\Omega^{\text{null}} = 0.196$–$0.213$, outperforming `MSE` trained directly on $R_H(40)$ at 0.280) and backward to R_E (0.110, $R^2_{\text{McShane}} = 0.947$) and $R_H(2)$ (0.209, $R^2_{\text{McShane}} = 0.952$), whereas `RNC`'s 1D curve specialized to $R_H(20)$ degrades under whitened McShane transfer to $R_H(2)$ ($\Omega/\Omega^{\text{null}} = 0.469$, $R^2_{\text{McShane}} = 0.840$).

6.2. Experiment G (Part 2): 90° rotation check and OOD-physics transfer ($\kappa \in \{5, 20\}$ and biaxial strain)

Because the encoder takes only the binary microstructure image x as input, changing the stiffness contrast $\kappa \in \{5, 20\}$ or applying biaxial macroscopic strain $\bar{\varepsilon} = (1.0, 0.5, 0.0)$ changes the response $R(x)$ without changing x. We therefore pose OOD-physics transfer by freezing the $\kappa = 10$, uniaxial e_1-trained encoder, evaluating $\Omega/\Omega^{\text{null}}$ (d_{white}) on the $n = 300$ `test1` cells, and fitting linear ridge and McShane readouts on $n_{\text{few}} \in \{50, 200\}$ adaptation cells from `val`. Simultaneously, we verify 90° load rotation ($\bar{\varepsilon} = e_2 \otimes e_2$ at $\kappa = 10$): continuum/FFT equivalence $R(x; e_2) = R(\text{rot}_{90}(x); e_1)$ holds to 7.6×10^{-14}, whereas feeding unrotated embeddings $E_\theta(x)$ to predict $R_H(x; e_2)$ collapses R^2_{ridge} to ≤ 0.059, confirming that 90° rotation is not a symmetry of an anisotropic cell under fixed load.

Table 17: Experiment G (Part 1): Cross-horizon transfer matrix on `test2` (n = 1,000, d = 8, 3-seed medians of $\Omega/\Omega^{\text{null}}$ and R^2_{McShane} under d_{white}).

Source Model	Source	$\Omega/\Omega^{\text{null}}$ (↓) on R_{tgt}				R^2_{McShane} (↑) on R_{tgt}			
	R_{src}	R_E	$R_H(2)$	$R_H(20)$	$R_H(40)$	R_E	$R_H(2)$	$R_H(20)$	$R_H(40)$
`MSE (GAP)`	R_E	**0.084**	0.208	0.480	0.534	**0.970**	0.973	0.809	0.763
`MSE (GeM)`	$R_H(2)$	0.129	**0.188**	0.510	0.531	0.941	**0.976**	0.768	0.714
`MSE (GeM)`	$R_H(40)$	0.168	0.269	0.267	0.280	0.891	0.919	0.938	0.934
`MSE (GeM)`	$R_H(20)$	0.162	0.258	0.273	0.302	0.904	0.916	0.947	0.930
`SBR-in` ($\lambda = 1$)	$R_H(20)$	0.170	0.265	0.303	0.338	0.913	0.918	0.926	0.915
`RankSim` ($\gamma = 1$)	$R_H(20)$	0.165	0.307	0.261	0.286	0.870	0.902	0.924	0.900
`RNC` ($\tau = 1.0$)	$R_H(20)$	0.171	0.469	0.225	0.306	0.873	0.840	0.915	0.908
`SBR-CF` ($\lambda = 1$)	$R_H(20)$	0.120	0.249	0.192	**0.196**	0.909	0.937	**0.962**	0.954
`SBR-CF` ($\lambda = 3$)	$R_H(20)$	**0.110**	**0.209**	**0.188**	0.213	**0.947**	**0.952**	0.961	**0.956**

Table 18: Experiment G (Part 2): Zero-shot 90° load-rotation check ($\bar{\varepsilon} = e_2 \otimes e_2$) and few-shot OOD-physics transfer across contrast $\kappa \in \{5, 20\}$ and biaxial strain $\bar{\varepsilon} = (1.0, 0.5, 0.0)$ on $R_H(\beta = 20)$ (n = 300 test cells, frozen $\kappa = 10$ uniaxial encoders, d = 8, 3-seed medians).

Regime	Metric / Adaptation	`MSE` ($\lambda = 0$)	`SBR-in` ($\lambda = 1$)	`RankSim` ($\gamma = 1$)	`RNC` ($\tau = 1$)	`SBR-CF` ($\lambda = 1$)	`SBR-CF` ($\lambda = 3$)
IID Ref. ($\kappa = 10, e_1$)	$\Omega/\Omega^{\text{null}}$ ($n = 300$, d_{white})	0.176	0.195	0.181	0.152	**0.121**	0.133
	Zero-shot R^2_{McShane} / R^2_{ridge}	0.942 / 0.956	0.922 / 0.942	0.937 / 0.954	0.931 / **0.978**	**0.965** / 0.947	0.958 / 0.959
Rotation ($\kappa = 10, e_2$)	$\Omega/\Omega^{\text{null}}$ on $\text{rot}_{90}(x)$	0.312	0.299	0.298	0.287	0.230	**0.181**
	Zero-shot R^2_{McShane} / R^2_{ridge}	0.944 / 0.970	0.922 / 0.963	0.940 / 0.976	0.927 / **0.991**	0.953 / 0.975	**0.957** / 0.974
	Unrotated control $\Omega/\Omega^{\text{null}}$ / R^2_{ridge}	0.605 / 0.042	0.720 / 0.059	0.509 / 0.024	0.823 / −0.010	0.459 / 0.015	0.522 / 0.017
OOD-Phys: $\kappa = 5$	$\Omega/\Omega^{\text{null}}$ ($n = 300$, d_{white})	0.197	0.189	0.186	0.185	**0.137**	0.143
	Few-shot $n_{\text{few}} = 50$: R^2_{Mc} / R^2_{ridge}	0.610 / 0.951	0.586 / 0.938	0.668 / 0.948	0.498 / **0.975**	0.620 / 0.921	**0.739** / 0.937
	Few-shot $n_{\text{few}} = 200$: R^2_{Mc} / R^2_{ridge}	0.871 / 0.963	0.818 / 0.951	0.859 / 0.953	0.831 / **0.976**	**0.887** / 0.950	0.885 / 0.960
OOD-Phys: $\kappa = 20$	$\Omega/\Omega^{\text{null}}$ ($n = 300$, d_{white})	0.153	0.186	0.170	0.157	**0.113**	0.149
	Few-shot $n_{\text{few}} = 50$: R^2_{Mc} / R^2_{ridge}	0.558 / 0.929	0.518 / 0.913	0.548 / 0.925	0.455 / **0.966**	0.574 / 0.914	**0.661** / 0.934
	Few-shot $n_{\text{few}} = 200$: R^2_{Mc} / R^2_{ridge}	0.824 / 0.941	0.772 / 0.924	0.817 / 0.940	0.804 / **0.970**	**0.865** / 0.927	0.850 / 0.946
OOD-Phys: Biaxial $\bar{\varepsilon} = (1, 0.5, 0)$	$\Omega/\Omega^{\text{null}}$ ($n = 300$, d_{white})	0.166	0.159	0.190	0.254	**0.118**	0.168
	Few-shot $n_{\text{few}} = 50$: R^2_{Mc} / R^2_{ridge}	0.557 / 0.915	0.531 / 0.897	0.582 / 0.908	0.507 / **0.948**	0.590 / 0.890	**0.695** / 0.904
	Few-shot $n_{\text{few}} = 200$: R^2_{Mc} / R^2_{ridge}	0.825 / 0.926	0.793 / 0.908	0.808 / 0.912	0.778 / **0.943**	**0.861** / 0.904	0.847 / 0.926

Table 18 highlights complementary strengths of `SBR-CF` and `RNC`: (i) `SBR-CF` achieves the lowest whitened worst-case insufficiency across all four shifted regimes (0.181 on 90° rotation at $\lambda_{\text{cf}} = 3$; $0.137, 0.113, 0.118$ on $\kappa = 5, \kappa = 20$, and biaxial loading at $\lambda_{\text{cf}} = 1$) and the highest few-shot McShane interpolation R^2_{McShane} at both $n_{\text{few}} = 50$ $(0.739, 0.661, 0.695)$ and $n_{\text{few}} = 200$ $(0.887, 0.865, 0.861)$; (ii) simultaneously, `RNC` ($\tau = 1.0$) achieves the highest linear ridge R^2_{ridge} across the contrast and biaxial shifts (0.976 on $\kappa = 5$, 0.970 on $\kappa = 20$, and 0.943 on biaxial strain at $n_{\text{few}} = 200$, vs. 0.950–0.960, 0.927–0.946, and 0.904–0.926 for `SBR-CF`), because $R_H(20)$ under modified contrast or biaxial load remains strongly correlated with the uniaxial scalar ordering captured by `RNC`'s leading axis, even though whitening its 7 orthogonal directions inflates its biaxial $\Omega/\Omega^{\text{null}}$ to 0.254.

6.3. Experiment H (Part 1): Seed-combined local-swap conditional permutation tests (`T1`–`T6`) and statistical power

To test whether pretrained embeddings carry incremental worst-case signal beyond explicit physics descriptors ($D_0 = E_1$ or $D_0 = E_1 + J_{1.5\ell}$) on the $n = 150$ multi-inclusion phase-field design cells, we apply the local-swap conditional permutation test [31, 32]. Because taking the median of three seed-level p-values is a descriptive summary rather than a single combined p-value, we construct a seed-combined local-swap conditional permutation test: for each local swap $b \in \{1, \ldots, B\}$ (B = 2,000, $k_{\text{nn}} = 5$ in standardized D_0), all three seeds $s \in \{0, 1, 2\}$ are permuted synchronously by the same swap permutation π_b, and the seed-aggregated statistic

$$\bar{\Omega}(q) = \frac{1}{3} \sum_{s=0}^{2} \Omega_s(q) \qquad (q = 1\%) \tag{40}$$

is compared against its permutation distribution $\{\bar{\Omega}^{(b)}(q)\}_{b=1}^{B}$, with Holm step-down correction [33] across all 6 tests `T1`–`T6`. While synchronous permutation across seeds accounts for cross-seed correlation without ad-hoc p-value medians, the conditional null calibration remains *approximate* because swapping $k_{\mathrm{nn}} = 5$ nearest neighbors in a continuous 6- or 7-dimensional descriptor space D_0 only approximates conditional exchangeability [32]. Table 19 reports the seed-combined ratio $\bar{\Omega}/\bar{\Omega}^{\mathrm{cnull}}$, the seed-combined p_{raw} (p_{Holm}), and the number of individual seeds achieving $p_{\mathrm{Holm}} < 0.05$ (individual seed p-values and ensemble-distance tests are reported in Appendix Appendix A.2).

To calibrate empirical Type-I error and finite-sample detection power on `T5` ($R_\sigma \mid E_1 + J_{1.5\ell}$), we generate synthetic responses from the linear-Gaussian conditional nuisance law

$$R_{\mathrm{syn}}^{(\eta,m)} = \widehat{R}_{\mathrm{OLS}}(D_0) + \eta\, \sigma_{R_\sigma}\, u_{\mathrm{orth}}(Z) + \varepsilon^{(m)}, \qquad \varepsilon^{(m)} \overset{\mathrm{iid}}{\sim} \mathcal{N}(0, \hat{\sigma}_{\mathrm{res}}^2 I_{150}), \tag{41}$$

where $\widehat{R}_{\mathrm{OLS}}(D_0)$ is the linear regression fit of R_σ on $D_0 = E_1 + J_{1.5\ell}$ ($\hat{\sigma}_{\mathrm{res}}$ is the empirical residual standard deviation) and $u_{\mathrm{orth}}(Z) \in \mathbb{R}^{150}$ is constructed in `scripts/run_phase5a_ml_and_stats_controls.py` by concatenating the column-standardized 8D embeddings of the three seeds of `SBR-CF` ($\lambda_{\mathrm{cf}} = 1.0$) into a 150×24 matrix Z_{concat}, projecting Z_{concat} orthogonally onto the complement of $\Phi = [1, \mathrm{zscore}(D_0)]$, extracting the first left singular vector of $(I - \Phi\Phi^{\dagger})Z_{\mathrm{concat}}$, and standardizing it to zero mean and unit variance. At $\eta = 0$, $R_{\mathrm{syn}}^{(0,m)} \perp\!\!\!\perp Z \mid D_0$ holds by construction; across $\eta \in \{0, 0.10, 0.15, 0.20, 0.25, 0.35, 0.50\}$, we evaluate 150 Monte Carlo replicates per η (Table A.24 in Appendix Appendix A.2). Because this Monte Carlo study simulates the single test `T5` in isolation rather than all 6 tests `T1`–`T6` jointly, Table A.24 reports unadjusted rejection rates at $\alpha \in \{0.05, 0.10\}$ together with the single-test Bonferroni threshold $\alpha/6 = 0.05/6 \approx 0.0083$ (the strictest first-step threshold of a 6-test Holm procedure).

Table 19: Experiment H (Part 1): Seed-combined local-swap conditional permutation tests (`T1`–`T6`, Eq. (40), $B = 2{,}000$, $q = 1\%$, $k_{\mathrm{nn}} = 5$ in standardized D_0) across the $n = 150$ multi-inclusion phase-field design cells. Columns report $\bar{\Omega}/\bar{\Omega}^{\mathrm{cnull}}$, seed-combined Holm-adjusted p_{Holm} (across the 6-test family `T1`–`T6`), and count of individual seeds with $p_{\mathrm{Holm}} < 0.05$ ($S_{\mathrm{sig}}/3$). Bold denotes $p_{\mathrm{Holm}} < 0.05$.

Test	Target	Cond. D_0	`MSE` ($R_H(20)$)		`Tail-MSE` ($\lambda = 3$)		`RankSim` ($\gamma = 1$)		`RNC` ($\tau = 1$)		`SBR-CF` ($\lambda = 1$ / $\lambda = 3$)	
			Ratio	p_{Holm} (S)	Ratio	p_{Holm} (S)	Ratio	p_{Holm} (S)	Ratio	p_{Holm} (S)	Ratio ($\lambda = 1, 3$)	p_{Holm} ($\lambda = 1, 3$)
`T1`	R_E	E_1 (neg. ctrl)	1.037	0.509 (0/3)	0.848	0.314 (0/3)	0.778	0.217 (0/3)	1.062	1.000 (0/3)	0.895 / 0.644	0.523 / **0.014** (0/3)
`T2`	$R_H(20)$	E_1	0.754	0.110 (1/3)	**0.563**	**0.003** (2/3)	**0.537**	**0.012** (1/3)	**0.184**	**0.005** (3/3)	**0.501** / **0.387**	**0.003** / **0.005** (2,3/3)
`T3`	$R_H(20)$	$E_1 + J_{1.5\ell}$	**0.640**	**0.009** (2/3)	**0.609**	**0.003** (2/3)	**0.637**	**0.020** (1/3)	**0.229**	**0.003** (3/3)	**0.550** / **0.448**	**0.003** / **0.003** (2,3/3)
`T4`	R_σ	E_1	0.707	0.110 (0/3)	**0.657**	**0.022** (0/3)	**0.612**	**0.012** (1/3)	0.627	0.376 (0/3)	0.726 / 0.668	0.262 / 0.139 (0/3)
`T5`	R_σ	$E_1 + J_{1.5\ell}$	0.893	0.411 (0/3)	**0.762**	**0.043** (1/3)	0.873	0.228 (0/3)	0.957	1.000 (0/3)	0.876 / 0.945	0.271 / 0.307 (1,0/3)
`T6`	R_{path}	$E_1 + J_{1.5\ell}$	0.837	0.411 (1/3)	0.931	0.314 (0/3)	0.809	0.228 (0/3)	1.119	1.000 (0/3)	0.899 / 0.669	0.539 / 0.164 (0/3)

Table 19 and the power calibration in Appendix Appendix A.2 (Tables A.23–A.24) establish three precise statistical conclusions:

1. **Low finite-sample detection power at $n = 150$ and non-exclusion of large effects:** Under the synthetic conditional null ($\eta = 0$, where η in Eq. (41) is the dimensionless signal amplitude in σ_{R_σ} units), the local-swap test on `T5` has empirical rejection rate 0.020 at nominal $\alpha = 0.05$ (0.060 at $\alpha = 0.10$, and 0.000 at the Bonferroni threshold $\alpha = 0.05/6$). However, because the $q = 1\%$ tail on $n = 150$ cells contains only $\binom{150}{2} \times 0.01 \approx 112$ pairs, statistical power is low even for substantial conditional signals: at $\eta = 0.35$ (signal amplitude $0.35\sigma_{R_\sigma}$, explaining 11.0% of total variance and 71.3% of residual variance), unadjusted $\alpha = 0.05$ power is 7.3% (0.7% at $\alpha = 0.05/6$), and at $\eta = 0.50$ (signal amplitude $0.50\sigma_{R_\sigma}$, explaining 20.1% of total variance and 83.5% of residual variance), unadjusted $\alpha = 0.05$ power is only **28.0**% (51.3% at $\alpha = 0.10$, and **2.7**% at the Bonferroni threshold $\alpha = 0.05/6$). Thus, at $\eta = 0.50$, 72% of replicates fail to detect the signal at unadjusted $\alpha = 0.05$ (97.3% at $\alpha = 0.05/6$). Consequently, *non-significant results on `T5` or `T6` cannot be used to rule out even large local conditional effects*, and the `T5` calibration does not calibrate `T6` ($R_{\mathrm{path}} \mid E_1 + J_{1.5\ell}$, where D_0 explains almost no variance).
2. **Approximate conditional exchangeability in 6D/7D D_0 and negative-control behavior (`T1`):** Local-swap permutations over $k_{\mathrm{nn}} = 5$ nearest neighbors in a 6- or 7-dimensional continuous

descriptor space D_0 provide only *approximate* conditional exchangeability [32]: when the true response varies across $k_{\text{nn}} = 5$ neighbors in D_0 and the learned embedding Z correlates with within-neighborhood variations of D_0, swapping neighbors can introduce a finite-sample local conditioning bias. Such finite-sample conditioning bias is a plausible explanation for the isolated seed-combined rejection on the negative control `T1` for `SBR-CF` ($\lambda_{\text{cf}} = 3$, $\bar{\Omega}/\bar{\Omega}^{\text{cnull}} = 0.644$, $p_{\text{Holm}} = 0.014$, although 0/3 individual seeds reject at $p_{\text{Holm}} < 0.05$ and the ensemble-distance test gives $p^{\text{ens}}_{\text{Holm}} = 0.183$; Table A.23), though a single test realization cannot isolate its cause. This approximate-exchangeability caveat applies symmetrically to both positive rejections and non-rejections across `T1`–`T6`.

3. **Positive conditional detections (`T2`–`T5`):** Subject to the approximate exchangeability of $k_{\text{nn}} = 5$ local swaps, on $R_H(20)$, `SBR-CF` ($\lambda_{\text{cf}} = 1, 3$), `Tail-MSE` ($\lambda_{\text{tail}} = 3$), `RankSim`, and `RNC` all reject the conditional null beyond both E_1 (`T2`: $p_{\text{Holm}} = 0.003$–0.012) and $E_1 + J_{1.5\ell}$ (`T3`: $p_{\text{Holm}} = 0.003$–0.020). On peak fracture strength R_σ, `Tail-MSE` ($\lambda_{\text{tail}} = 3$) and `RankSim` ($\gamma = 1$) reject `T4` ($R_\sigma \mid E_1$, $p_{\text{Holm}} = 0.022$ and 0.012), and `Tail-MSE` ($\lambda_{\text{tail}} = 3$) also reaches nominal significance on `T5` ($R_\sigma \mid E_1 + J_{1.5\ell}$, $\bar{\Omega}/\bar{\Omega}^{\text{cnull}} = 0.762$, $p_{\text{Holm}} = 0.043$, 1/3 individual seeds; Table A.23), whereas `MSE`, `SBR-CF` ($\lambda_{\text{cf}} = 1, 3$), `RankSim`, and `RNC` do not reject `T5` ($p_{\text{Holm}} = 0.228$–1.000) or `T6` ($p_{\text{Holm}} = 0.164$–1.000).

6.4. Experiment H (Part 2): Frozen AT1 phase-field fracture transfer, extreme stress errors, and strip-shift barrier b_S

We fit linear ridge readouts by 5-fold cell-grouped cross-validation on the $n = 180$ non-laminate design cells and evaluate frozen out-of-sample predictions across `Set B` ($n = 45$), `Set O` ($\alpha = 6$, $n = 12$), `Set C1` ($n = 30$), and `Set C2` ($n_{\text{inc}} \in \{16, 25\}$, $n = 12$). For clarity, we separate the evaluation into two tables: Table 20 reports out-of-sample R^2 across all five fracture splits for R_σ and R_{path}, while Table 21 reports detailed tail and physical stress error metrics—RMSE, 90th-percentile error $P_{90}|\Delta R_\sigma|$, maximum absolute error $\max|\Delta R_\sigma|$, and mean and maximum relative error in physical peak stress

$$\delta_{\sigma,\text{rel}} = \left| \frac{\widehat{\sigma}_{\text{peak}} - \sigma_{\text{peak}}}{\sigma_{\text{peak}}} \right| = \left| \exp(\widehat{R}_\sigma - R_\sigma) - 1 \right| \times 100\% \tag{42}$$

—on Design CV ($n = 180$), blind Set B ($n = 45$), and prospective Set C2 ($n = 12$).

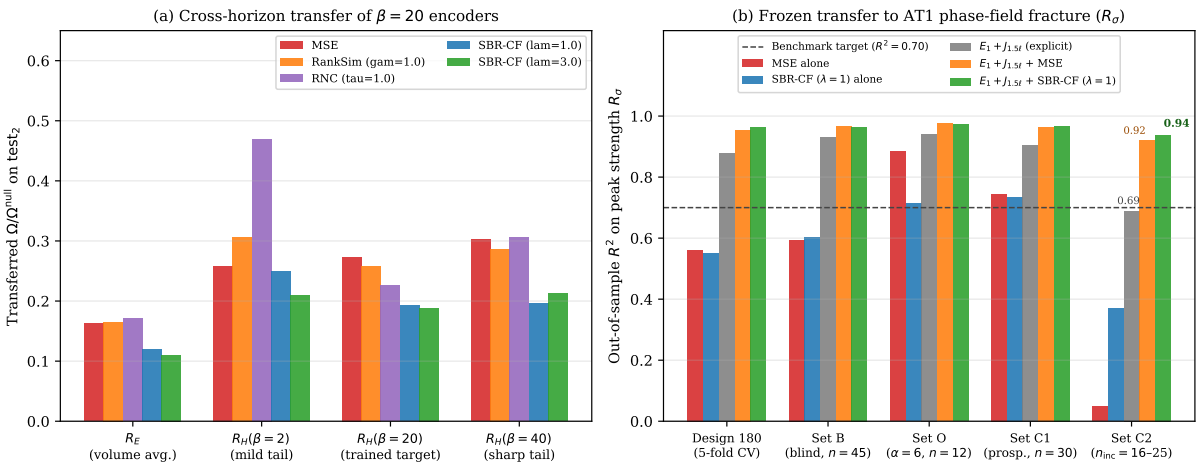


Figure 7: Left: Cross-horizon transfer matrix ($\Omega/\Omega^{\text{null}}$ on `test2`). Right: Out-of-sample R^2 on peak fracture strength R_σ across Design CV ($n = 180$) and frozen Sets B, O, C1, C2, comparing $E_1 + J_{1.5\ell}$ alone (0.68–0.69 on Set C2) against $E_1 + J_{1.5\ell} +$ `MSE` (0.92) and $E_1 + J_{1.5\ell} +$ `SBR-CF` ($\lambda_{\text{cf}} = 1$, 0.94).

Table 20: Experiment H (Part 2a): Out-of-sample R^2 (↑, 3-seed medians) on frozen AT1 phase-field fracture splits across Design CV ($n = 180$), blind Set B ($n = 45$ for R_σ, $n_{\text{eff}} = 44$ for R_{path}), high-aspect-ratio Set O ($\alpha = 6$, $n = 12$ for R_σ, $n_{\text{eff}} = 11$ for R_{path}), prospective Set C1 ($n = 30$), and high-inclusion-count prospective Set C2 ($n = 12$). Bold entries mark the best model within each group (physics-only, standalone image-only, and hybrid solved-field + embedding).

Target	Representation	FFT Solve at Test?	Design CV_{180}	Blind Set B	OOD Set O	Prosp. Set C1	Prosp. Set C2
R_σ	`E1` (mean strains + gaps)	Yes (FFT)	0.319	0.298	−1.365	0.536	−3.424
	`E1 + RH_b20` (scalar hazard)	Yes (FFT)	0.511	0.454	0.538	0.707	−1.407
	`E1 + J_1.5` (critical distance)†	Yes (FFT)	**0.881**	**0.930**	**0.945**	**0.906**	**0.679**
	`MSE` ($R_H(20)$) alone	**No (Img)**	0.562	0.593	**0.884**	0.744	0.048
	`Tail-MSE` ($\lambda = 1$) alone	**No (Img)**	**0.661**	0.606	0.784	0.766	**0.398**
	`Tail-MSE` ($\lambda = 3$) alone	**No (Img)**	0.637	0.615	0.828	0.757	0.360
	`RankSim` ($\gamma = 1$) alone	**No (Img)**	0.604	0.563	0.864	0.789	0.019
	`RNC` ($\tau = 1$) alone	**No (Img)**	0.557	0.543	0.685	0.751	−1.342
	`SBR-CF` ($\lambda_{\text{cf}} = 1$) alone	**No (Img)**	0.551	0.604	0.714	0.735	0.369
	`SBR-CF` ($\lambda_{\text{cf}} = 3$) alone	**No (Img)**	0.592	**0.628**	0.790	**0.812**	0.111
	`E1 + J_1.5 + MSE` ($R_H(20)$)	Yes (FFT)	0.954	**0.968**	0.976	0.964	0.922
	`E1 + J_1.5 + Tail-MSE` ($\lambda = 1$)	Yes (FFT)	0.942	0.965	0.958	0.966	0.886
	`E1 + J_1.5 + Tail-MSE` ($\lambda = 3$)	Yes (FFT)	0.960	0.964	0.973	**0.973**	0.905
	`E1 + J_1.5 + RankSim` ($\gamma = 1$)	Yes (FFT)	0.946	0.967	0.970	0.964	0.901
	`E1 + J_1.5 + RNC` ($\tau = 1$)	Yes (FFT)	0.883	0.948	0.945	0.920	0.766
	`E1 + J_1.5 + SBR-CF` ($\lambda = 1$)	Yes (FFT)	**0.963**	0.964	0.973	0.967	**0.939**
	`E1 + J_1.5 + SBR-CF` ($\lambda = 3$)	Yes (FFT)	0.955	0.967	**0.982**	0.971	0.918
$R^{\ddagger}_{\text{path}}$	`E1 + J_1.5`	Yes (FFT)	−0.078	0.046	−2.658	—	—
	`E1 + J_1.5 + SBR-CF` ($\lambda = 1$)	Yes (FFT)	0.105	**0.112**	−0.915	—	—
	`E1 + J_1.5 + SBR-CF` ($\lambda = 3$)	Yes (FFT)	**0.120**	0.062	−1.355	—	—

†Under unregularized OLS [1], `E1+J_1.5` attains $R^2 = 0.879$ (CV), 0.932 (Set B), 0.941 (Set O), 0.905 (Set C1), and 0.689 (Set C2); under the common 5-fold CV ridge grid $\alpha \in \{10^{-3}, \ldots, 20\}$ used for all rows above, `E1+J_1.5` attains $0.881, 0.930, 0.945, 0.906, 0.679$.

‡For R_{path} (Eq. (34)), all rows use the identical valid-cell mask with effective sample sizes $n_{\text{eff}} = 180$ (Design CV), $n_{\text{eff}} = 44$ (Set B, excluding non-percolating en-echelon cell `bid` = 72), and $n_{\text{eff}} = 11$ (Set O, excluding `bid` = 106); see Section 3.2 and `data/frozen/blind_pf.csv`.

Table 21: Experiment H (Part 2b): Detailed tail and physical peak-stress error metrics (↓, 3-seed medians) on Design CV ($n = 180$), blind Set B ($n = 45$), and high-inclusion-count prospective Set C2 ($n = 12$). Columns report RMSE, $P_{90}|\Delta R_\sigma|$, $\max|\Delta R_\sigma|$, and mean/maximum relative peak-stress error $\delta_{\sigma,\text{rel}}$ (%).

Representation (R_σ)	Design CV ($n = 180$)		Blind Set B ($n = 45$)		Prospective Set C2 ($n = 12$, $n_{\text{inc}} \in \{16, 25\}$)				
	Mean $\delta_{\sigma,\text{rel}}$	Max $\delta_{\sigma,\text{rel}}$	Mean $\delta_{\sigma,\text{rel}}$	Max $\delta_{\sigma,\text{rel}}$	RMSE	$P_{90}\lvert\Delta R_\sigma\rvert$	$\max\lvert\Delta R_\sigma\rvert$	Mean $\delta_{\sigma,\text{rel}}$	Max $\delta_{\sigma,\text{rel}}$
`E1` (mean strains + gaps)	7.12%	23.24%	8.05%	21.06%	0.0989	0.1337	0.1356	8.95%	12.68%
`E1 + RH_b20` (scalar hazard)	6.08%	15.47%	7.21%	16.75%	0.0729	0.0961	0.1158	6.43%	10.94%
`E1 + J_1.5` (critical distance)	**2.84%**	**7.61%**	**2.26%**	**4.53%**	**0.0266**	**0.0396**	**0.0417**	**2.35%**	**4.09%**
`MSE` ($R_H(20)$) alone	5.58%	15.09%	6.01%	14.73%	0.0459	0.0700	0.0921	3.69%	8.80%
`Tail-MSE` ($\lambda = 1$) alone	**4.93%**	**13.26%**	5.91%	15.35%	**0.0365**	0.0619	0.0753	**2.72%**	7.82%
`Tail-MSE` ($\lambda = 3$) alone	5.10%	16.68%	5.83%	15.67%	0.0376	**0.0575**	**0.0731**	3.21%	**7.04%**
`RankSim` ($\gamma = 1$) alone	5.31%	18.20%	6.23%	13.54%	0.0466	0.0737	0.0899	3.60%	8.59%
`RNC` ($\tau = 1$) alone	5.65%	19.56%	6.38%	**13.23%**	0.0719	0.0953	0.1225	5.91%	11.53%
`SBR-CF` ($\lambda_{\text{cf}} = 1$) alone	5.64%	16.86%	5.92%	17.04%	0.0373	0.0632	0.0792	2.97%	7.61%
`SBR-CF` ($\lambda_{\text{cf}} = 3$) alone	5.39%	20.95%	**5.73%**	13.77%	0.0443	0.0759	0.1004	3.44%	9.55%
`E1 + J_1.5 + MSE` ($R_H(20)$)	1.75%	5.17%	**1.53%**	3.90%	0.0131	0.0193	0.0230	1.10%	2.27%
`E1 + J_1.5 + Tail-MSE` ($\lambda = 1$)	1.98%	5.32%	1.59%	4.14%	0.0159	0.0221	0.0338	1.20%	3.32%
`E1 + J_1.5 + Tail-MSE` ($\lambda = 3$)	1.64%	**4.79%**	1.62%	4.25%	0.0145	0.0211	0.0287	1.14%	2.91%
`E1 + J_1.5 + RankSim` ($\gamma = 1$)	1.90%	4.93%	1.55%	3.35%	0.0148	0.0219	0.0245	1.26%	2.42%
`E1 + J_1.5 + RNC` ($\tau = 1$)	2.81%	7.92%	1.95%	4.52%	0.0228	0.0315	0.0435	1.88%	4.26%
`E1 + J_1.5 + SBR-CF` ($\lambda = 1$)	**1.58%**	5.44%	1.62%	3.57%	**0.0116**	**0.0191**	0.0250	**0.94%**	2.47%
`E1 + J_1.5 + SBR-CF` ($\lambda = 3$)	1.74%	5.26%	1.55%	**3.23%**	0.0135	0.0206	**0.0225**	1.13%	**2.26%**

Note: On Set C2, Seed 2 of `E1+J_1.5+SBR-CF` ($\lambda = 1$) reaches $R^2 = 0.940$, $\max|\Delta R_\sigma| = 0.0197$, and $\max \delta_{\sigma,\text{rel}} = 1.95\%$.

Cell-block bootstrap uncertainty and balanced comparison across fracture models.. Because predictions from the 3 random seeds on the same microstructure are correlated, all paired uncertainty intervals in Experiment H are computed via a *cell-level block bootstrap* ($B = 2{,}000$ resamples of the n_s physical cells with replacement, averaging the paired squared-error difference across the 3 seeds

within each cell so that the effective sample size is $n_s = 12$ on `Set C2` and $n_s = 180$ on `Design CV`). Tables 20–21 support three nuanced conclusions:

1. **Substantial gain of hybrid embeddings over $E_1 + J_{1.5\ell}$ alone on Set C2:** Augmenting $E_1 + J_{1.5\ell}$ ($R^2 = 0.679$–0.689, mean/max $\delta_{\sigma,\mathrm{rel}} = 2.35\%/4.09\%$) with any multi-dimensional $R_H(20)$ embedding (`MSE`, `Tail-MSE`, `RankSim`, or `SBR-CF`) lifts prospective `Set C2` R^2 to 0.886–0.939 and cuts mean relative peak-stress error to 0.94%–1.26%. For $E_1 + J_{1.5\ell} +$ `SBR-CF` ($\lambda_{\mathrm{cf}} = 1$) vs. $E_1 + J_{1.5\ell}$ alone, the cell-block paired MSE reduction is $\Delta\mathrm{MSE} = -6.10 \times 10^{-4}$ on `Set C2` (90% cell-block bootstrap CI $[-9.04 \times 10^{-4}, -3.28 \times 10^{-4}]$) and $\Delta\mathrm{MSE} = -5.57 \times 10^{-4}$ on `Design CV` (90% CI $[-6.56 \times 10^{-4}, -4.63 \times 10^{-4}]$).
2. **Comparison between +`SBR-CF` and +`MSE` on Set C2:** Relative to the strong hybrid baseline $E_1 + J_{1.5\ell} +$ `MSE` ($R^2 = 0.922$, RMSE 0.0131, $P_{90} = 0.0193$, mean $\delta_{\sigma,\mathrm{rel}} = 1.10\%$), $E_1 + J_{1.5\ell} +$ `SBR-CF` ($\lambda_{\mathrm{cf}} = 1$) improves average and 90th-percentile accuracy ($R^2 = \mathbf{0.939}$, RMSE $\mathbf{0.0116}$, $P_{90} = \mathbf{0.0191}$, mean $\delta_{\sigma,\mathrm{rel}} = \mathbf{0.94}\%$), with a modest, borderline cell-block paired MSE difference on the $n_s = 12$ cells of $\Delta\mathrm{MSE} = -5.26 \times 10^{-5}$ (90% cell-block bootstrap CI $[-1.09 \times 10^{-4}, -1.59 \times 10^{-6}]$). At the same time, the 3-seed median worst-case error on `Set C2` is slightly lower for +`SBR-CF` ($\lambda_{\mathrm{cf}} = 3$, $\max \delta_{\sigma,\mathrm{rel}} = \mathbf{2.26}\%$) and +`MSE` (2.27%) than for +`SBR-CF` ($\lambda_{\mathrm{cf}} = 1$, 2.47%).
3. **Standalone image-only transfer (no FFT solve at test time):** When predicting R_σ directly from the binary image without an elastic solve at test time, `Tail-MSE` ($\lambda_{\mathrm{tail}} = 1, 3$) achieves the strongest standalone generalization on `Design CV` ($R^2 = \mathbf{0.661}$) and `Set C2` ($R^2 = \mathbf{0.398}$, mean $\delta_{\sigma,\mathrm{rel}} = \mathbf{2.72}\%$, max $\delta_{\sigma,\mathrm{rel}} = \mathbf{7.04}\%$ at $\lambda_{\mathrm{tail}} = 3$), closely followed by standalone `SBR-CF` ($\lambda_{\mathrm{cf}} = 1$, $R^2 = 0.369$, mean/max $\delta_{\sigma,\mathrm{rel}} = 2.97\%/7.61\%$), both outperforming standalone `MSE` ($R^2 = 0.048$, $\Delta\mathrm{MSE} = -3.21 \times 10^{-4}$ for `SBR-CF` $\lambda_{\mathrm{cf}} = 1$ vs. `MSE`) and `RNC` ($R^2 = -1.342$).

Direct measurement of the AT1 strip-translation barrier $b_S(x) \le b_{G_{x_1}}(x)$.. Finally, to test whether the periodic translation invariance built into `E-circ` conflicts with the 2% weakened vertical localization strip at $x_1 = 0.95$ in the AT1 phase-field solver (Section 3.2), we evaluated 8 representative design cells (3 laminates, 3 multi-disk cells, and 2 multi-ellipse cells) across the 4-shift grid subset $S_{\mathrm{grid}} = \{0, 32, 64, 95\}$ pixels and the up-to-5-shift subset $S_{\mathrm{all}} = S_{\mathrm{grid}} \cup \{\Delta x_1^{\mathrm{adv}}\}$ that includes the cell-specific adversarial shift $\Delta x_1^{\mathrm{adv}}$ aligning each cell's maximum elastic driving hotspot $\arg\max_x(\psi_+/\psi_c)$ directly with the weakened strip $x_1 = 0.95$ (pixel 120). Because $\Delta x_1^{\mathrm{adv}} = 0 \in S_{\mathrm{grid}}$ for laminate cells 16 and 28 ($|S_{\mathrm{all}}| = 4$) while $\Delta x_1^{\mathrm{adv}} \notin S_{\mathrm{grid}}$ for the remaining 6 cells ($|S_{\mathrm{all}}| = 5$), this design comprises 38 distinct shifted cell configurations in total: the 8 unshifted $\Delta x_1 = 0$ solutions from `design_cells.csv` plus 30 newly solved $\Delta x_1 \neq 0$ AT1 trajectories (`results/pf_shifts/`), all computed with the identical `PhaseFieldRVE.run` protocol ($\delta_\varepsilon =$ `dE_rel` $= 0.01$, Eq. (32)) and all achieving complete post-peak separation (`separated = True`). Table 22 reports the empirical shift barriers $b_{S_{\mathrm{grid}}}(x)$ and $b_{S_{\mathrm{all}}}(x)$ (Eq. (36)), together with both the minimax constant-prediction relative stress error $\tanh(b_S) \approx b_S$ and the full peak-to-trough relative peak-stress spread $(\sigma_{\max} - \sigma_{\min})/\sigma_{\min} = \exp(2b_S) - 1$.

Table 22 reveals two concrete findings: (i) Across the four non-adversarial grid translations $S_{\mathrm{grid}} = \{0, 32, 64, 95\}$ px on the eight solved cells, the computed AT1 trajectories exhibit an empirical finite-subset peak-strength barrier $b^{\mathrm{grid}}_{\mathrm{shift}}(R_\sigma) \le 9.5 \times 10^{-6}$ (for R_{path}, $b^{\mathrm{grid}}_{\mathrm{shift}}(R_{\mathrm{path}}) \le 0.00182$ across the 5 disk and ellipse cells and reaches 0.0101 on laminate cell 16, where $\Delta x_1^{\mathrm{adv}} = 0 \in S_{\mathrm{grid}}$ so $S_{\mathrm{all}} = S_{\mathrm{grid}}$). We treat this strictly as an empirical observation on the tested cells and shifts rather than a general theoretical guarantee: although the undamaged threshold $\psi_c(x) = \frac{3G_c(x)}{16\ell}$ (Eq. (31)) requires only 2% less strain energy density in the strip, whether the strip remains undamaged ($d = 0$) up to peak load depends on the full nonlinear stress redistribution of the coupled elastodamage trajectory rather than on the initial elastic threshold alone. (ii) When the adversarial shift $\Delta x_1^{\mathrm{adv}}$ aligns the critical inter-inclusion ligament directly with the weakened strip $x_1 = 0.95$, every disk and ellipse cell exhibits a strictly positive finite-subset barrier $b^{\mathrm{all}}_{\mathrm{shift}}(R_\sigma) \in [2.0 \times 10^{-4}, 1.90 \times 10^{-3}]$ (corresponding to a minimax

Table 22: Direct AT1 phase-field evaluation of the finite-subset horizontal translation barrier $b_S(x) = \frac{1}{2}[\max_{T\in S} R(Tx) - \min_{T\in S} R(Tx)] \le b_{G_{x_1}}(x)$ (Eq. (36)) across 8 representative cells (38 distinct configurations: 8 base $\Delta x_1 = 0$ solutions plus 30 shifted trajectories, all solved with $\delta_\varepsilon = 0.01$) under grid shifts $S_{\text{grid}} = \{0, 32, 64, 95\}$ px ($b^{\text{grid}}_{\text{shift}}$) and $S_{\text{all}} = S_{\text{grid}} \cup \{\Delta x_1^{\text{adv}}\}$ ($b^{\text{all}}_{\text{shift}}$, up to 5 shifts per cell). Columns report $b^{\text{grid}}_{\text{shift}}(R_\sigma)$, $b^{\text{all}}_{\text{shift}}(R_\sigma)$, the minimax relative peak-stress error ($\approx b^{\text{all}}_{\text{shift}}$) and full peak-to-trough relative spread ($\exp(2b^{\text{all}}_{\text{shift}}) - 1$), and $b^{\text{all}}_{\text{shift}}(R_{\text{path}})$.

Cell ID	Family (n_{inc}, f)	Δx_1^{adv}	$\bar{R}_\sigma$	$b^{\text{grid}}_{\text{shift}}(R_\sigma)$	$b^{\text{all}}_{\text{shift}}(R_\sigma)$	Minimax / Spread $\delta_{\sigma,\text{rel}}$	$\bar{R}_{\text{path}}$	$b^{\text{all}}_{\text{shift}}(R_{\text{path}})$
13	Laminate (2, 0.252)	126 px	+0.5641	$< 10^{-15}$	$< 10^{-15}$	0.000% / 0.000%	0.4010	0.0000
16	Laminate (2, 0.252)	0 px	0.0000	$< 10^{-15}$	$< 10^{-15}$	0.000% / 0.000%	0.0101	**0.0101**[†]
28	Laminate (2, 0.378)	0 px	0.0000	$< 10^{-15}$	$< 10^{-15}$	0.000% / 0.000%	0.0202	0.0000
49	Disks (4, 0.100)	55 px	−0.4132	$< 10^{-14}$	0.00111	0.111% / 0.222%	0.0232	0.00186
96	Disks (9, 0.200)	50 px	−0.2926	7.6×10^{-9}	0.00020	0.020% / 0.041%	0.0313	0.0000
100	Disks (9, 0.200)	39 px	−0.2767	9.5×10^{-6}	0.00173	0.173% / 0.346%	0.1585	0.00444
190	Ellipses (9, 0.199)	14 px	−0.5019	2.9×10^{-8}	0.00165	0.165% / 0.331%	0.0370	0.00411
205	Ellipses (4, 0.300)	79 px	−0.2253	2.9×10^{-8}	**0.00190**	**0.190%** / **0.381%**	0.0401	0.00182

[†]For laminate cell 16, $\Delta x_1^{\text{adv}} = 0 \in S_{\text{grid}}$, so $S_{\text{all}} = S_{\text{grid}}$ and $b^{\text{grid}}_{\text{shift}}(R_{\text{path}}) = b^{\text{all}}_{\text{shift}}(R_{\text{path}}) = 0.0101$ ($R_{\text{path}} = 0.0202$ at $\Delta x_1 \in \{0, 64\}$ px vs. 0.0000 at $\Delta x_1 \in \{32, 95\}$ px); across the 5 disk and ellipse cells, $b^{\text{grid}}_{\text{shift}}(R_{\text{path}}) \le 0.00182$ and $b^{\text{all}}_{\text{shift}}(R_{\text{path}}) \le 0.00444$.

constant-prediction relative stress error of up to 0.190% and a full peak-to-trough relative peak-stress spread $\exp(2b^{\text{all}}_{\text{shift}}) - 1$ of up to 0.381%, while $b^{\text{all}}_{\text{shift}}(R_{\text{path}}) \le 0.00444$ across the 5 disk/ellipse cells and ≤ 0.0101 across all 8 cells). Because $S_{\text{all}} \subset G_{x_1}$, $b^{\text{all}}_{\text{shift}}(x) \le b_{G_{x_1}}(x)$ is a rigorous lower bound on the worst-case error of any translation-invariant model over the full horizontal translation orbit. While a lower bound on a finite subset S_{all} does not mathematically upper-bound the full orbit G_{x_1}, its small observed magnitude (≤ 0.0019 in R_σ across both grid and ligament-aligned adversarial shifts, an order of magnitude smaller than the cross-cell prediction RMSE of 0.0116–0.0266) is empirically compatible with the strong performance of periodic translation-invariant architectures on the fracture benchmark.

7. Discussion, Limitations, and Conclusions

7.1. Synthesis across the three levels of representation evaluation

By separating functional sufficiency (Level 1), metric regularity (Level 2), and readout/interpolant accuracy (Level 3), our theoretical and empirical results support five main conclusions:

1. **Metric regularity vs. functional information:** As shown in the synthetic pilot (Section 2.5) and the neural dimension sweep $d \in \{4, 8, 16\}$ (Table 13), adding weakly constrained coordinates preserves linear readout accuracy ($R^2_{\text{ridge}} = 0.967$) while degrading isotropic Mahalanobis regularity ($\Omega/\Omega^{\text{null}} = 0.433$) and nearest-anchor McShane interpolation ($R^2_{\text{McShane}} = 0.834$). Evaluating $\Omega/\Omega^{\text{null}}$, exact barrier curves $e^\star_\theta(\Lambda)$, and cross-fitted McShane errors across multiple covariance normalizations ($d_{\text{white}}, d^\star_{\text{reg}}, d_{\text{trunc}}, d_{\text{std}}, d_{\mathcal{N}}$) diagnoses both high-dimensional dilution and low-rank collapse.

2. **Morphological reconstruction vs. failure regularity:** Even when unsupervised residual autoencoders (`ConvAE-HiFi`, $d = 128$) are trained with positive-weighted BCE and soft Dice loss to achieve 93.7% pixel accuracy, IoU = 0.728, Dice = 0.825, and boundary F1 = 0.857 (far above the 79.7% all-matrix baseline), their latent distances still exhibit high worst-case insufficiency on $R_H(20)$ ($\Omega/\Omega^{\text{null}} = 0.882$, plausibly because reconstruction losses prioritize global inclusion placement over narrow inter-inclusion stress ligaments).

3. **Partial support of Hypothesis H4 and distinct geometric roles of Ω, e^{cf}, and $e^\star_\theta(\Lambda)$:** Relative to standard `MSE` ($\Omega/\Omega^{\text{null}} = 0.273$), `SBR-CF` ($\lambda_{\text{cf}} = 1, 3$) lowers whitened 1%-ball worst-case insufficiency to 0.188–0.192 on `test2` (98.3%–99.7% paired half-sample wins) and 0.169 on OOD morphologies, raises R^2_{McShane} to 0.962, lowers $\text{CF}_{\max}/\sigma_R$ to 1.165, halves the $\widetilde{\Lambda}_{\text{cert}} = 4.0$ training head excess gap (0.513 → 0.248), expels 85.5%–94.7% of severe `MSE` collisions outside

the 1% ball (with a 2.55% median relative change, max 8.13%, upon doubling the FFT grid resolution on the fixed voxelized masks), and achieves the lowest exact barriers $e^\star_\theta(\Lambda)$ on `test2` in $d_{\mathcal{N}}$ at higher budgets $\Lambda/\sigma_R \in \{2.0, 3.0\}$. At the same time, under d_{white} on `test2`, `RNC` ($\tau = 1.0$) and `Tail-MSE` ($\lambda_{\text{tail}} = 3.0$) achieve lower exact global barriers $e^\star_\theta(\Lambda)$ than `SBR-CF`, demonstrating that local 1%-ball regularity ($\Omega/\Omega^{\text{null}}$), cross-fitted midpoint error (e^{cf}), and global single-pair barriers ($e^\star_\theta(\Lambda)$) capture distinct geometric properties.

4. **Trade-off between `RNC` and `SBR-CF`:** Under coordinate standardization d_{std}, frozen training standardization $d_{\mathcal{N}}$, and validation-regularized whitening $d^\star_{\text{reg}}$, Rank-N-Contrast (`RNC`, $\tau = 1.0$) achieves the strongest single-target alignment ($\Omega/\Omega^{\text{null}} = 0.085$–$0.148$, $R^2_{\text{McShane}} = 0.989$–$0.991$) and the highest linear ridge R^2_{ridge} under contrast and biaxial shifts (0.943–0.976) by compressing the embedding onto a 1D curve ($d_{\text{eff}} = 1.004$). Conversely, `SBR-CF` maintains a multi-dimensional representation ($d_{\text{eff}} = 5.21$–5.77) that is empirically stable across all tested whitening and spectral-truncation regimes and transfers better across distinct mechanical horizons $(R_E, R_H(2))$ under whitened McShane evaluation (d_{white}), few-shot McShane interpolation, and standalone prospective fracture.

5. **Transfer to AT1 phase-field fracture:** Augmenting the solved-field critical-distance descriptor $E_1 + J_{1.5\ell}$ with frozen `SBR-CF` ($\lambda_{\text{cf}} = 1$) embeddings lifts prospective high-multiplicity `Set C2` R^2 from 0.679–0.689 to 0.939 (reducing mean relative peak-stress error from 2.35% to 0.94% and maximum error from 4.09% to 2.47%, vs. 0.922 and 1.10%/2.27% for +`MSE`), while standalone `Tail-MSE` ($\lambda_{\text{tail}} = 1, 3$) and `SBR-CF` ($\lambda_{\text{cf}} = 1$) achieve the strongest image-only transfer on `Set C2` ($R^2 = 0.398$ and 0.369).

7.2. Scope conditions, limitations, and future directions

Four scope limitations bound the present conclusions: (i) the cross-fitted McShane error $e^{\text{cf}}_{A\to Q}(\Lambda)$ evaluates a specific central extension rule M_A whose extrapolation outside the anchor support flattens to constant prediction in 1D and becomes a localized distance-offset competition in $\mathbb{R}^d$ (Remark 2), so out-of-distribution evaluations should always pair e^{cf} with $\Omega/\Omega^{\text{null}}$ and parametric readouts; (ii) on the $n = 150$ multi-inclusion phase-field design cells, the local-swap conditional permutation test has low finite-sample power (28.0% at unadjusted $\alpha = 0.05$ and 2.7% at the Bonferroni threshold $\alpha = 0.05/6$ for a signal explaining 20.1% of total variance at dimensionless amplitude $\eta = 0.50$, i.e., $0.50\sigma_{R_\sigma}$), and $k_{\text{nn}} = 5$ neighbor swaps in 6D/7D D_0 provide only approximate conditional exchangeability (which affects both positive rejections—such as the negative-control `T1` rejection for `SBR-CF` $\lambda_{\text{cf}} = 3$—and non-rejections on `T5` and `T6`); consequently, non-significant results do not rule out even large local conditional effects, and positive rejections should be interpreted together with the explicit predictive transfer evaluations in Section 6.4; (iii) exact circular translation invariance incurs a positive Level-1 translation barrier $b_S(R_\sigma) \le b_{G_{x_1}}(R_\sigma)$ (with observed $b^{\text{all}}_{\text{shift}}(R_\sigma) \le 0.0019$, or 0.190% minimax relative peak-stress error and 0.381% peak-to-trough spread across the tested grid and ligament-aligned adversarial shifts) when the weakened localization strip is aligned adversarially with the dominant inter-inclusion ligament, which could be eliminated in future work by feeding the strip coordinate Channel 2 to the encoder; and (iv) under frozen pre-critical elastic embeddings paired with linear ridge readouts, the normalized minimum-cost percolating crack-path cost R_{path} (Eq. (34)) is not captured on the valid percolating cells ($R^2 \le 0.12$; note that this evaluates the extracted path-cost functional $C_{\text{path}}(d_{\text{final}})$ rather than the bulk AT1 fracture energy $\Gamma_\ell(d_{\text{final}})$), motivating multi-stage damage-evolution surrogates, nonlinear readouts, and contrast/load-conditioned neural operators.

7.3. Data and code availability

All source modules, execution and verification scripts, dataset split manifests, and CSV/JSON artifacts required to reproduce every table and figure in this manuscript are openly available in the companion GitHub repository at `https://github.com/aegovalle/sufficiency-ml`

(building on the frozen phase-field validation protocol archived at `https://github.com/aegovalle/energy-failure-descriptors`) and packaged in the self-contained verification archive `Sufficiency_ML_Referee_Artifacts.zip`:

- **Core library (`src/suffml/`):** `metrics/metric.py` (`WhiteningTransform`, `RegWhiteningTransform`, `TruncatedWhiteningTransform`, `StdTransform`, `effective_rank`), `metrics/sufficiency.py` and `analysis/subsample.py` ($\Omega(q)$, $\Omega/\Omega^{\mathrm{null}}$, half-sample split bootstrap), `metrics/lipschitz_barrier.py` ($e^\star(\Lambda)$, McShane envelopes U_A, L_A, M_A, $e^{\mathrm{cf}}_{A\to Q}(\Lambda)$), `models/encoder_circ.py`, `models/normalization.py`, and `models/autoencoder.py` (`Encoder`, `EncoderFNO`, `LatentNorm`, `ConvAutoencoder`, `ConvAutoencoderHiFi`), `models/lipschitz_head.py` (`SpectralLinear`, `LipschitzHead`, `finalize_spectral`, `certified_lambda`), `train/train_objectives.py` (`MSE`, `Tail-MSE`, `RNC`, `SBR-CF`, memory queue, mining strategies), `physics/fft_rve.py` (Moulinec–Suquet solver), `physics/pf_fft.py` (`PhaseFieldRVE` AT1 staggered solver), and `metrics/conditional_test.py` (local-swap permutation tests).
- **Execution and verification scripts (`scripts/`):** `run_gate.py`, `run_experiments_de.py`, `run_consolidation_eval.py`, `run_phase1_counterexamples_abl7_transfer.py`, `run_phase2_pf_transfer_and_cond_test.py`, `run_phase3_ablations.py`, `run_phase4_ood_physics_and_rotation.py`, `run_phase5a_ml_and_stats_controls.py`, `run_phase5b_pf_strip_shifts.py`, `make_paper_figures.py`, and `build_manuscript_v2.py`.
- **Tabular and per-trajectory results (`results/` and `data/`):** `robust_metrics.csv`, `de_metrics.csv`, `consolidation_metrics.csv`, `ablation7_metrics.csv`, `ablations_phase3_metrics.csv`, `transfer_matrix_i16.csv`, `ood_physics_rotation_metrics.csv`, `pf_transfer_i19.csv`, `phase5_whitening_and_tailmse.csv`, `phase5_barrier_curves.csv`, `phase5_ae_reconstruction.csv`, `phase5_counterexample_convergence_254.csv`, `phase5_pf_extreme_metrics.csv`, `phase5_pf_strip_shifts.csv`, `phase5_pf_strip_barrier_summary.csv`, `pf_shifts/*.json` (all 30 shifted AT1 solver histories), `phase5_cond_test_combined.csv`, `phase5_cond_power_t5.csv`, `data/stage0/design_cells.csv`, `data/frozen/blind_pf.csv` (including the two non-percolating en-echelon records `bid` $= 72$ and `bid` $= 106$), `data/frozen/tcd_pf.csv`, and `data/processed/splits.json`.

Appendix A. Mathematical Proofs, Counterexample Derivations, and Extended Tables

Appendix A.1. Proof of Theorem 1 and Proposition 2

Proof of Theorem 1 and Proposition 2. Let $g : \mathbb{R}^d \to \mathbb{R}$ be any Λ-Lipschitz function with respect to d_z, and let $e_\infty(g) = \max_{1\le i\le n} |R_i - g(z_i)|$. For any pair of sample indices $i, j \in \{1, \dots, n\}$, the triangle inequality and the Λ-Lipschitz condition give

$$|R_i - R_j| \le |R_i - g(z_i)| + |g(z_i) - g(z_j)| + |g(z_j) - R_j| \le 2e_\infty(g) + \Lambda d_z(i, j).$$

Rearranging and taking the positive part yields $e_\infty(g) \ge \frac{1}{2}\left[|R_i - R_j| - \Lambda d_z(i, j)\right]_+$ for all i, j; maximizing over i, j establishes the universal lower bound $e_\infty(g) \ge e^\star(\Lambda)$.

For attainment on the anchor set $A = \{1, \dots, n\}$, consider $U_A(z) = \min_{i\in A}[R_i + \Lambda d_z(z, z_i)]$, $L_A(z) = \max_{i\in A}[R_i - \Lambda d_z(z, z_i)]$, and $M_A(z) = \frac{1}{2}(U_A(z) + L_A(z))$. Because each map $z \mapsto R_i \pm \Lambda d_z(z, z_i)$ is Λ-Lipschitz by the reverse triangle inequality $|d_z(z, z_i) - d_z(z', z_i)| \le d_z(z, z')$, the pointwise minimum U_A, maximum L_A, and their average M_A are globally Λ-Lipschitz on $\mathbb{R}^d$. At any anchor z_k ($k \in A$), setting $i = k$ in the definitions gives $U_A(z_k) \le R_k \le L_A(z_k)$, with

$$R_k - U_A(z_k) = \max_{i\in A}\left[R_k - R_i - \Lambda d_z(i, k)\right] \ge 0, \qquad L_A(z_k) - R_k = \max_{i\in A}\left[R_i - R_k - \Lambda d_z(i, k)\right] \ge 0.$$

Since $M_A(z_k) - R_k = \frac{1}{2}([L_A(z_k) - R_k] - [R_k - U_A(z_k)])$ and the difference of two non-negative numbers is bounded in magnitude by their maximum, we obtain

$$|M_A(z_k) - R_k| \le \frac{1}{2}\max\left(R_k - U_A(z_k),\ L_A(z_k) - R_k\right) = \frac{1}{2}\max_{i\in A}\left[|R_i - R_k| - \Lambda d_z(i,k)\right]_+ \le e_A^\star(\Lambda).$$

Taking the maximum over $k \in A$ and combining with the universal lower bound $e_\infty(M_A) \ge e_A^\star(\Lambda)$ yields $\max_{k\in A}|M_A(z_k) - R_k| = e_A^\star(\Lambda)$. □

Appendix A.2. Detailed seed-level conditional permutation p-values and power calibration

Table A.23 reports the individual seed-level Holm-corrected p-values ($p_{\mathrm{Holm},s0}, p_{\mathrm{Holm},s1}, p_{\mathrm{Holm},s2}$) and the root-mean-square ensemble-distance test $p^{\mathrm{ens}}_{\mathrm{Holm}}$ across `T1`–`T6` for `MSE`, `Tail-MSE` ($\lambda_{\mathrm{tail}} = 3$), and `SBR-CF` ($\lambda_{\mathrm{cf}} = 1, 3$), complementing the seed-combined test in Table 19. Table A.24 reports the Monte Carlo Type-I error and power calibration of the local-swap conditional permutation test on `T5` ($R_\sigma \mid E_1 + J_{1.5\ell}$, Eq. (41), $n = 150$ multi-inclusion cells, 150 independent replicates per signal amplitude η), including both unadjusted rejection rates ($\alpha = 0.05, 0.10$) and the single-test Bonferroni rejection rate ($\alpha_{\mathrm{Bonf}} = 0.05/6 \approx 0.00833$, matching the strictest step-down threshold of the 6-test Holm family).

Table A.23: Seed-level and ensemble-distance local-swap conditional permutation tests (`T1`–`T6`, $B = 2{,}000$, $q = 1\%$, $k_{\mathrm{nn}} = 5$) on $n = 150$ multi-inclusion phase-field design cells. Columns report the seed-combined $p^{\mathrm{comb}}_{\mathrm{Holm}}$, the ensemble-distance $p^{\mathrm{ens}}_{\mathrm{Holm}}$, and individual seed Holm-corrected p-values ($s = 0, 1, 2$).

Model Family	Test	$\bar\Omega/\bar\Omega^{\mathrm{cnull}}$	$p^{\mathrm{comb}}_{\mathrm{raw}}$	$p^{\mathrm{comb}}_{\mathrm{Holm}}$	$\Omega_{\mathrm{ens}}/\Omega^{\mathrm{cnull}}_{\mathrm{ens}}$	$p^{\mathrm{ens}}_{\mathrm{Holm}}$	$p_{\mathrm{Holm},s0}$	$p_{\mathrm{Holm},s1}$	$p_{\mathrm{Holm},s2}$
`MSE` ($R_H(20)$)	`T1`: $R_E \mid E_1$	1.037	0.509	0.509	1.050	0.692	1.000	1.000	0.722
	`T2`: $R_H(20) \mid E_1$	0.754	0.024	0.110	0.549	**0.017**	0.210	1.000	**0.024**
	`T3`: $R_H(20) \mid E_1 + J_{1.5\ell}$	0.640	0.0015	**0.009**	0.598	**0.012**	0.225	**0.030**	**0.047**
	`T4`: $R_\sigma \mid E_1$	0.707	0.022	0.110	0.658	0.122	0.225	1.000	0.088
	`T5`: $R_\sigma \mid E_1 + J_{1.5\ell}$	0.893	0.194	0.411	0.868	0.655	0.804	1.000	0.290
	`T6`: $R_{\mathrm{path}} \mid E_1 + J_{1.5\ell}$	0.837	0.137	0.411	0.784	0.655	1.000	1.000	**0.047**
`Tail-MSE` ($\lambda_{\mathrm{tail}} = 3$)	`T1`: $R_E \mid E_1$	0.848	0.160	0.314	0.893	0.525	0.664	0.742	0.752
	`T2`: $R_H(20) \mid E_1$	0.563	0.0005	**0.003**	0.506	**0.024**	**0.009**	0.225	**0.003**
	`T3`: $R_H(20) \mid E_1 + J_{1.5\ell}$	0.609	0.0005	**0.003**	0.562	**0.012**	**0.005**	0.369	**0.006**
	`T4`: $R_\sigma \mid E_1$	0.657	0.0055	**0.022**	0.576	**0.036**	0.126	0.369	0.091
	`T5`: $R_\sigma \mid E_1 + J_{1.5\ell}$	0.762	0.0145	**0.043**	0.779	0.236	0.157	0.742	**0.039**
	`T6`: $R_{\mathrm{path}} \mid E_1 + J_{1.5\ell}$	0.931	0.157	0.314	0.963	0.525	0.664	0.561	0.752
`SBR-CF` ($\lambda_{\mathrm{cf}} = 1$)	`T1`: $R_E \mid E_1$	0.895	0.261	0.523	0.754	0.298	0.665	1.000	0.596
	`T2`: $R_H(20) \mid E_1$	0.501	0.0005	**0.003**	0.436	**0.015**	**0.045**	0.294	**0.003**
	`T3`: $R_H(20) \mid E_1 + J_{1.5\ell}$	0.550	0.0005	**0.003**	0.479	**0.017**	**0.045**	0.294	**0.005**
	`T4`: $R_\sigma \mid E_1$	0.726	0.065	0.262	0.573	**0.042**	0.396	0.664	0.596
	`T5`: $R_\sigma \mid E_1 + J_{1.5\ell}$	0.876	0.090	0.271	0.842	0.298	**0.045**	1.000	0.616
	`T6`: $R_{\mathrm{path}} \mid E_1 + J_{1.5\ell}$	0.899	0.539	0.539	0.537	0.111	0.674	1.000	0.638
`SBR-CF` ($\lambda_{\mathrm{cf}} = 3$)	`T1`: $R_E \mid E_1$	0.644	0.0035	**0.014**	0.661	0.183	0.212	0.082	0.242
	`T2`: $R_H(20) \mid E_1$	0.387	0.0010	**0.005**	0.388	**0.018**	**0.015**	**0.006**	**0.009**
	`T3`: $R_H(20) \mid E_1 + J_{1.5\ell}$	0.448	0.0005	**0.003**	0.430	**0.020**	**0.012**	**0.007**	**0.042**
	`T4`: $R_\sigma \mid E_1$	0.668	0.046	0.139	0.739	0.294	0.463	0.231	0.242
	`T5`: $R_\sigma \mid E_1 + J_{1.5\ell}$	0.945	0.307	0.307	0.735	0.078	0.463	0.710	0.616
	`T6`: $R_{\mathrm{path}} \mid E_1 + J_{1.5\ell}$	0.669	0.082	0.164	0.723	0.359	0.463	0.710	0.351

Appendix A.3. Top-10 consensus counterexample 254×254 *FFT grid-convergence table*

Table A.25 reports the 127×127 vs. 254×254 Moulinec–Suquet FFT solver stability verification (via 2×2 sub-voxel splitting of the canonical binary masks) across all top-10 consensus `MSE` counterexample pairs on `test2`.

Table A.24: Empirical Type-I error ($\eta = 0$) and statistical power curve of the local-swap conditional permutation test on `T5` ($R_\sigma \mid E_1 + J_{1.5\ell}$, Eq. (41), $n = 150$ cells, $q = 1\%$, $k_{\mathrm{nn}} = 5$, 150 Monte Carlo replicates per η) as a function of the dimensionless signal parameter η (representing a conditional signal amplitude of $\eta\sigma_{R_\sigma}$). Even at $\eta = 0.50$ ($0.50\sigma_{R_\sigma}$, 20.1% of total variance), unadjusted $\alpha = 0.05$ power is 28.0% (2.7% at the single-test Bonferroni threshold $\alpha_{\mathrm{Bonf}} = 0.05/6$), confirming that non-significant results cannot rule out large conditional effects.

Signal η	Total Var. Share	Residual Var. Share	Mean $\Omega/\Omega^{\mathrm{cnull}}$	Reject ($\alpha = 0.05$)	Reject ($\alpha = 0.10$)	Reject (Bonf. $\alpha = 0.05/6$)	Median p_{raw}
0.00 (Null)	0.0%	0.0%	0.968	**0.020**	**0.060**	**0.000**	0.384
0.10	1.0%	16.9%	0.973	0.000	0.027	0.000	0.380
0.15	2.2%	31.3%	0.967	0.033	0.067	0.000	0.365
0.20	3.9%	44.8%	0.965	0.020	0.067	0.000	0.336
0.25	5.9%	55.9%	0.965	0.027	0.073	0.007	0.350
0.35	11.0%	71.3%	0.930	0.073	0.200	0.007	0.226
0.50	20.1%	83.5%	0.869	**0.280**	**0.513**	**0.027**	0.098

Table A.25: FFT solver stability verification (127×127 vs. 254×254 elasticity solves via 2×2 sub-voxel splitting on the fixed voxelized masks) across the top-10 consensus `MSE` counterexample pairs on `test2`. Median relative difference in $|\Delta R_H(20)|$ is 2.55% (max 8.13%).

Rank	Pair (i, j)	$R_H(20)_i$ ($127^2/254^2$)	$R_H(20)_j$ ($127^2/254^2$)	$\lvert\Delta R_H\rvert_{127}$	$\lvert\Delta R_H\rvert_{254}$	Rel. Diff. (%)	$\lvert\Delta R_E\rvert_{127}$	$\lvert\Delta R_E\rvert_{254}$
1	(2751, 3270)	13.777/13.925	18.289/18.502	4.512	4.577	1.42%	0.1046	0.1058
2	(2723, 3270)	13.802/13.988	18.289/18.502	4.487	4.514	0.58%	0.1672	0.1686
3	(2340, 3207)	13.207/12.974	9.115/9.134	4.092	3.840	6.56%	0.0224	0.0232
4	(2370, 2464)	5.535/5.619	9.514/9.694	3.980	4.075	2.33%	0.1248	0.1258
5	(2505, 2856)	13.810/14.017	9.863/9.993	3.947	4.023	1.90%	0.1156	0.1163
6	(2370, 2376)	5.535/5.619	9.464/9.589	3.929	3.970	1.02%	0.1019	0.1027
7	(2745, 3028)	13.136/13.358	9.369/9.484	3.767	3.874	2.77%	0.0644	0.0655
8	(2527, 3105)	9.235/9.541	12.998/13.439	3.763	3.898	3.47%	0.1298	0.1304
9	(2659, 2781)	9.539/9.744	13.287/13.823	3.748	4.080	8.13%	0.1452	0.1470
10	(2462, 3105)	9.258/9.553	12.998/13.439	3.740	3.887	3.77%	0.1435	0.1444

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